

10. 光的衍射

(第13章)

本章内容

问题？

- 10.1 光波衍射基本理论
- 10.2 菲涅耳衍射近似
- 10.3 夫琅和费衍射近似
- 10.4 光学成像系统的衍射和分辨本领
- 10.5 多缝夫琅和费衍射
- 10.6 衍射光栅
- 10.7 二元光学
- 10.8 衍射图像生成

问题？

■ 现象

光波：通过有限空域被限束 → 弯折？弥散？汇聚？图案化排布？
出现：图案化的？扩展的？压缩的？ → 光场(光能空间分布)

✓ 经典表征

波动光学：长时间、较大尺度空域 → 累积、时空平均效果 → 强度(平方振幅)可视
振幅叠加 → 强度变动：同相 → 振幅增大，反相 → 振幅减小/抵消

✓ 量子表征

概率表征：某时刻、某空间点 → 光子出现概率

(所携带的：能量、动量 → 概率出现)

测不准原理 $(\Delta x \cdot \Delta p \geq \frac{\hbar}{2}, \Delta E \cdot \Delta T \geq \frac{\hbar}{2})$

■ 条件

✓ 远场

仪器、装置、元件、感测结构、等，特征尺寸 → 波长级 (衍射)

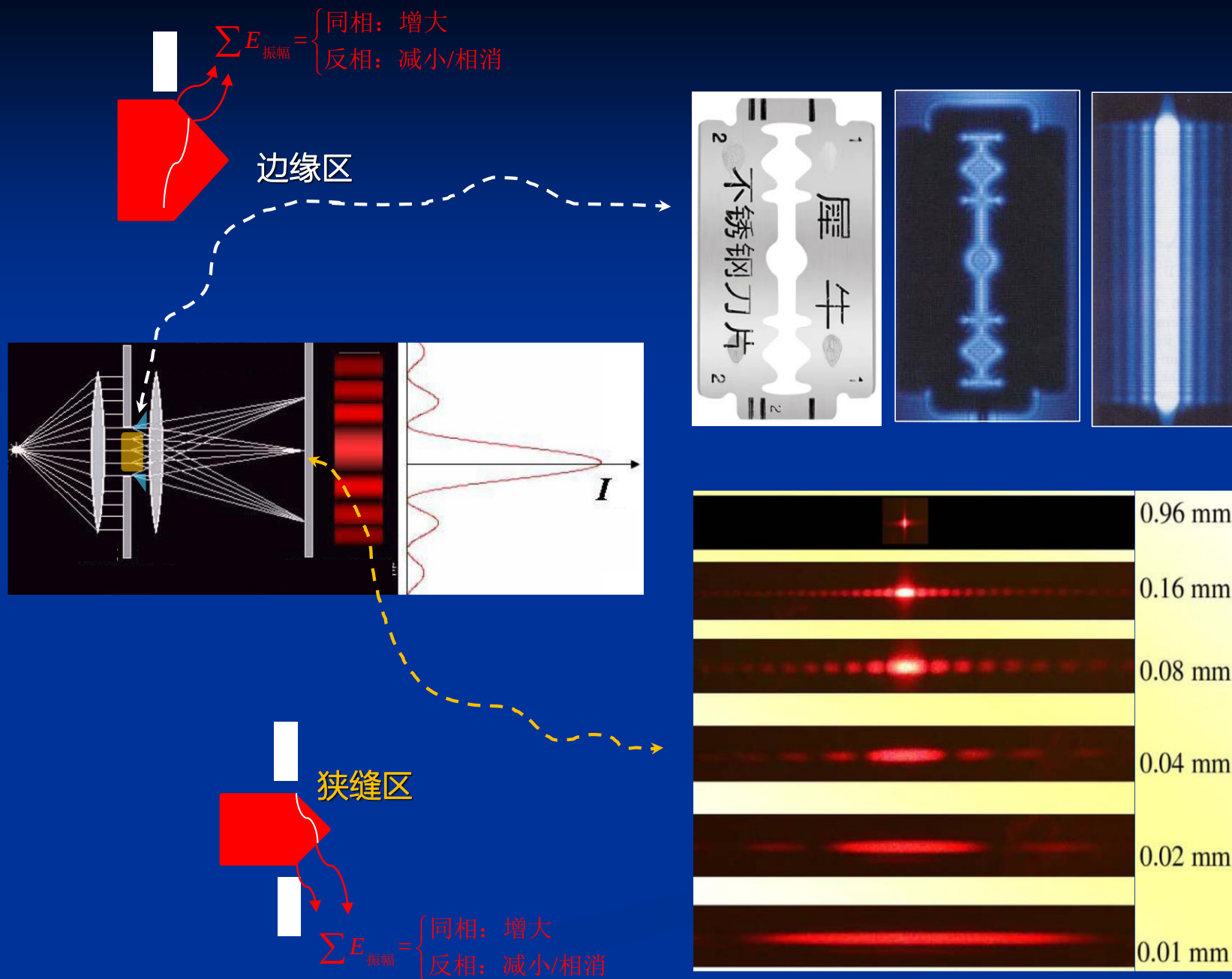
特征尺寸 $\gg$ 波长级 (不考虑衍射)

✓ 临近场

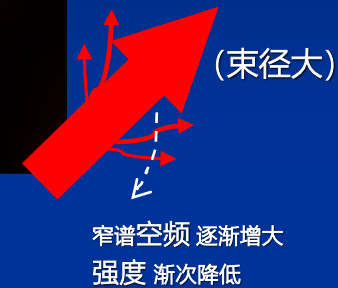
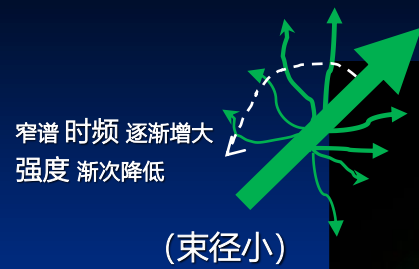
✓ 近场

微米级 (红外、THz)

纳米级 (可见光)

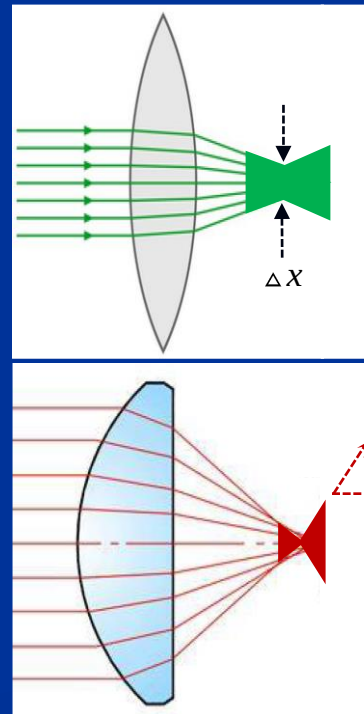


光束粗细



$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

(最小约: 2/3波长)



$$\Delta p \downarrow \Rightarrow \Delta x \uparrow$$

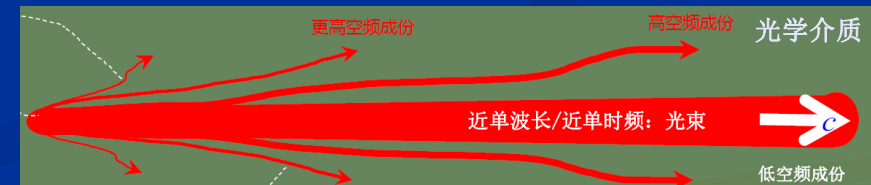
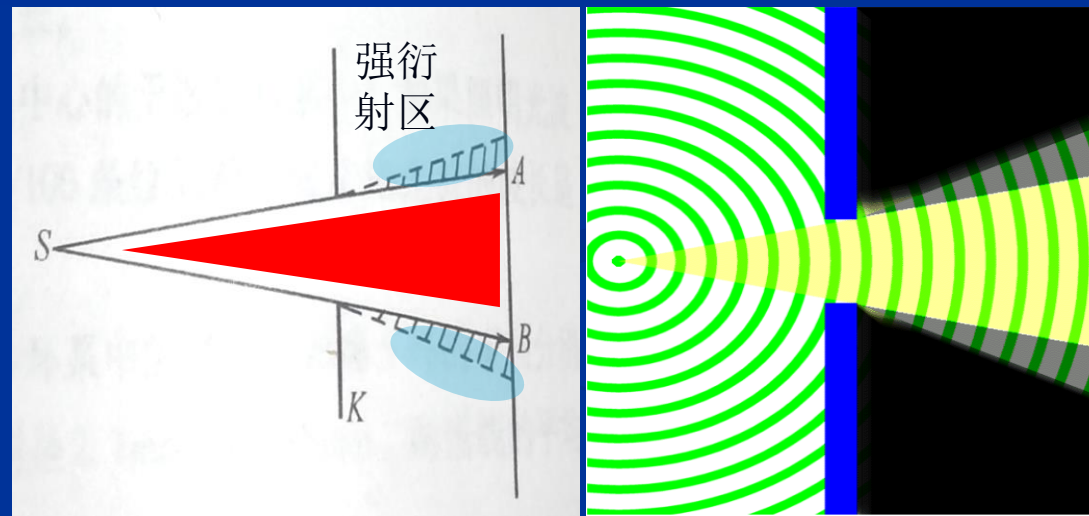
$$\Delta p \uparrow \Rightarrow \Delta x \downarrow$$

焦斑大小

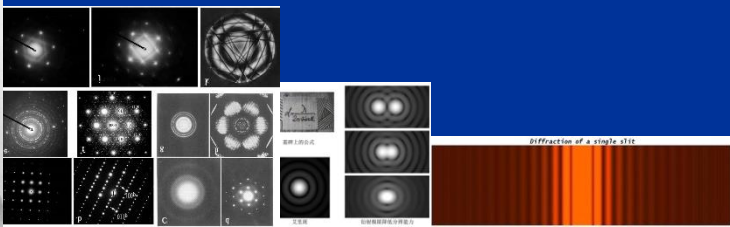
10.1 光波衍射基本理论

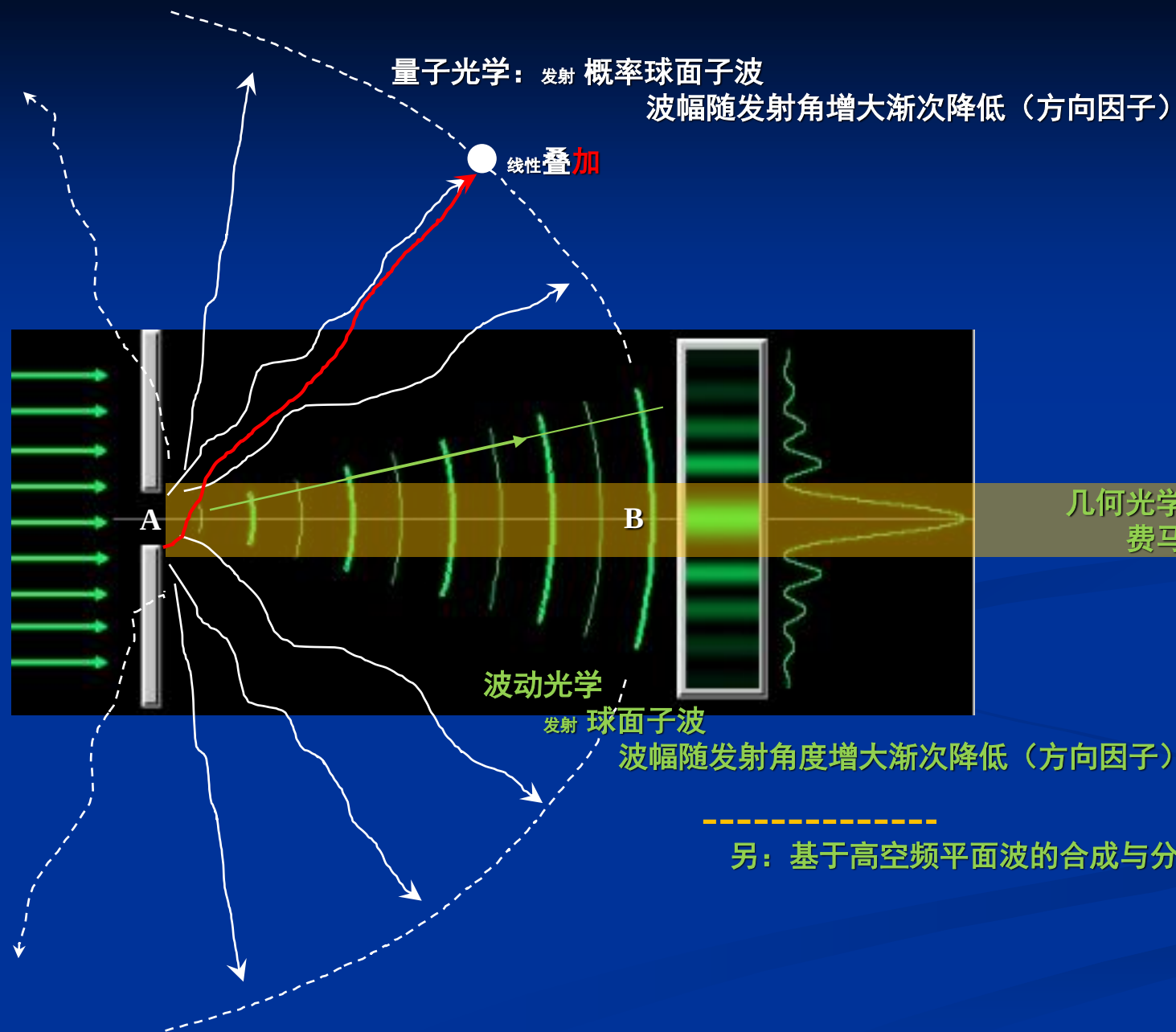
问题 { 反射、折射无法解释的：光线偏离直线光路、高低空频(时频)成份渐次分离 (索莫菲唯像定义)
光波与介质：介质边缘出现模糊→图案、花纹 (光能/振幅：图案化分布)

(1) 球面波 合成与分解



(2) 平面波 合成与分解



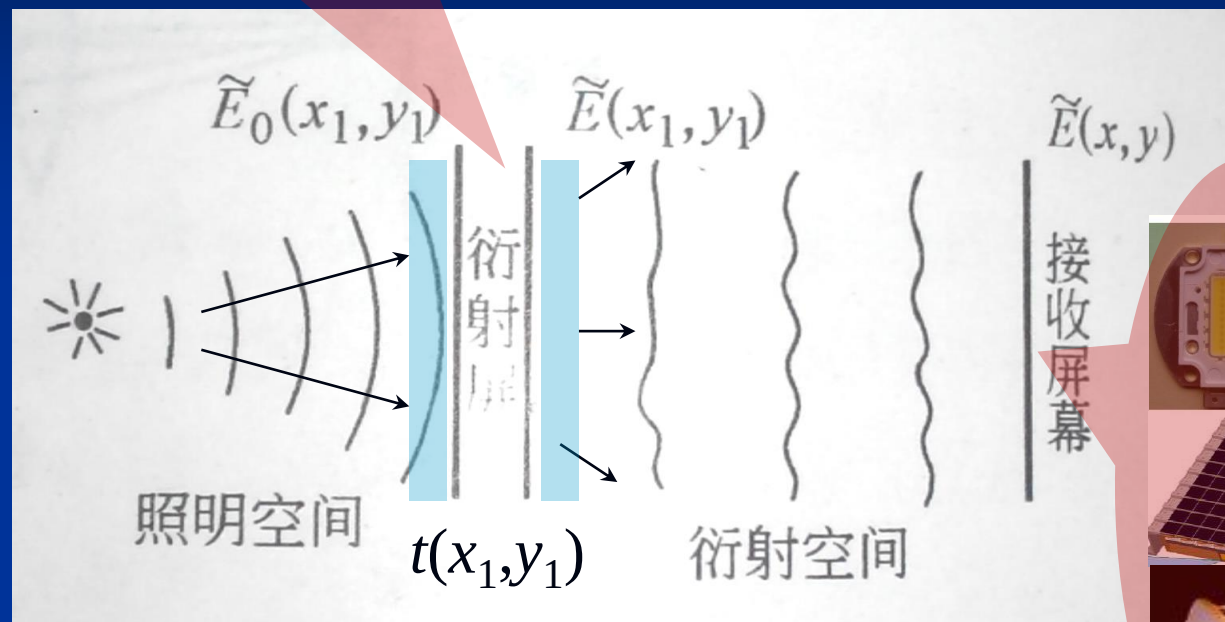
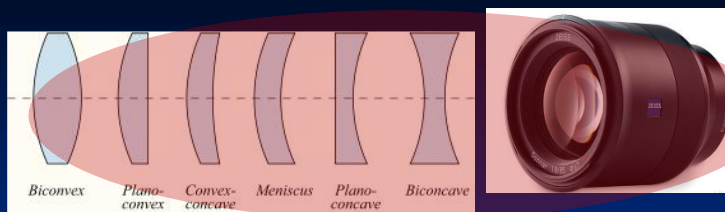


几何光学
费马原理

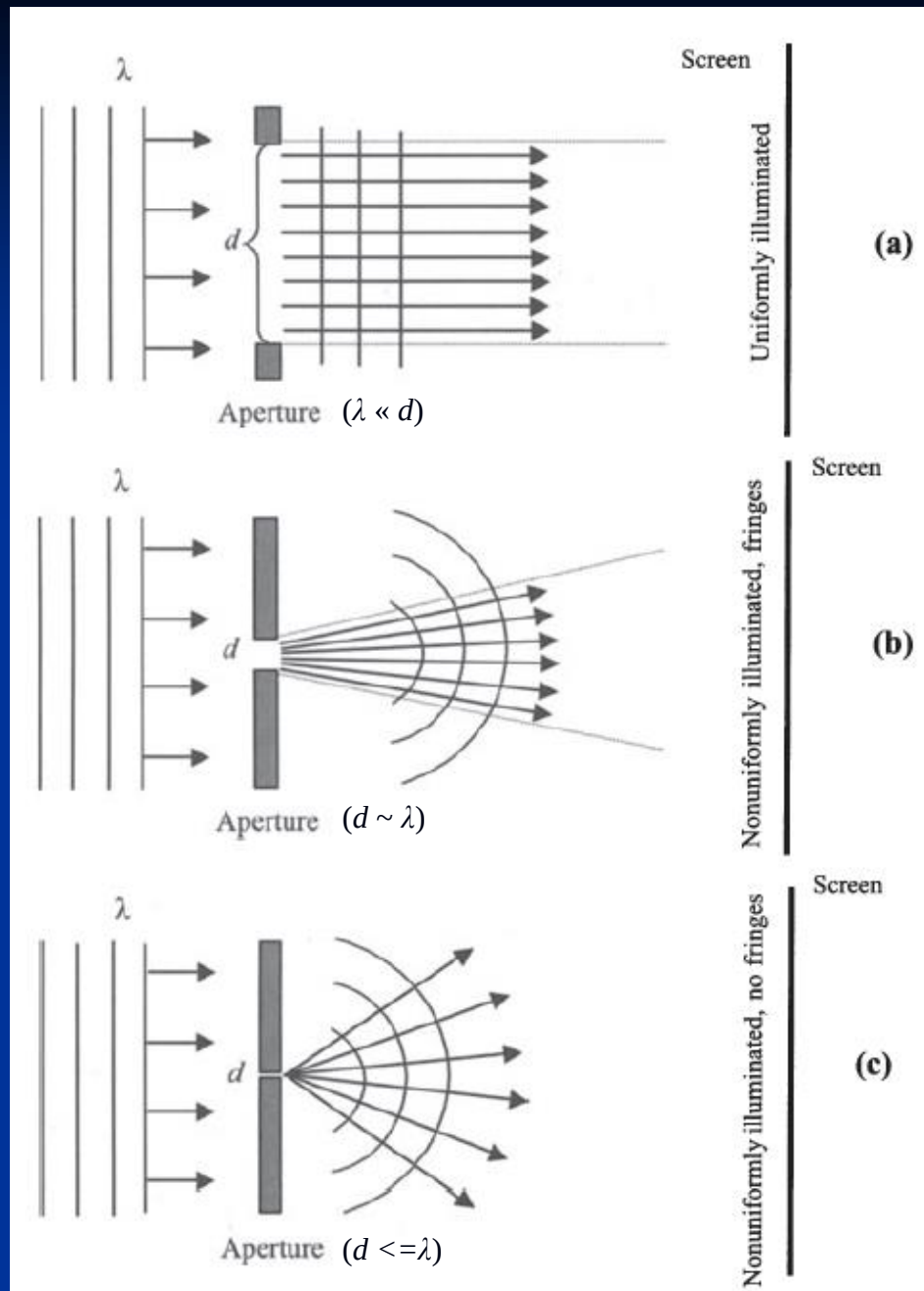
$$\delta \int_A^B n dl = 0$$

能量占比: >80%

另: 基于高空频平面波的合成与分解 → 傅里叶变换

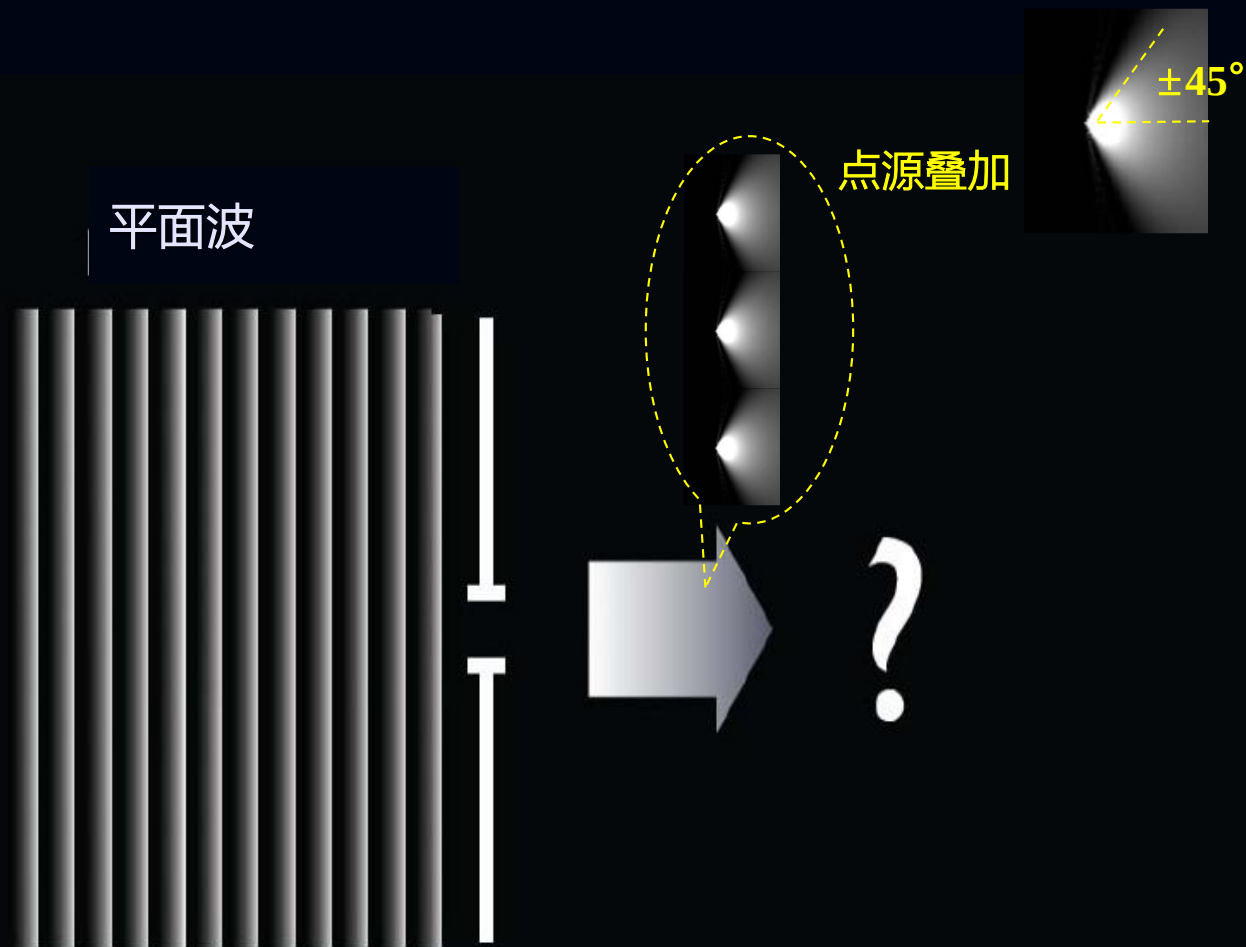


$$\begin{aligned} \text{屏复透射系数: } t(x_1, y_1) &= \frac{\tilde{E}(x_1, y_1)}{\tilde{E}_0(x_1, y_1)} \\ &= A(x_1, y_1) \exp[j\varphi(x_1, y_1)] \end{aligned}$$



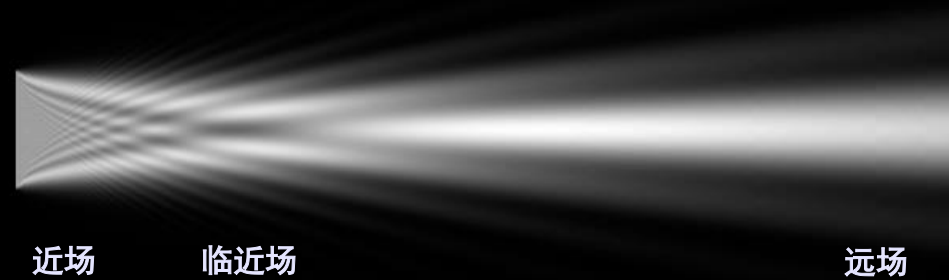
位置、方向

(经典波动光学)



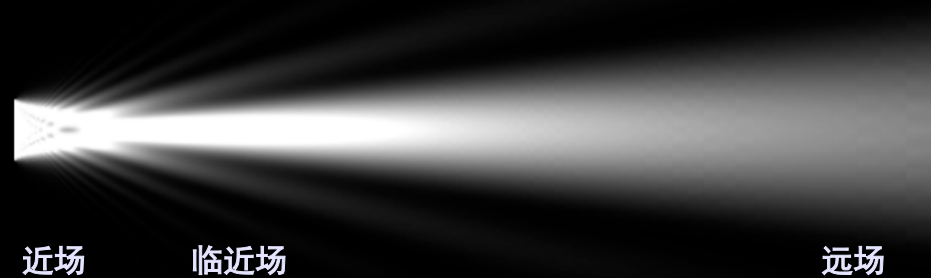
位置、方向

(经典波动光学)



孔径=128 波长

位置、方向 (经典波动光学)



孔径=64 波长

位置、方向 (经典波动光学)



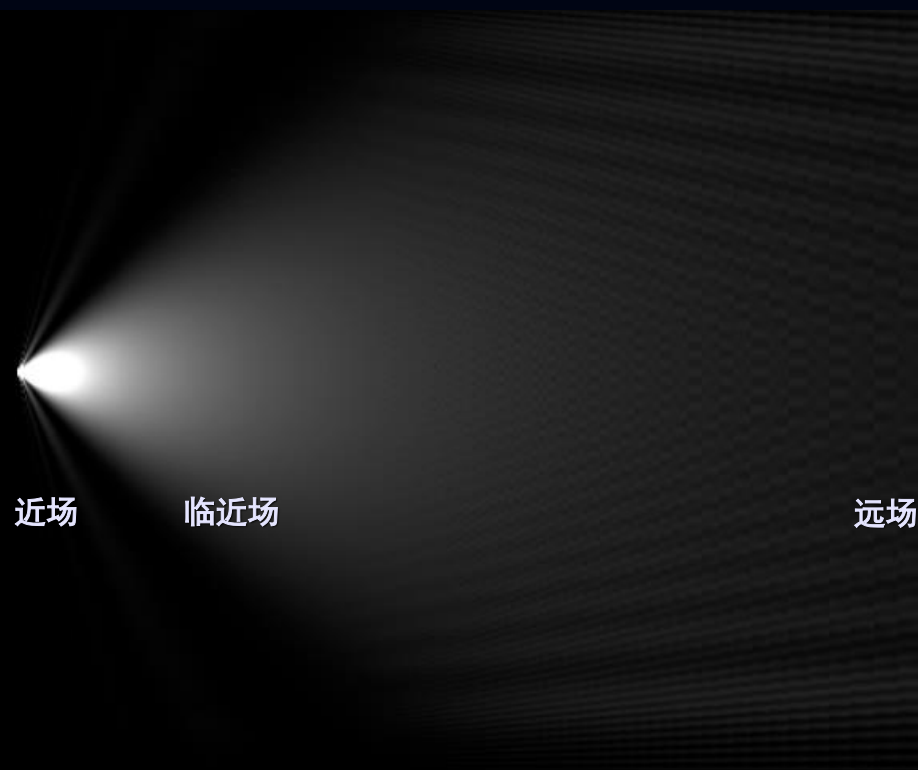
孔径=32 波长

位置、方向 (经典波动光学)



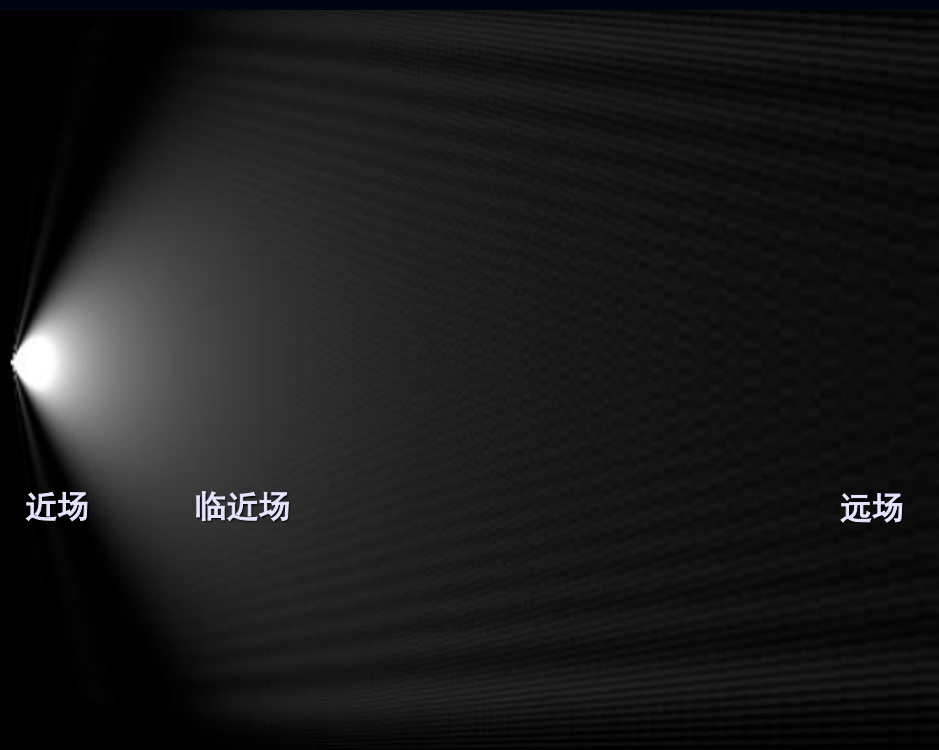
孔径=16 波长

位置、方向 (经典波动光学)



孔径=8 波长

位置、方向 (经典波动光学)



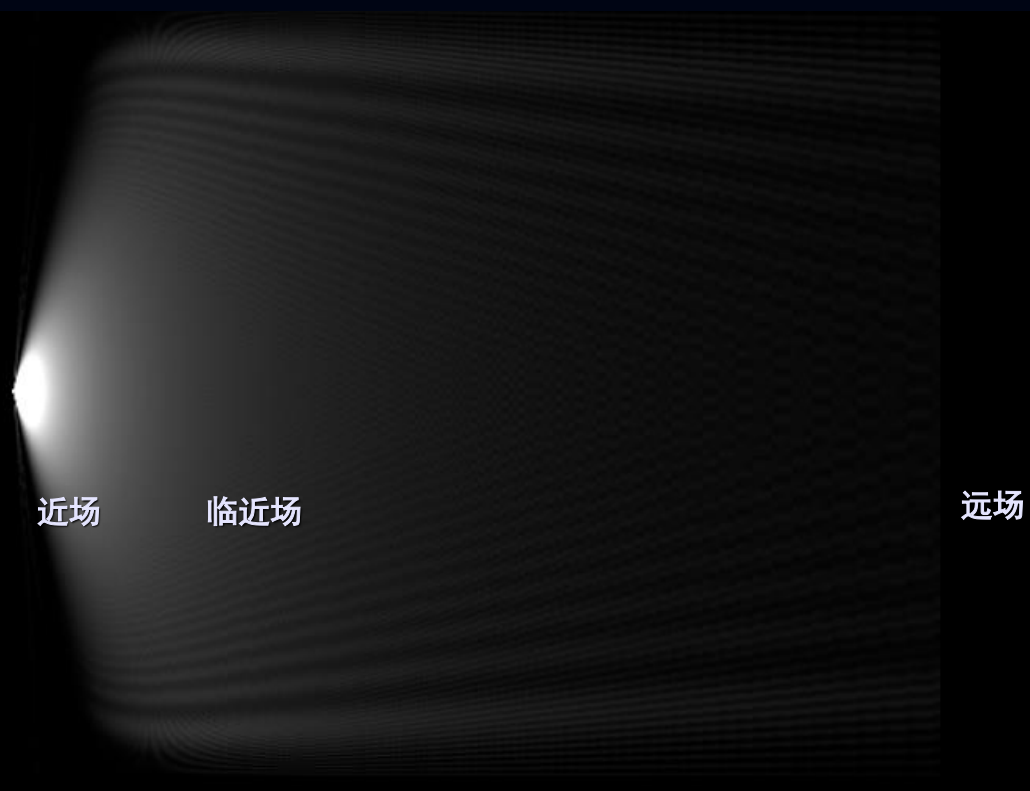
近场

临近场

远场

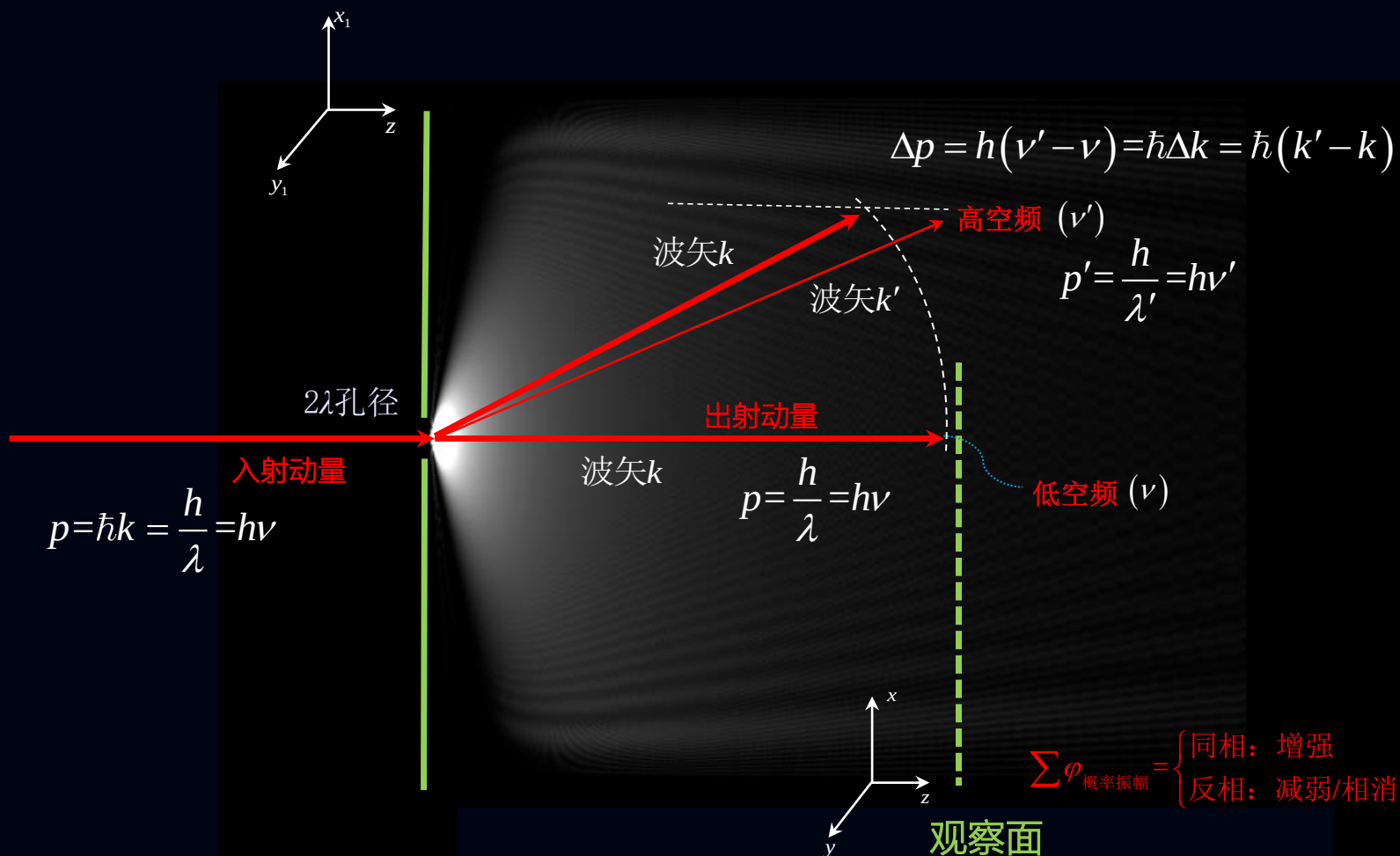
孔径=4 波长

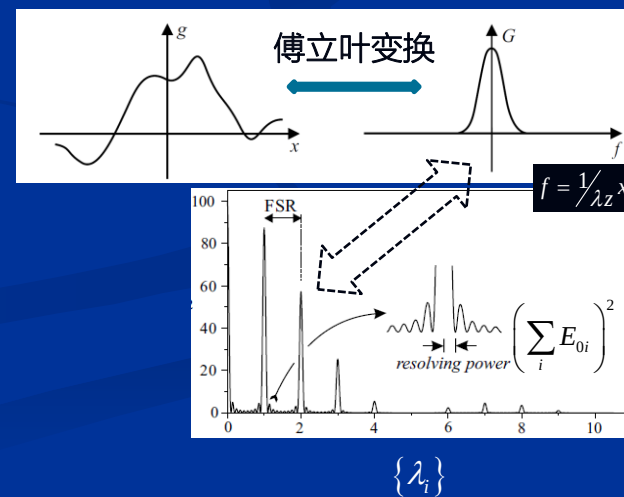
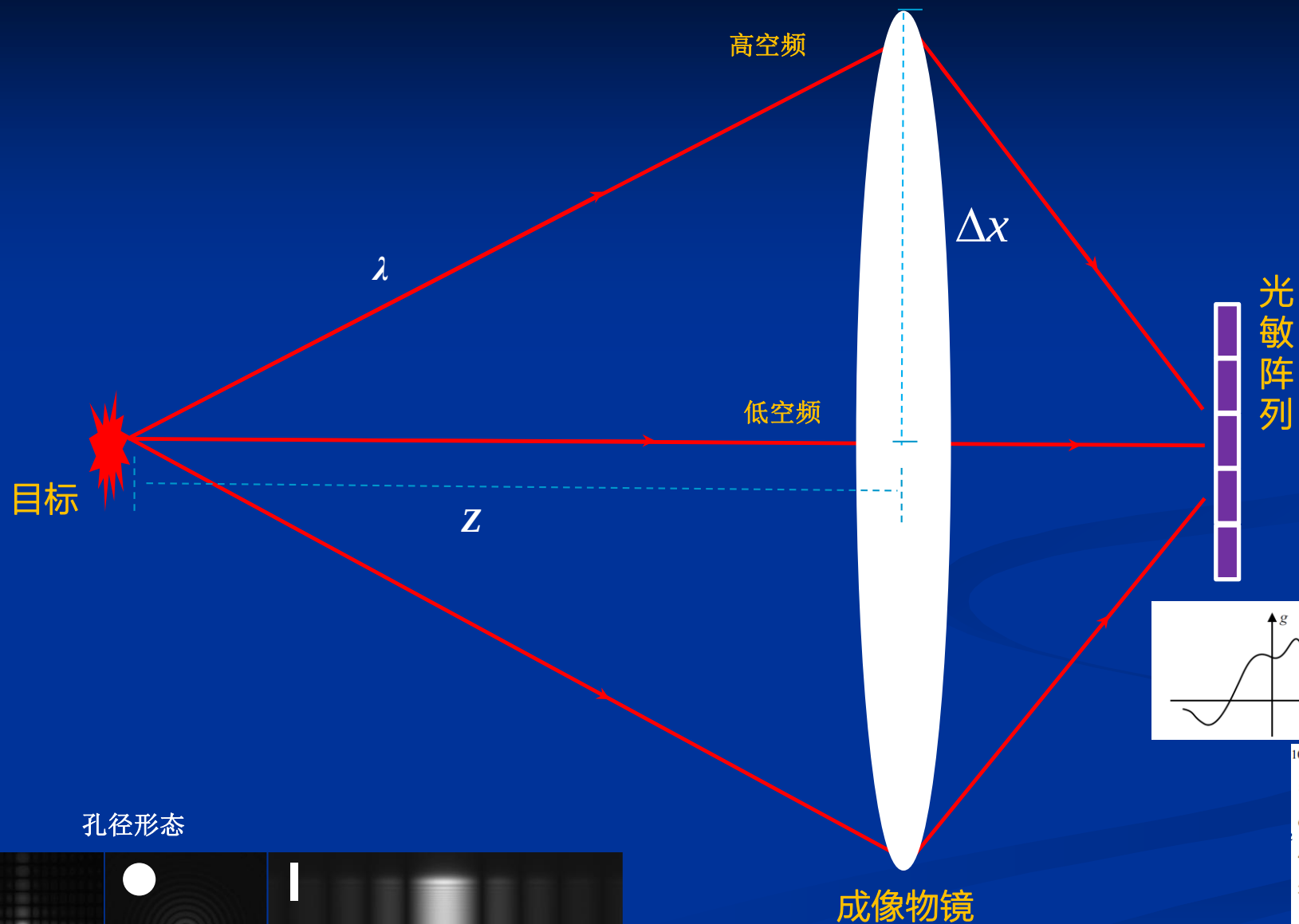
位置、方向 (经典波动光学)



孔径=2 波长

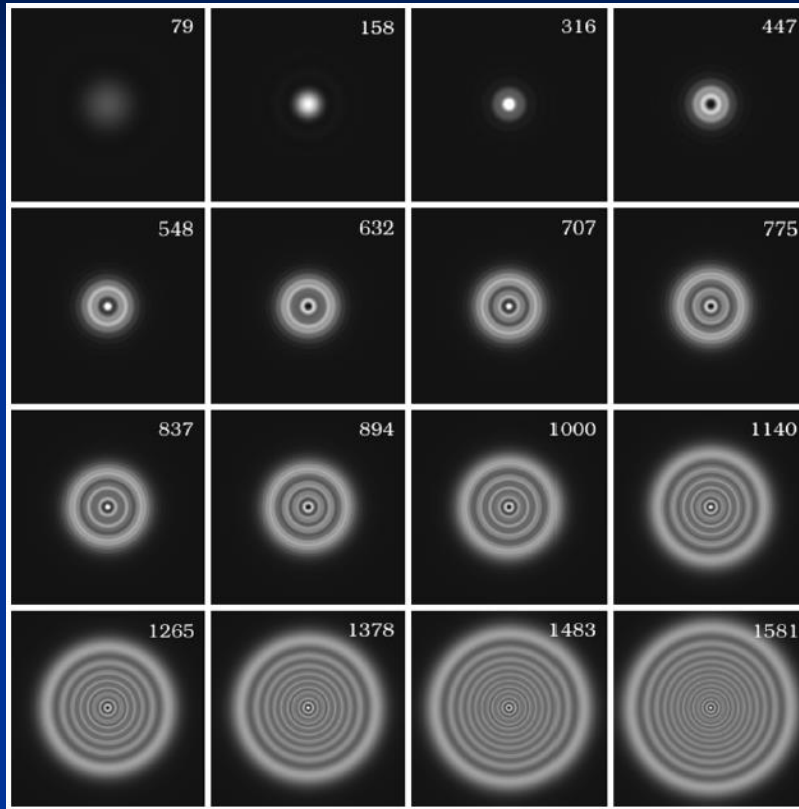
观察位置、光子动量、光波空频





孔径形态

衍射光场



Diffraction figures for circular apertures of increasing radius.

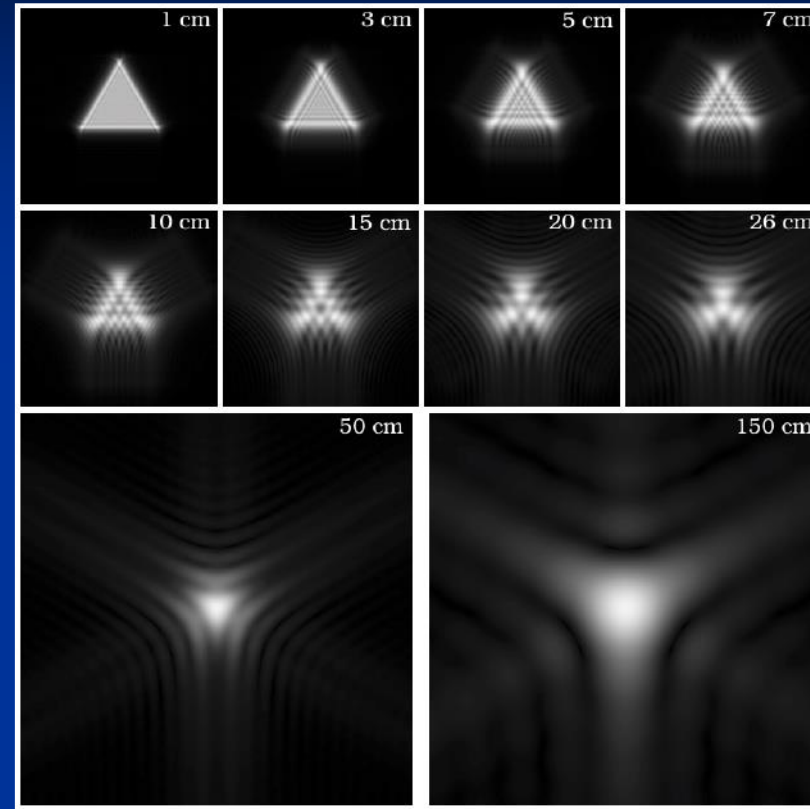
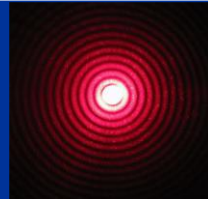
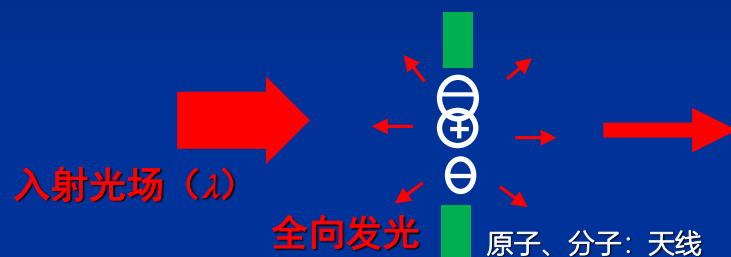


Fig. 4.5 Simulation of the diffraction pattern produced by a triangular aperture on a monochromatic plane wave. The height of the triangle is 1.5 mm and the wavelength is 1 μm . The figures are taken at distances from the opening indicated at the top, and their width is 4mm $\times$ 4mm for the first eight and 8mm $\times$ 8mm for the other two. The figure

光波(光子)衍射 根源?

1) 经典: # 惠更斯原理 (朦胧的点空间光子全向子波发射)

光场激励介质电荷体系振动→原子分子天线→全向发光→叠加(衍射)出射

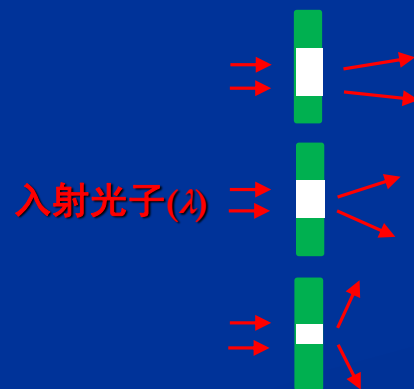


A) 球面子波叠加



2) 量子: 光子的点空间全向概率波发射

位置、动量→不能同时精确测量



B) 傅里叶子平面波展开

$$\Delta x \cdot \Delta p_x \geq \frac{1}{2} \hbar$$

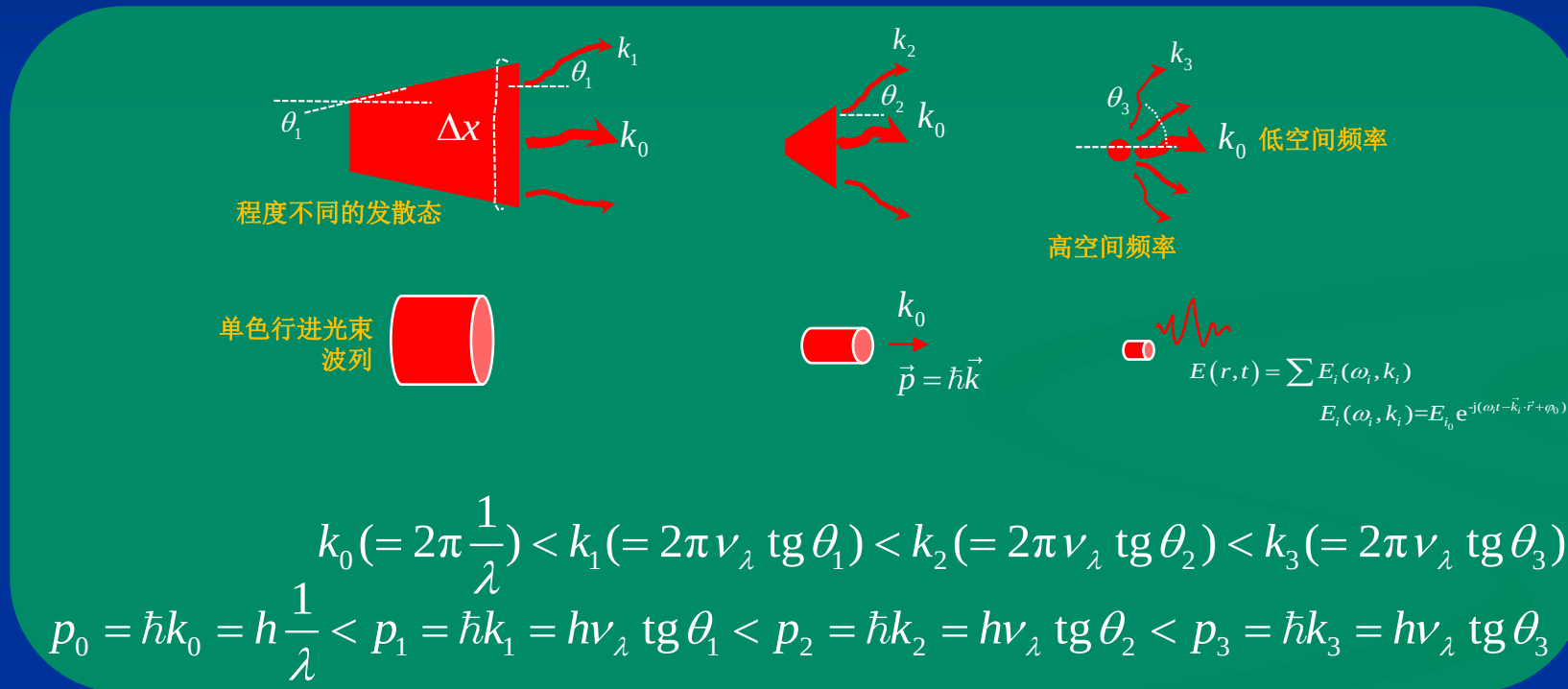
$$\Delta x \rightarrow 0, \Delta p_x \rightarrow \infty$$

$$\Delta p_x \rightarrow 0, \Delta x \rightarrow \infty$$

测不准原理 (第〇步)

$$\left(\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \right)$$

(A): $\Delta x \searrow, \Delta p \nearrow$ 



程度不同的发散态

单色行进光束波列

高空间频率

低空间频率

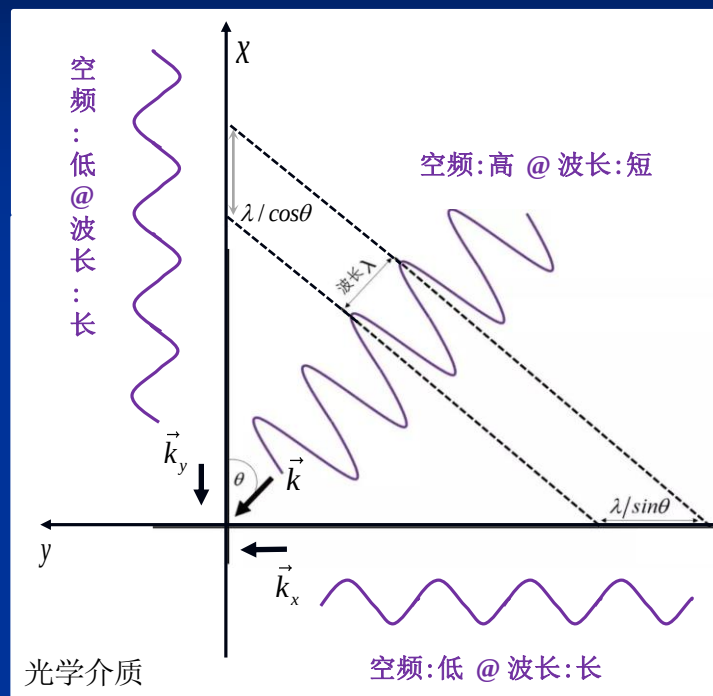
$\vec{p} = \hbar \vec{k}$

$E(r, t) = \sum E_i(\omega_i, k_i)$
 $E_i(\omega_i, k_i) = E_{i0} e^{-j(\omega_i t - \vec{k}_i \cdot \vec{r} + \phi_0)}$

$k_0 (= 2\pi \frac{1}{\lambda}) < k_1 (= 2\pi \nu_\lambda \text{tg } \theta_1) < k_2 (= 2\pi \nu_\lambda \text{tg } \theta_2) < k_3 (= 2\pi \nu_\lambda \text{tg } \theta_3)$
 $p_0 = \hbar k_0 = h \frac{1}{\lambda} < p_1 = \hbar k_1 = h \nu_\lambda \text{tg } \theta_1 < p_2 = \hbar k_2 = h \nu_\lambda \text{tg } \theta_2 < p_3 = \hbar k_3 = h \nu_\lambda \text{tg } \theta_3$

 (B): $\Delta x \nearrow, \Delta p \searrow$

(A) 可分解性



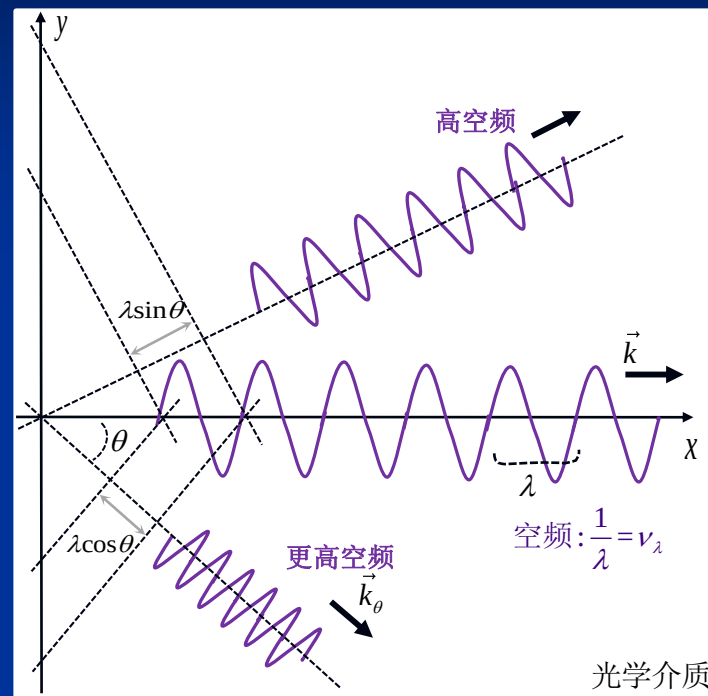
$$\vec{k} = \vec{k}_x + \vec{k}_y$$

$$k^2 = k_x^2 + k_y^2$$

$$* k > k_x (= k \cos \theta), k_y (= k \sin \theta)$$

$$* \omega_\lambda > \omega_{\lambda_x}, \omega_{\lambda_y} \quad (v_\lambda > v_{\lambda_x}, v_{\lambda_y})$$

(B) 辐射特性



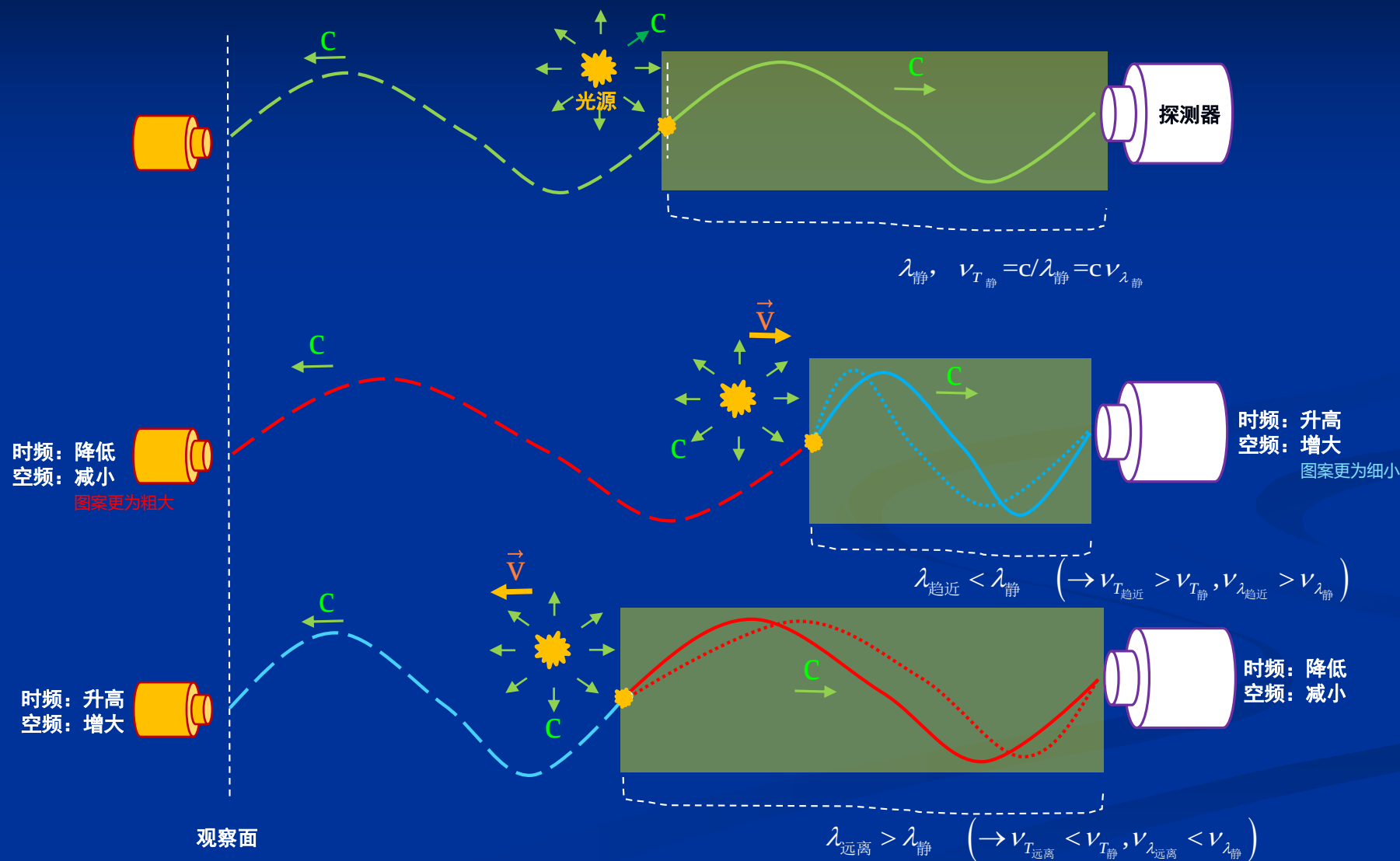
$$k_\theta = k(\theta)$$

$$* k < k_\theta, \quad \omega_\lambda < \omega_{\lambda_\theta}$$

$$* \theta \nearrow \Rightarrow \omega_{\lambda_\theta} \nearrow, k \nearrow, \omega_{T\theta} \nearrow$$

光学多普勒效应

能量、动量：守恒



惠更斯原理 (第一步)

➤ 波前/波面

点源集合

波前/波面上的想象点源→发射球面子波

➤ 某时刻波前/波面

前某时刻波前/波面上的点源球面次波→包络面/切面

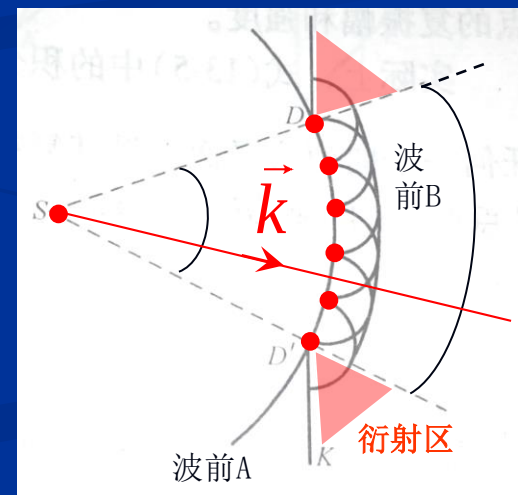
➤ 波前/波面：法向

* 光线方向 (不一定有能量运输)

* 波前/波面(相位空间分布)

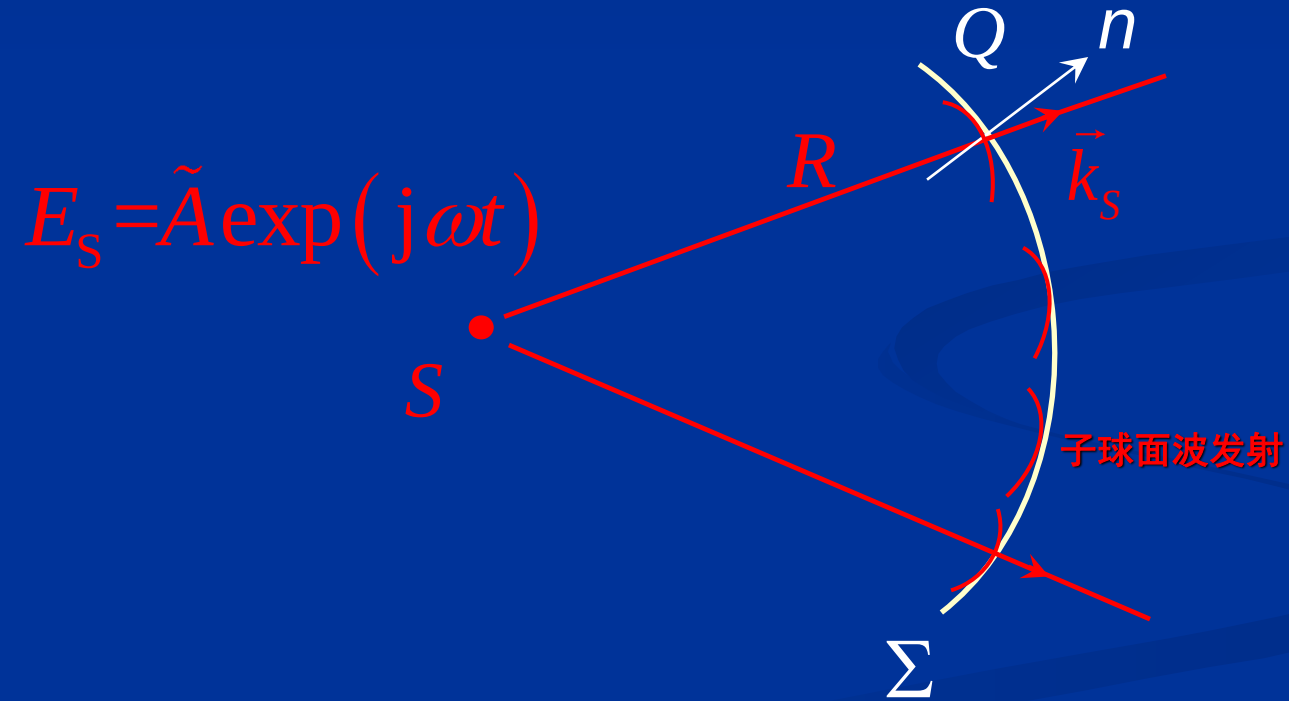
* 不涉及时序能流/振幅空间分布

* 适用所有波场 (水波、声波、...)

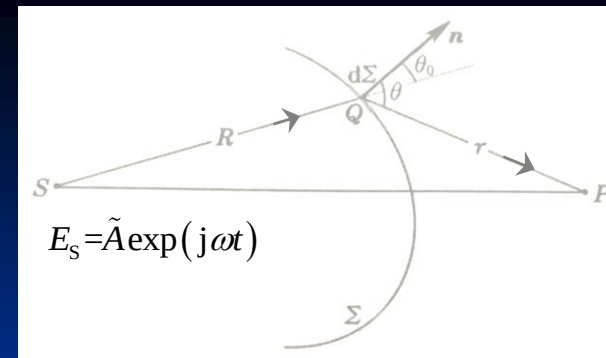
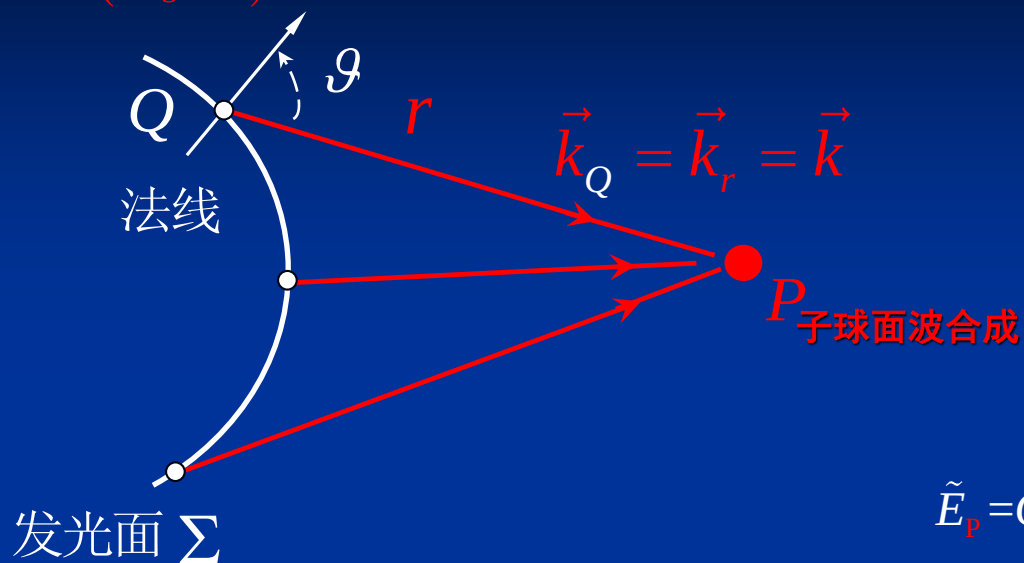


惠更斯-菲涅耳原理 (第二步)

$$\tilde{E}_Q = \frac{E_s}{R} \exp(j\vec{k}_s \cdot \vec{R}) = \frac{E_s}{R} \exp(jk_s \cdot R)$$



$$\tilde{E}_Q = \frac{E_s}{R} \exp(jk_s \cdot R)$$



$$\tilde{E}_P = C \iint_{\Sigma} K(\theta) \frac{\tilde{E}_Q}{r} \exp(jk_Q \cdot r) d\sigma$$

$$= C \iint_{\Sigma} K(\theta) \frac{\frac{\tilde{A}}{R} \exp(jk_s \cdot R)}{r} \exp(jk_Q \cdot r) d\sigma$$

$$= \boxed{C} \frac{\tilde{A} \exp(jk_s \cdot R)}{R} \iint_{\Sigma} \boxed{K(\theta)} \frac{\exp(jk_Q \cdot r)}{r} d\sigma$$

?

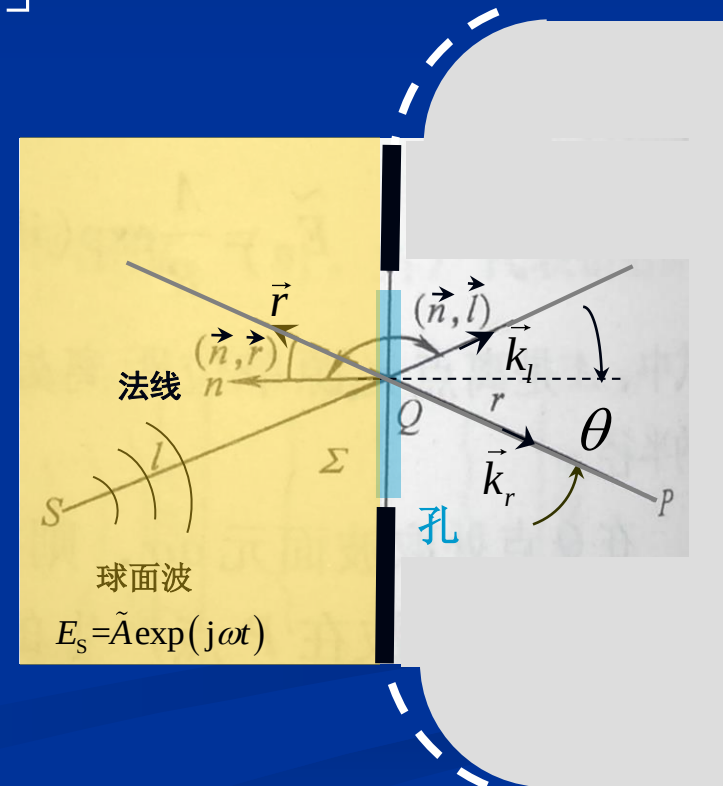
?

$$d\tilde{E}_P = C \cdot K(\theta) \frac{\frac{\tilde{A}}{R} \exp(jk_s \cdot R)}{r} \exp(jk_Q \cdot r) d\sigma$$

菲涅耳-基尔霍夫衍射 (第三步)



$$\tilde{E}_p = \frac{1}{j\lambda} \iint_{\Sigma} \frac{\tilde{E}_Q}{r} \exp(jk_r \cdot r) \left[\frac{\cos(\vec{n}, \vec{r}) - \cos(\vec{n}, \vec{l})}{2} \right] d\sigma$$
$$\left(\tilde{E}_Q = \frac{\tilde{A}}{l} \exp(jk_l \cdot l) \right)$$

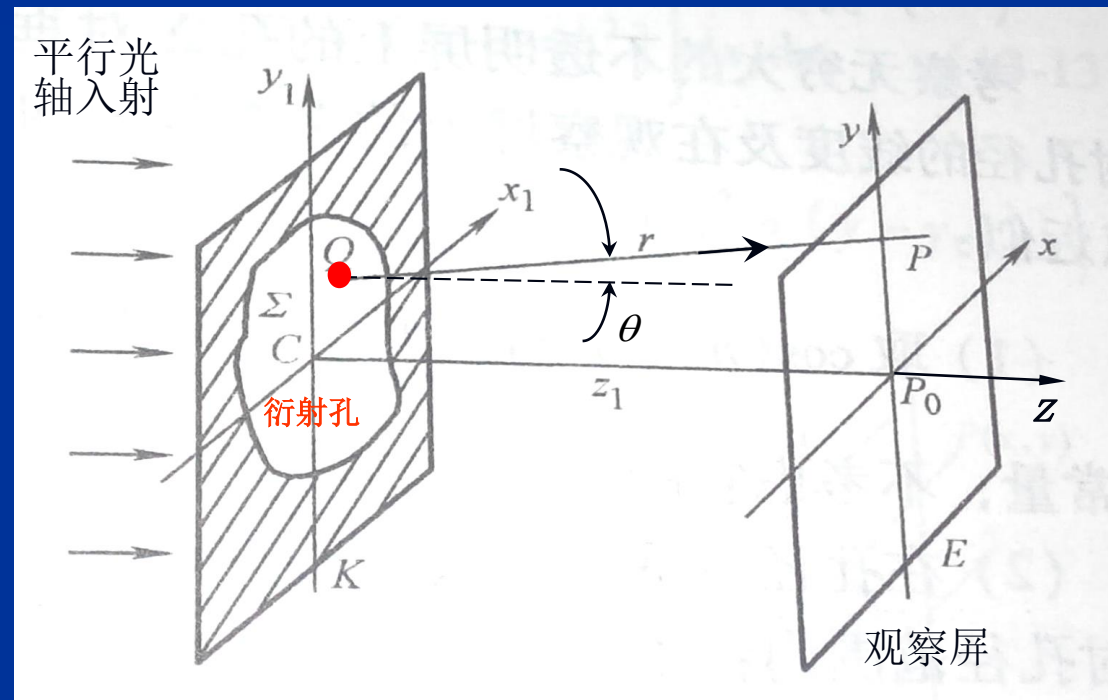


被封闭成无限大空域

$$C = \frac{1}{j\lambda}$$
$$K(\theta) = \frac{\cos(\vec{n}, \vec{r}) - \cos(\vec{n}, \vec{l})}{2}$$

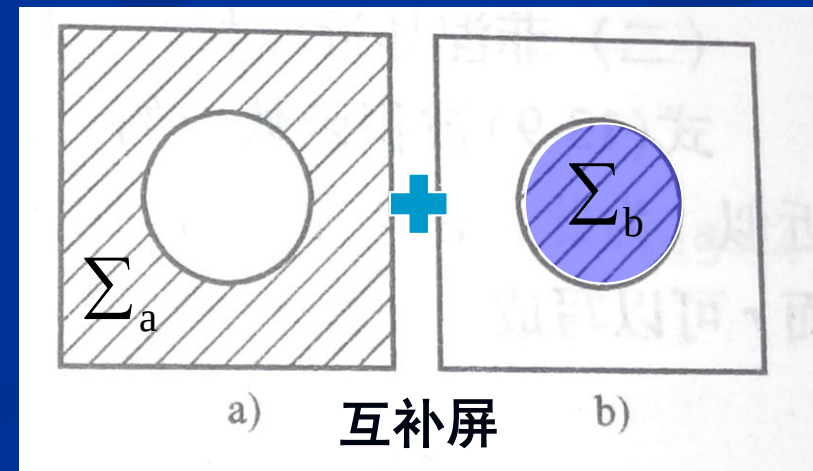
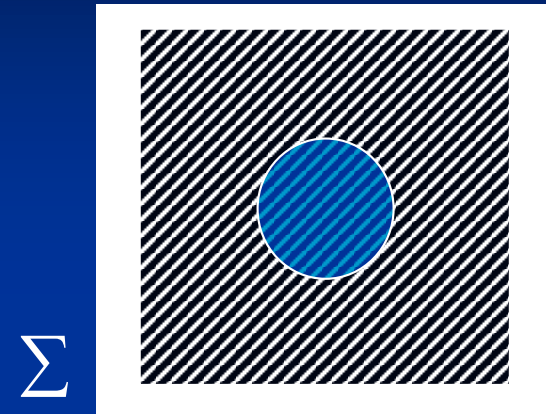
一般性规则

$$\begin{aligned} K(\theta) &= \frac{\cos(\vec{n}, \vec{r}) - \cos(\vec{n}, \vec{l})}{2} \\ &= \frac{1 + \cos(\vec{n}, \vec{r})}{2} \\ &= \frac{1 + \cos \theta}{2} \end{aligned}$$



巴比涅原理

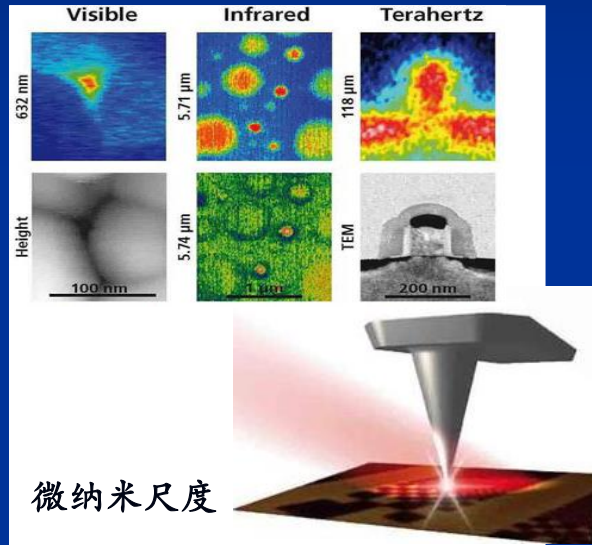
$$\begin{aligned}\tilde{E}_P &= \frac{1}{j\lambda} \iint_{\Sigma} [+] d\sigma \\ &= \frac{1}{j\lambda} \iint_{\Sigma_a} [+] d\sigma + \frac{1}{j\lambda} \iint_{\Sigma_b} [+] d\sigma \\ &= \tilde{E}_P(a) + \tilde{E}_P(b)\end{aligned}$$



$$\Sigma = \Sigma_a + \Sigma_b$$

$$\begin{aligned}\tilde{E}_P = 0 &= \tilde{E}_P(a) + \tilde{E}_P(b) \\ \tilde{E}_P(a) &= -\tilde{E}_P(b)\end{aligned}$$

衍射场典型划分



近场

临近场 (mm级间距)

远场 (>cm级间距)

基尔霍夫衍射近似

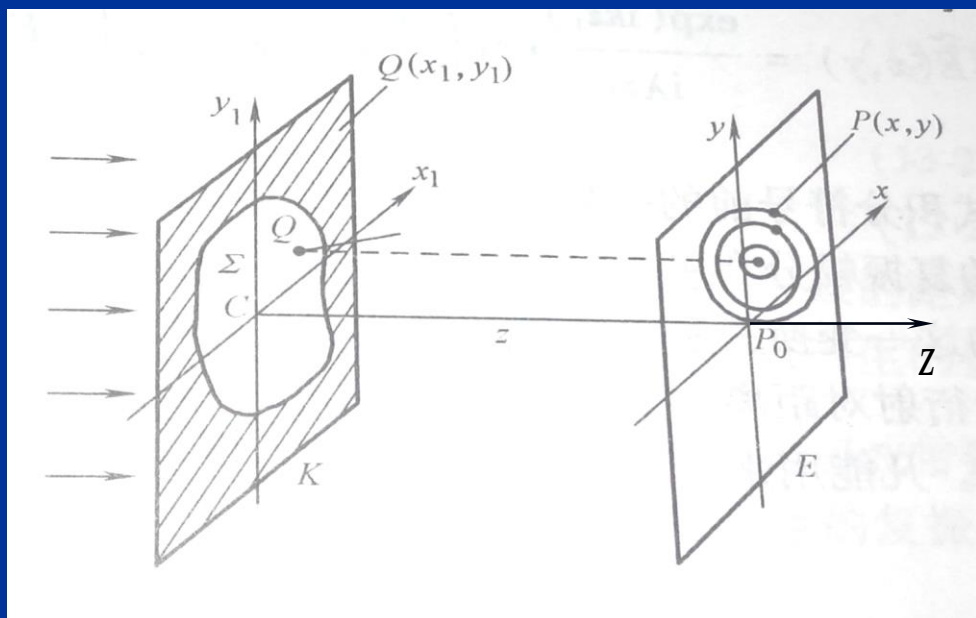
1) 观察屏上的考察范围: $\ll$ 屏到孔的距离
 衍射孔的线度: $\ll$ 屏到孔的距离

$$K(\theta) = \frac{1 + \cos \theta_{\text{极小}}}{2} = 1$$

2) $r \approx z$

$$\tilde{E}_p = \frac{1}{j\lambda} \iint_{\Sigma} \tilde{E}_Q \frac{\exp(j\vec{k} \cdot \vec{r})}{r} d\sigma_1 = \frac{1}{j\lambda z} \iint_{\Sigma} \tilde{E}_Q \exp(j\vec{k} \cdot \vec{r}) d\sigma_1$$

$$kr = \frac{2\pi}{\lambda} r: \lambda \text{ 在微米/纳米级, } r \neq z$$



$$r = |QP| = \sqrt{(z)^2 + (x - x_1)^2 + (y - y_1)^2} = z \left[1 + \left(\frac{x - x_1}{z} \right)^2 + \left(\frac{y - y_1}{z} \right)^2 \right]^{\frac{1}{2}}$$

$$= z + \frac{1}{2z} \left[(x - x_1)^2 + (y - y_1)^2 \right]$$

$$= z + \frac{x^2 + y^2}{2z} + \frac{x_1^2 + y_1^2}{2z} - \frac{xx_1 + yy_1}{z}$$

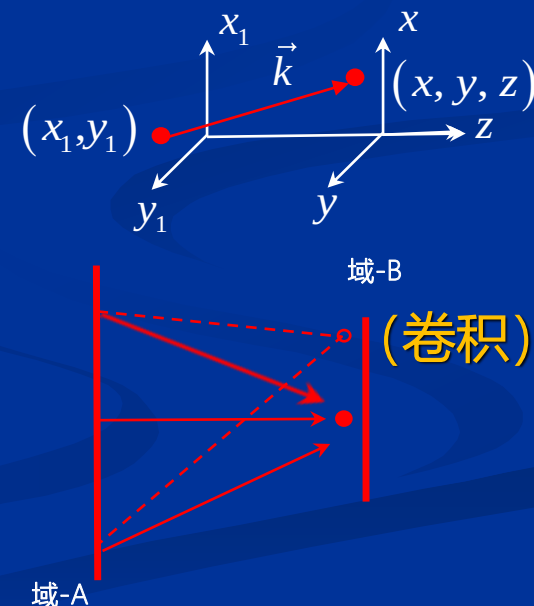
小量

小量

小量

$$\Rightarrow r \approx z$$

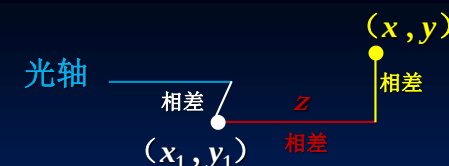
$$\begin{aligned}
 \tilde{E}_p &= \frac{1}{j\lambda z} \iint_{\Sigma} \tilde{E}_Q \exp(jkr) d\sigma_1 \\
 &= \frac{\exp(jkz)}{j\lambda z} \iint_{\Sigma} \tilde{E}(x_1, y_1) \exp\left[j\frac{k}{2z} \left\{ (x-x_1)^2 + (y-y_1)^2 \right\}\right] d\sigma_1 \\
 &= \frac{\exp(jkz)}{j\lambda z} \tilde{E}(x, y) * \exp\left[j\frac{k}{2z} (x^2 + y^2)\right]
 \end{aligned}$$



$$\hat{\mathbf{k}} = (\cos\alpha, \cos\beta, \cos\gamma)$$

$$\mathbf{r} = (x-x_1)\hat{\mathbf{e}}_x + (y-y_1)\hat{\mathbf{e}}_y + z\hat{\mathbf{e}}_z$$

$$\cos\alpha = \frac{x-x_1}{r}, \cos\beta = \frac{y-y_1}{r}, \cos\gamma = \frac{z}{r}$$



$$\tilde{E}_p = \tilde{E}(x, y) = \frac{1}{j\lambda z} \iint_{\Sigma} \tilde{E}_Q \exp(jkr) d\sigma_1$$

$$= \frac{1}{j\lambda z} \iint_{\Sigma} \tilde{E}(x_1, y_1) \exp \left[jk \left(z + \frac{x^2 + y^2}{2z} + \frac{x_1^2 + y_1^2}{2z} - \frac{xx_1 + yy_1}{z} \right) \right] d\sigma_1$$

$$= \frac{\exp(jkz)}{j\lambda z} \exp \left[\frac{jk}{2z} (x^2 + y^2) \right] \iint_{\Sigma} \tilde{E}(x_1, y_1) \exp \left[j\pi \left(\frac{x_1^2}{\lambda z} + \frac{y_1^2}{\lambda z} \right) \right] \exp \left[-j2\pi \left(x_1 \frac{1}{\lambda z} x + y_1 \frac{1}{\lambda z} y \right) \right] d\sigma_1$$

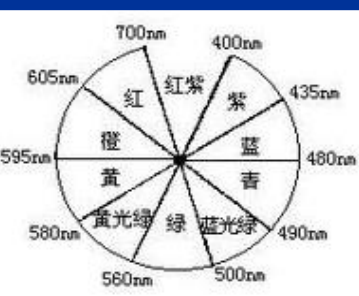
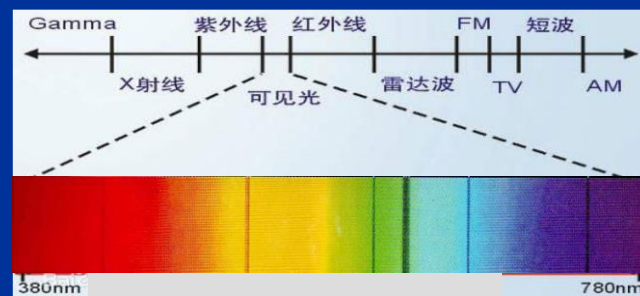
$$= \iint_{\Sigma} \frac{\exp(jkz)}{j\lambda z} \exp \left[\frac{jk}{2z} (x^2 + y^2) \right] \tilde{E}(x_1, y_1) \exp \left[j\pi \left(\frac{x_1^2}{\lambda z} + \frac{y_1^2}{\lambda z} \right) \right] \exp \left[-j2\pi \left(x_1 \frac{\sin \theta_x}{\lambda} + y_1 \frac{\sin \theta_y}{\lambda} \right) \right] d\sigma_1$$

隐含时频的谱空频振幅或权重

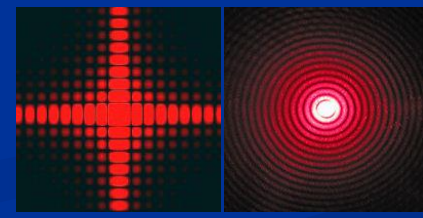
谱空频

时频 (ω/ν)

光波颜色

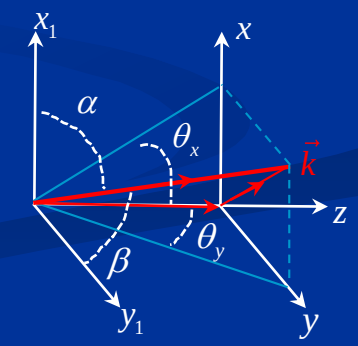


结构尺寸微细程度

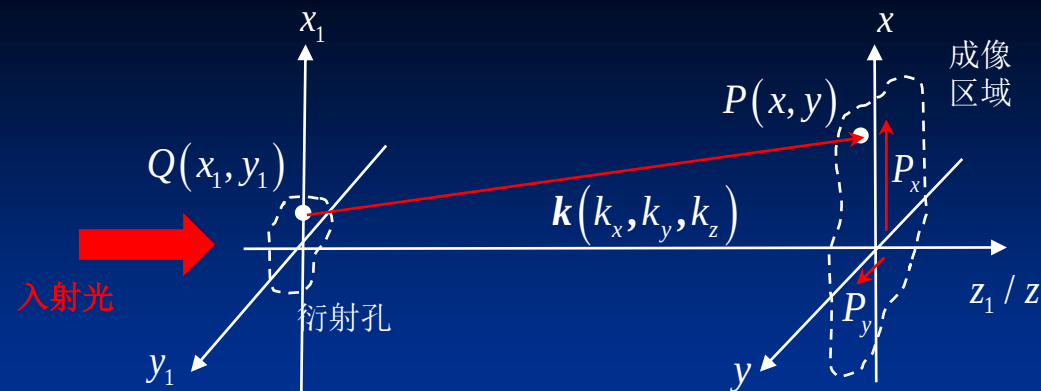


$$\tilde{E}_p \cdot \tilde{E}_p^*$$

$$\tilde{E}(x, y) \cdot \tilde{E}^*(x, y)$$



动量 (空频) 空间展开



$$\tilde{E}(x, y) = \tilde{E}(p_x, p_y) = \tilde{E}(p_{x_1}, p_{y_1})$$

$$= \frac{\exp(jkz)}{j\lambda z} \exp\left[\frac{jk}{2z}(x^2 + y^2)\right] \iint_{\Sigma} \tilde{E}(x_1, y_1) \exp\left[j\pi\left(\frac{x_1^2}{\lambda z} + \frac{y_1^2}{\lambda z}\right)\right] \exp\left[-\frac{j}{\hbar}\{x_1 p_x + y_1 p_y\}\right] d\sigma_1 \\ \exp\left[-\frac{j}{\hbar}\{x_1 p_{x_1} + y_1 p_{y_1}\}\right] d\sigma_1$$

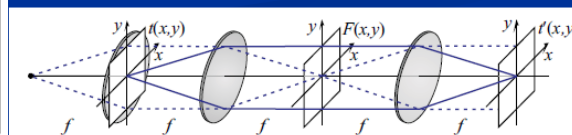
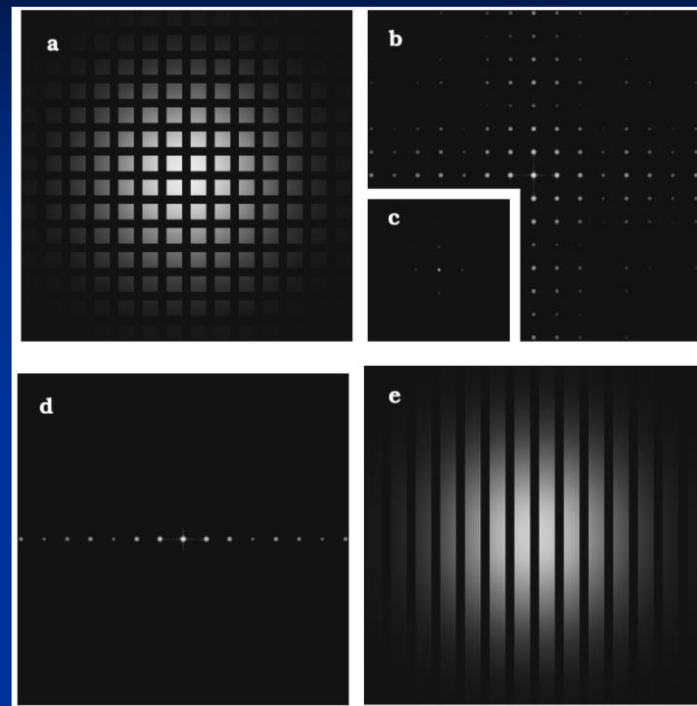
光子: $\hbar\omega$ (能量)

$\hbar\mathbf{k}$ (动量)

$$\psi_0 \exp\left[-\frac{j}{\hbar}\{\hbar\omega - \hbar\mathbf{k}\} + \varphi_0\right] \quad (\text{波函数})$$

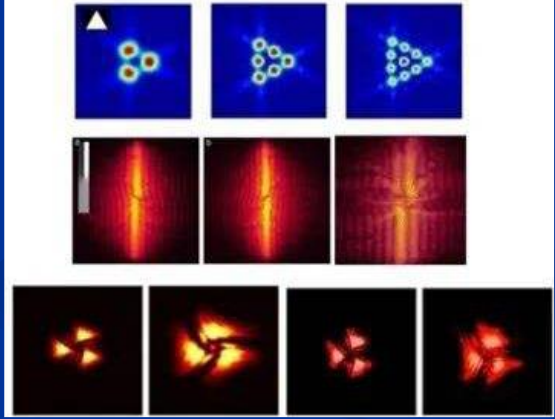
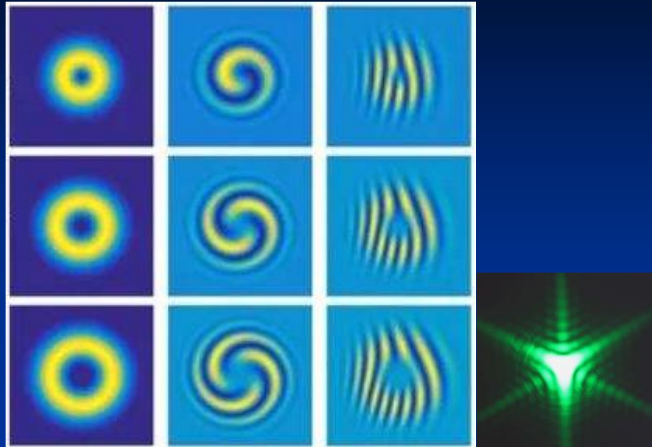
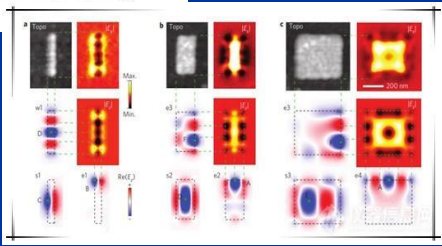
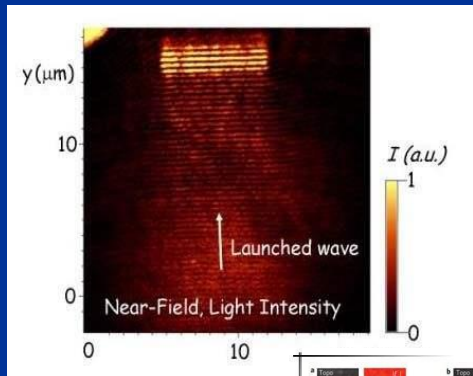
$$\exp\left[-j2\pi\left(x_1 \frac{1}{\lambda z} x + y_1 \frac{1}{\lambda z} y\right)\right] = \exp\left[-\frac{j}{\hbar}\left\{x_1 \hbar\left(\frac{2\pi}{\lambda z/x}\right) + y_1 \hbar\left(\frac{2\pi}{\lambda z/y}\right)\right\}\right] \\ = \exp\left[-\frac{j}{\hbar}\{x_1 \hbar k_x + y_1 \hbar k_y\}\right] = \exp\left[-\frac{j}{\hbar}\{x_1 p_x + y_1 p_y\}\right] \\ = \exp\left[-\frac{j}{\hbar}\{x_1 p_{x_1} + y_1 p_{y_1}\}\right]$$

基于空频选择与变换的图像信息处理



输入图像	频 谱 变 换			输出图像
	$\tilde{U}_\lambda$	$\tilde{T}$	$\tilde{U}_\text{出}$	
a	b	c	d	e

临近场、远场



极小距离：几~百纳米级波长：小量？
否→菲涅耳衍射近似
是→夫琅和费衍射近似

傅里叶时频变换

成像面：图像(光强/亮度)的三角函数分解，宽谱域内色光展开

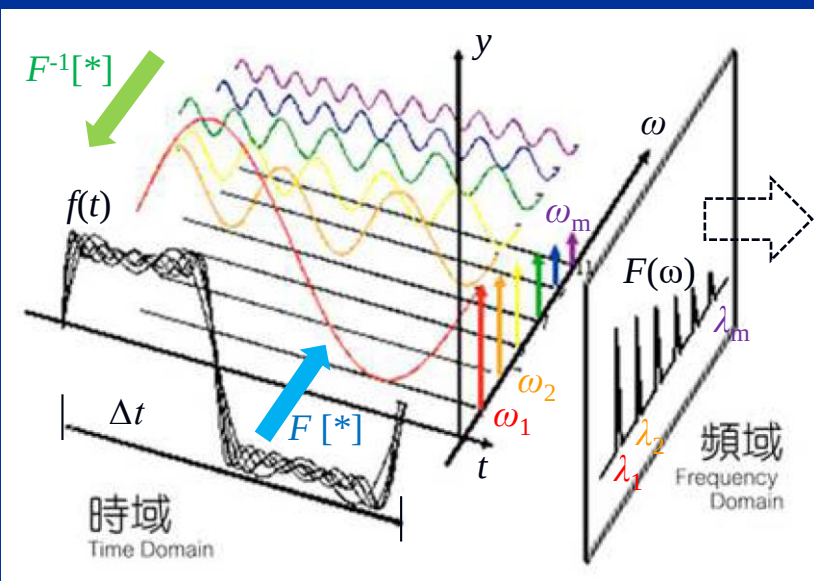
$$F(\omega) = F[f(t)] = \int_{-\infty}^{+\infty} f(t) \exp(-j\omega t) dt$$

傅里叶逆变换

叠加三角函数拟合波列与图像

$$f(t) = F^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) \exp(j\omega t) d\omega$$

傅里叶(时频)变换?



- * 图像映射
- * 图像压缩
- * 图像信息隐藏、加密
- * 基于频谱变换的图像处理

$$E_i = E_{i_0} \exp(-j\omega_i t)$$

$$E_i = E_{i_0} \exp(-jk_i r)$$

$$E_i = E_{i_0} \exp[-j(\omega_i t - k_i r)]$$

10.2 菲涅耳衍射近似

$$\tilde{E}_{P_0} = |\tilde{E}_1| - |\tilde{E}_2| + |\tilde{E}_3| - |\tilde{E}_4| + \dots + (-1)^{k-1} |\tilde{E}_k| \approx \frac{|\tilde{E}_1|}{2} \pm \frac{|\tilde{E}_k|}{2}$$

$$\sim \frac{1}{2} \lambda$$

	i	$(-1)^{i-1} \tilde{E}_i $
	3	$ \tilde{E}_3 $
	2	$- \tilde{E}_2 $
	1	$ \tilde{E}_1 $

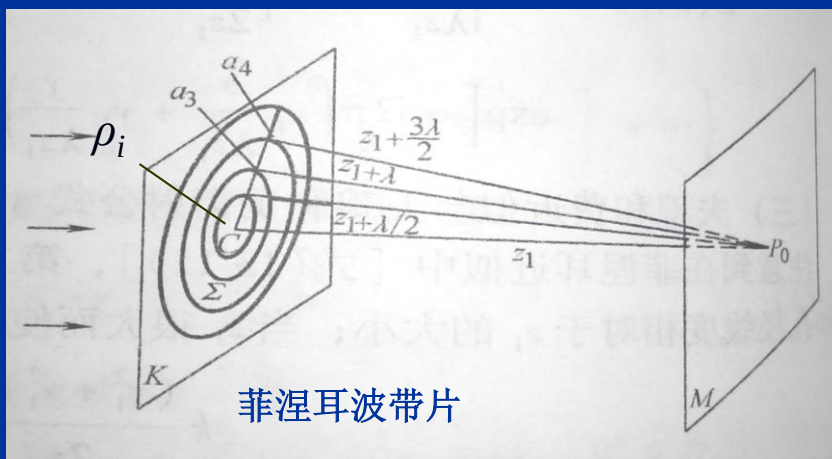
$$|\tilde{E}_i| = C \frac{A_i}{r_i} \frac{1 + \cos \vartheta}{2}$$

$$|\tilde{E}_2| = \frac{|\tilde{E}_1| + |\tilde{E}_3|}{2} ?$$

$$|\tilde{E}_4| = \frac{|\tilde{E}_3| + |\tilde{E}_5|}{2} ?, \dots$$

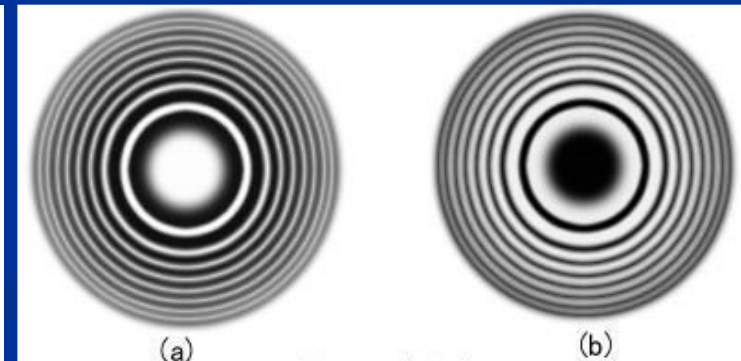
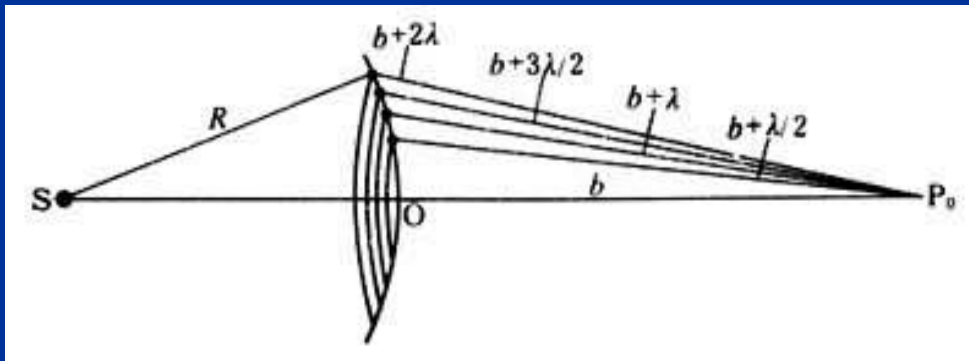
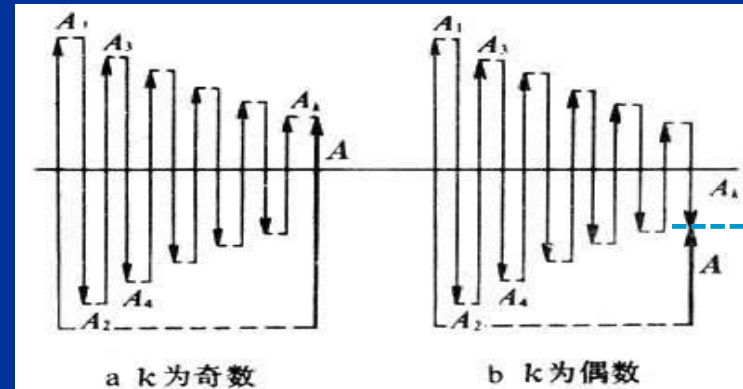
$$A_i = \pi (\rho_i^2 - \rho_{i-1}^2) \approx \pi z_1 \lambda$$

$$\rho_i = \left[\left(z_1 + i \frac{\lambda}{2} \right)^2 - z^2 \right]^{1/2} \approx \sqrt{iz_1 \lambda}$$

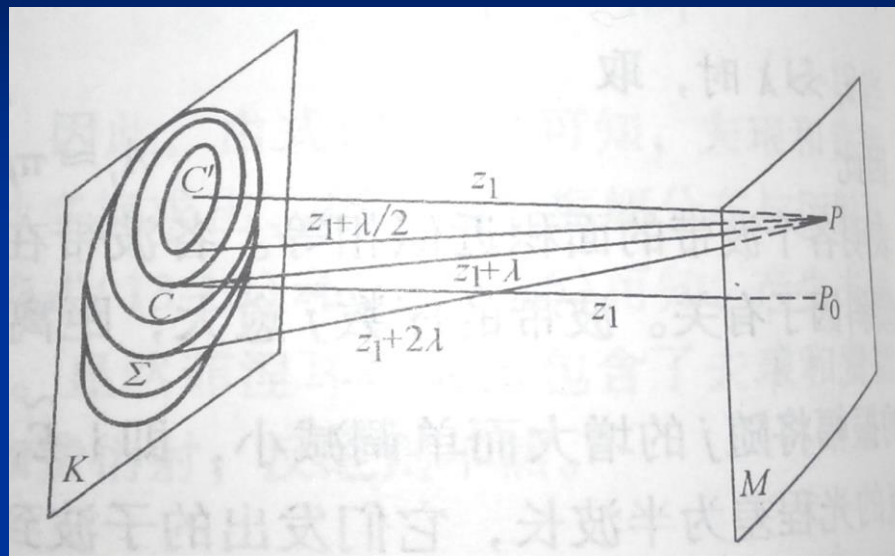


1) 菲涅耳波带片：轴上点

- 1): $\tilde{E}_{P_0} \approx \frac{|\tilde{E}_1|}{2} + \frac{|\tilde{E}_k|}{2},$ k 为奇数, 亮斑
- 2): $\tilde{E}_{P_0} \approx \frac{|\tilde{E}_1|}{2} - \frac{|\tilde{E}_k|}{2},$ k 为偶数, 暗斑
- 3): $\tilde{E}_{P_0} \approx \frac{|\tilde{E}_1|}{2},$ 亮斑 (k 足够大)



2) 菲涅耳波带片：轴外点



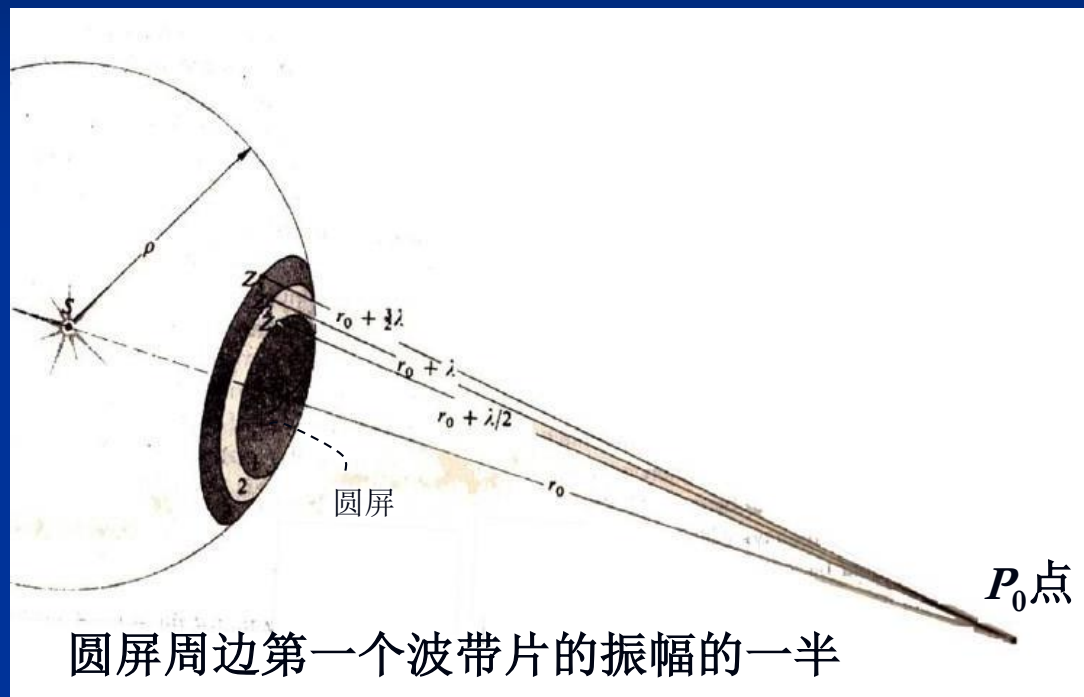
1) 轴上点随距离的增大呈现：…亮点…暗点…较亮点…暗点…

轴上多焦点

2) 轴外点随距离增大呈现：…亮环…暗环…较亮环…暗环…

轴外同心环状焦斑

3) 菲涅耳波带片：圆屏

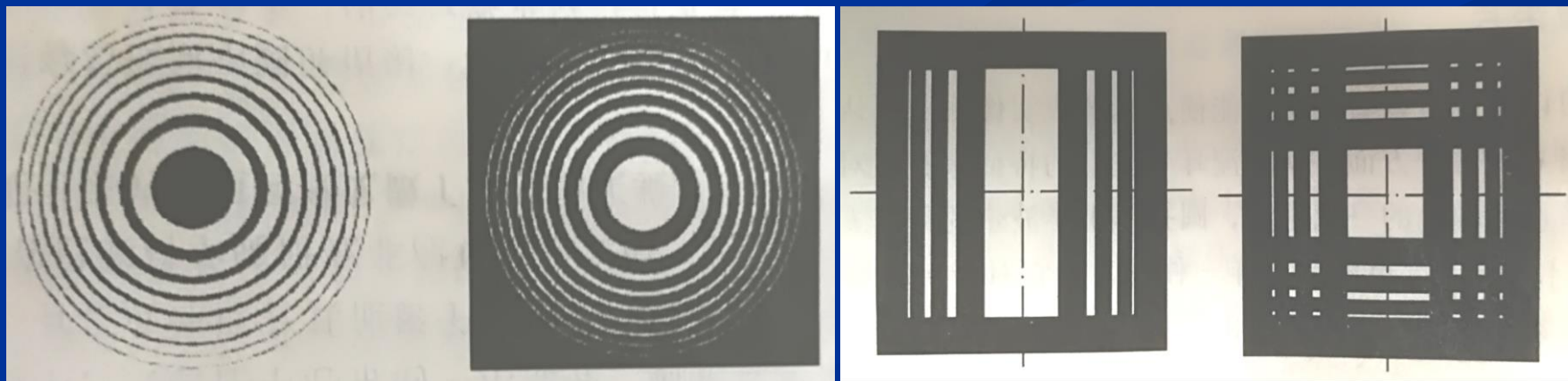


4) 菲涅耳波带片：奇数波带、偶数波带

$$\tilde{E}_{P_0} = |\tilde{E}_1| - |\tilde{E}_2| + |\tilde{E}_3| - |\tilde{E}_4| + \dots + (-1)^{k-1} |\tilde{E}_n| \approx \frac{|\tilde{E}_1|}{2} \pm \frac{|\tilde{E}_k|}{2}$$

1) 透光区: $\tilde{E}_{P_0} = |\tilde{E}_1| + |\tilde{E}_3| + \dots + (-1)^{2k-1} |\tilde{E}_{2k}|$

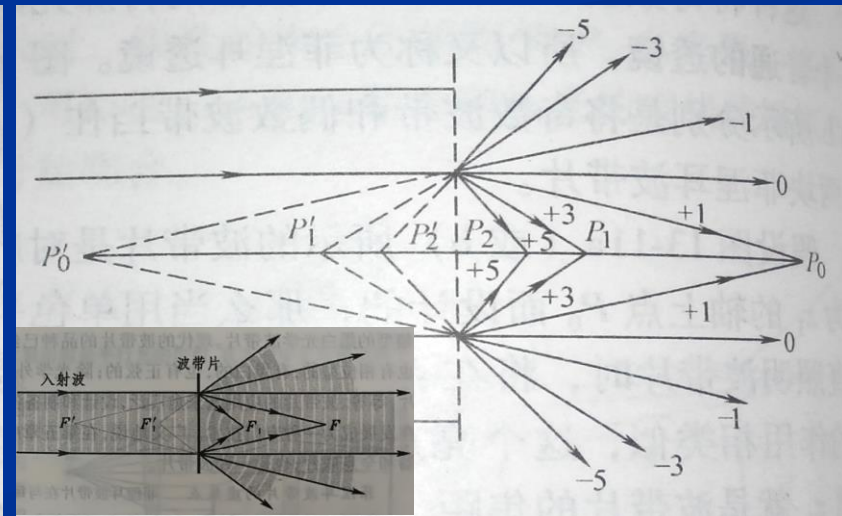
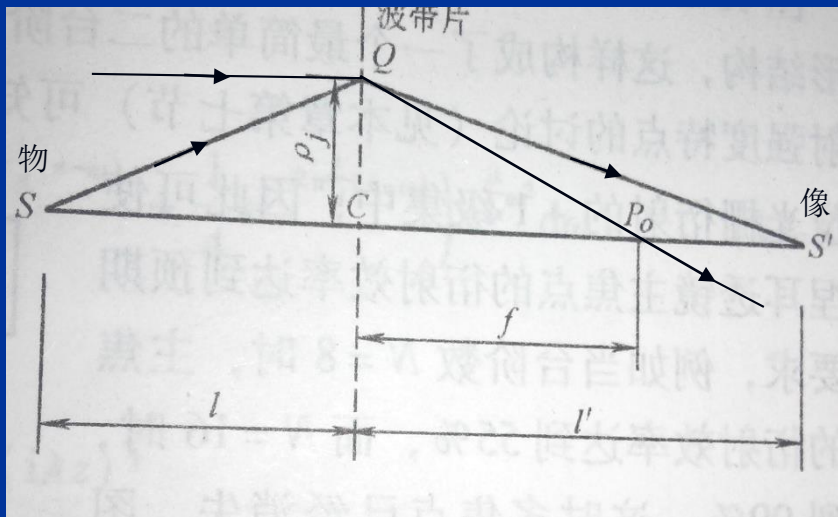
2) 背面反射区: $\tilde{E}_{P_0} = -|\tilde{E}_2| - |\tilde{E}_4| - \dots - (-1)^{2k-1} |\tilde{E}_{2k}|$



5) 菲涅耳波带片：成像、焦距

$$\frac{1}{l'} + \frac{1}{l} = \frac{1}{f}$$

$$f_1 = z_1 = \frac{\rho_1}{i\lambda}$$



6) 菲涅耳波带片：二元光学结构与微透镜

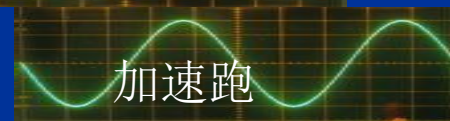
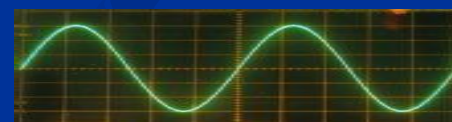
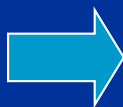
$$\tilde{E}_{P_0} = |\tilde{E}_1| - |\tilde{E}_2| + |\tilde{E}_3| - |\tilde{E}_4| + \dots + (-1)^{k-1} |\tilde{E}_n| \approx \frac{|\tilde{E}_1|}{2} \pm \frac{|\tilde{E}_k|}{2}$$



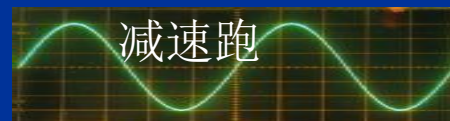
$$\tilde{E}_{P_0} = |\tilde{E}_1| + |\tilde{E}_2| + |\tilde{E}_3| + |\tilde{E}_4| + \dots + (-1)^{k-1} |\tilde{E}_n| \approx \frac{|\tilde{E}_1|}{2} \pm \frac{|\tilde{E}_k|}{2}$$



反相



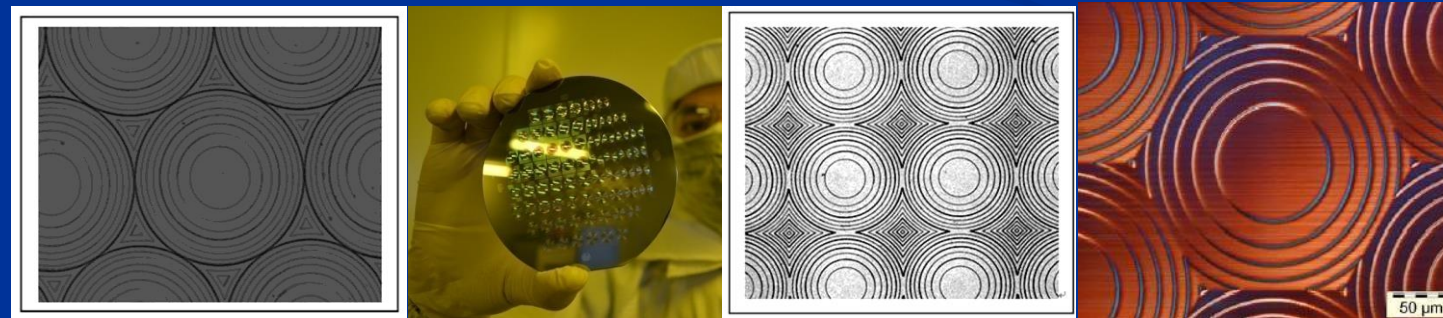
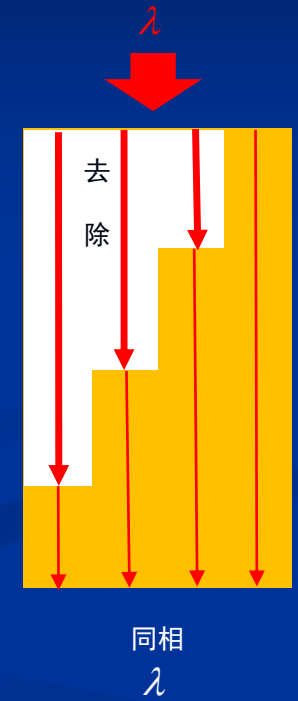
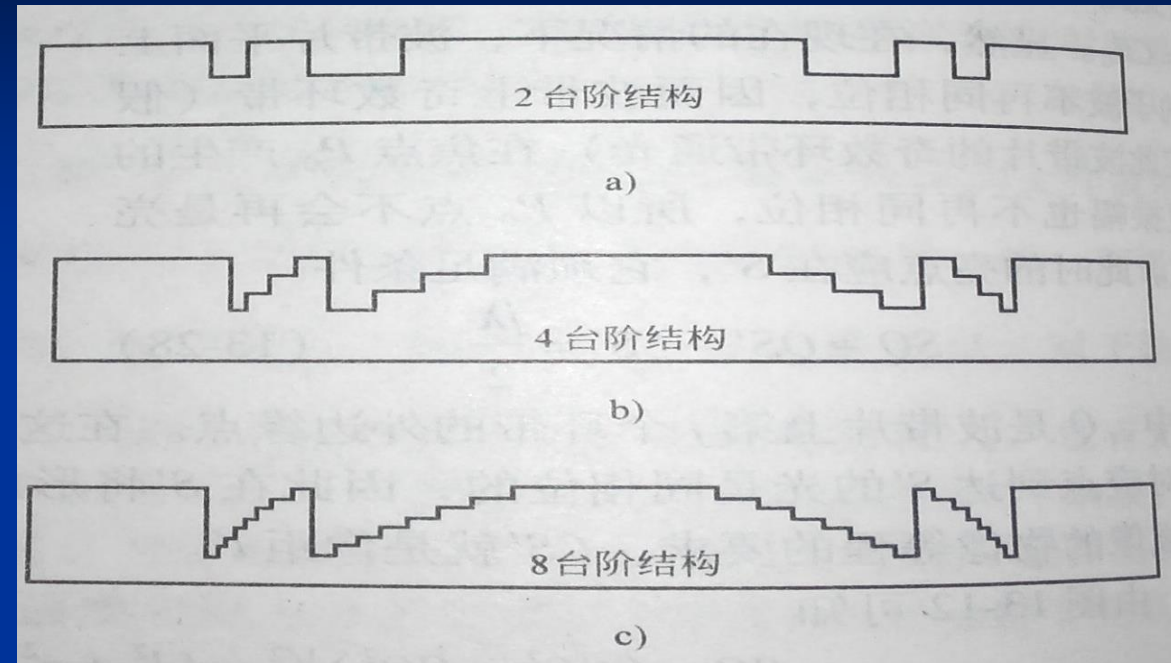
加速跑



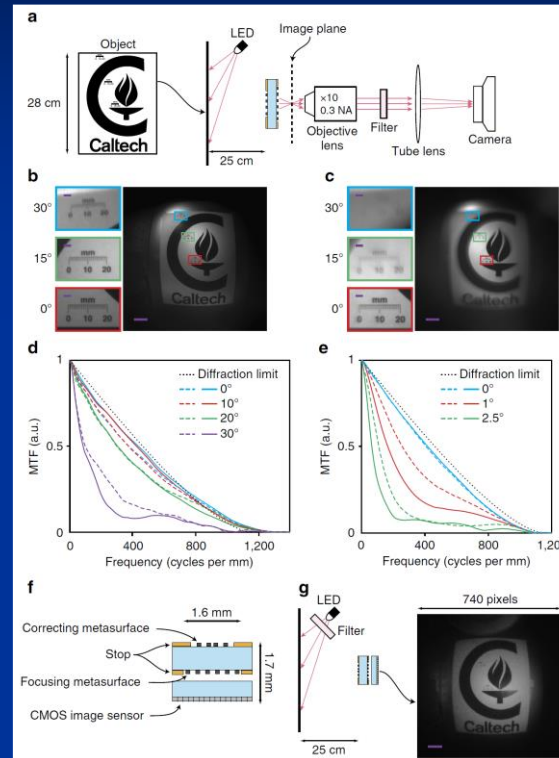
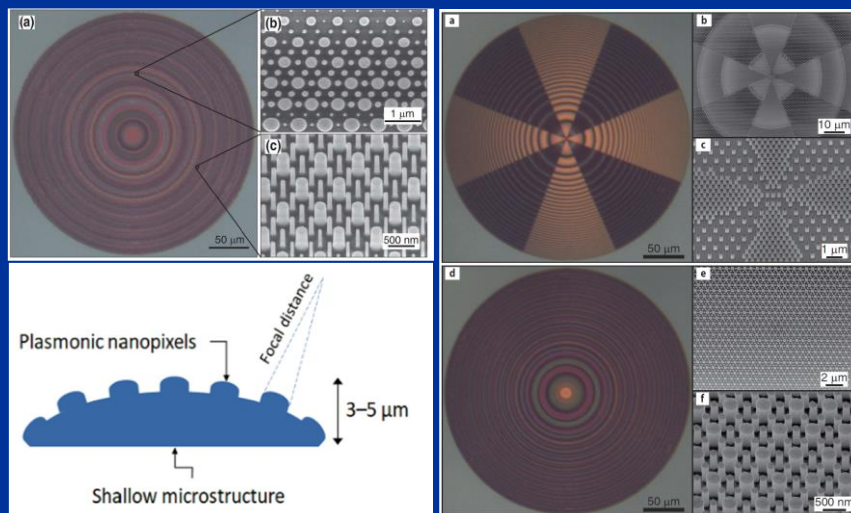
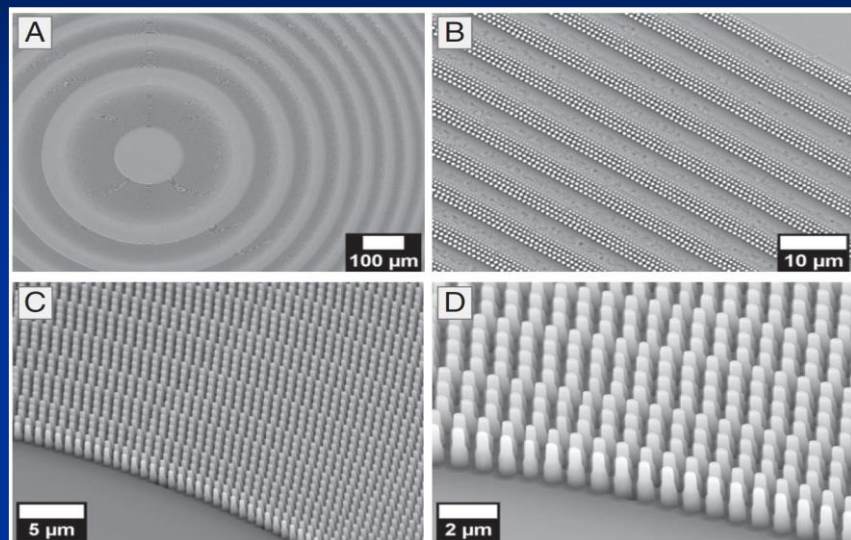
减速跑



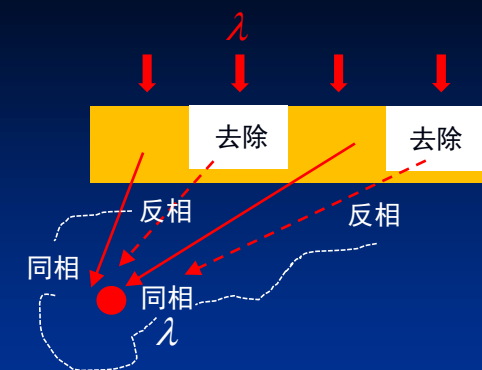
同相



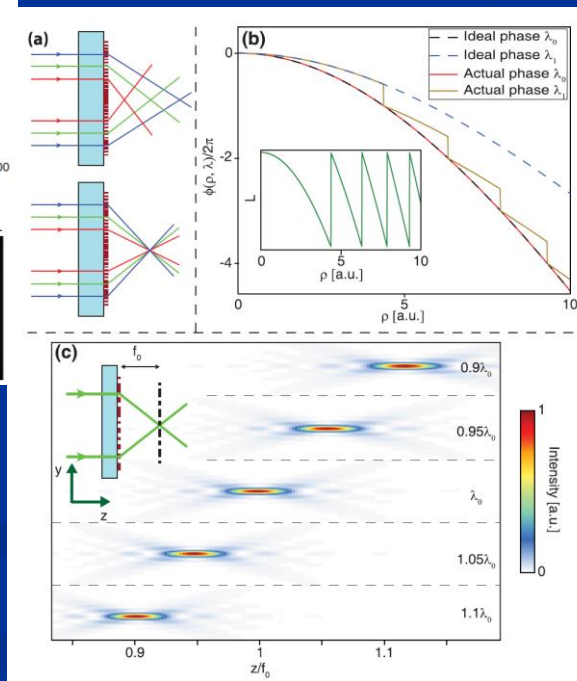
7) 类/近 超表面



成像

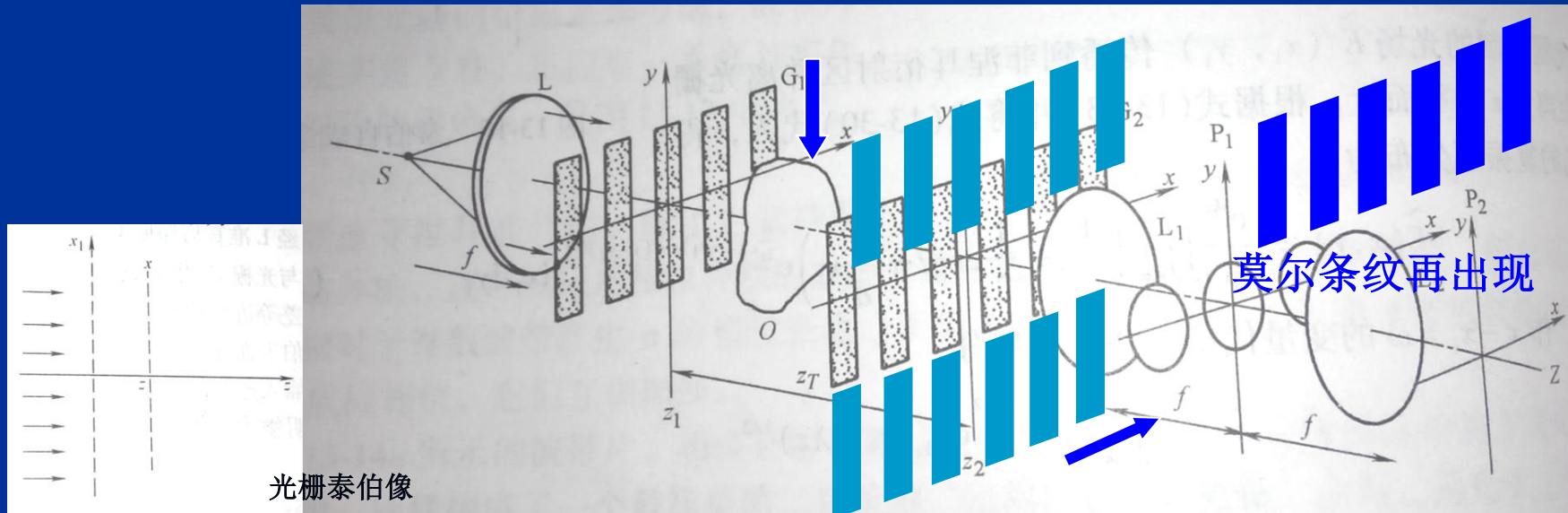


聚束



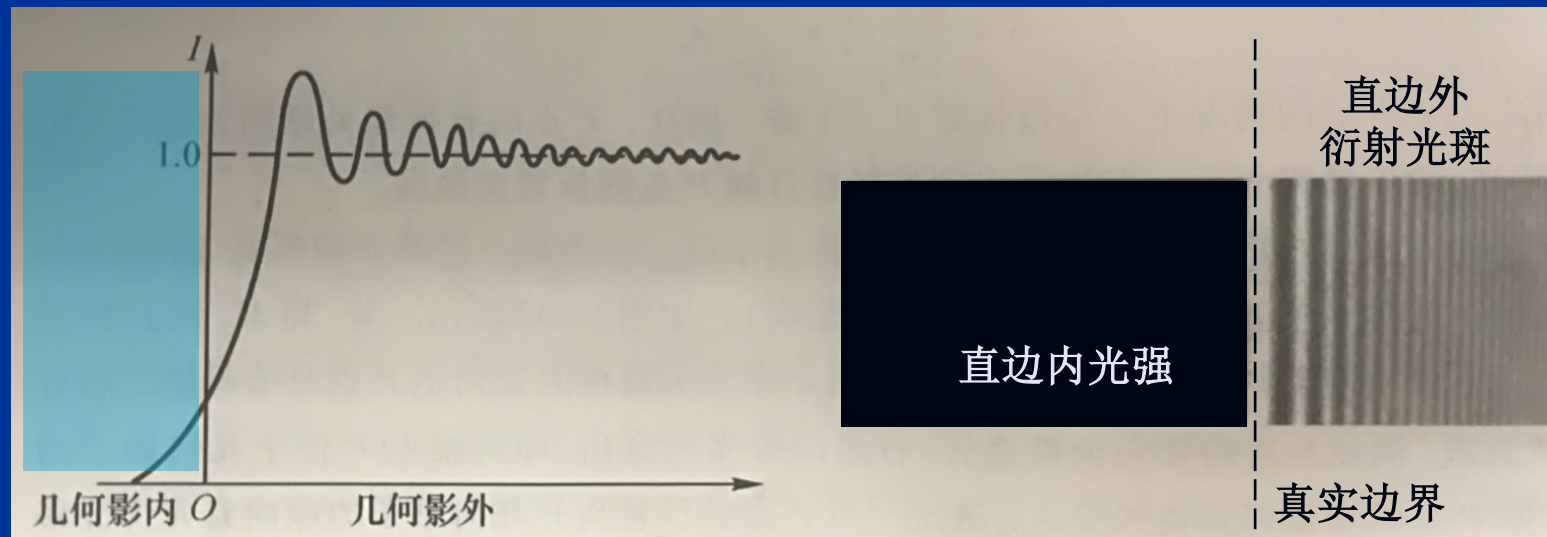
泰伯 (Talbot) 效应

- 单色平面波垂直照射：基于周期性结构的衍射屏→衍射屏后出现周期性结构的几何像
- 泰伯干涉仪：光栅G1泰伯像与光栅G2耦合→生成莫尔条纹
- 精密测量：波前



直边衍射

- 边界外：近场衍射→菲涅耳衍射
- 边界处：衍射光场模糊了边界
- 边界内：衍射光场略微深入边界内

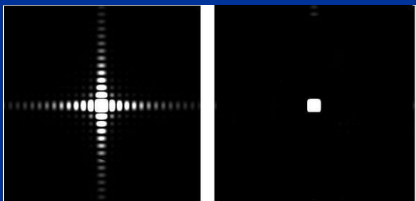


10.3 夫琅和费衍射近似

细束径光入射、极小孔径衍射：

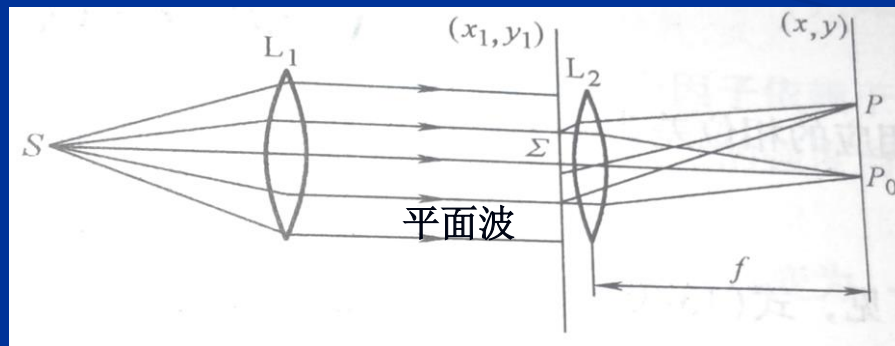
$$\frac{(x_1^2 + y_1^2)}{2z} \ll \lambda, \quad \frac{(x_1^2 + y_1^2)}{2z\lambda} \ll 1 \text{ 或 } k \frac{(x_1^2 + y_1^2)}{2z} \ll \pi$$

$$\begin{aligned}\tilde{E}_P &= \frac{1}{j\lambda z} \iint_{\Sigma} \tilde{E}_Q \exp(jkz) d\sigma \\ &= \frac{1}{j\lambda z} \exp(jkz) \exp jk \left(\frac{x^2 + y^2}{2z} \right) \iint_{\Sigma} \tilde{E}_Q \exp \left[-jk \frac{xx_1 + yy_1}{z} \right] d\sigma \\ &= \frac{1}{j\lambda z} \exp(jkz) \exp jk \left(\frac{x^2 + y^2}{2z} \right) \iint_{\infty} \tilde{E}_Q \exp \left[-jk \frac{xx_1 + yy_1}{z} \right] d\sigma\end{aligned}$$



$$\begin{aligned}
\tilde{E}(x, y) &= C \frac{1}{z} \exp(jkz) \exp jk \left(\frac{x^2 + y^2}{2z} \right) \iint_{\Sigma} \tilde{E}_Q \exp \left[-jk \frac{xx_1 + yy_1}{z} \right] d\sigma \\
&= C' \frac{1}{f} \exp(jkf) \exp jk \left(\frac{x^2 + y^2}{2f} \right) \iint_{\Sigma} \tilde{E}(x_1, y_1) \exp \left[-jk \frac{xx_1 + yy_1}{f} \right] d\sigma \\
&= C' \frac{1}{f} \exp jk \left(f + \frac{x^2 + y^2}{2f} \right) \iint_{\Sigma} A' \exp \left[-jk \left(\frac{x}{f} x_1 + \frac{y}{f} y_1 \right) \right] dx_1 dy_1
\end{aligned}$$

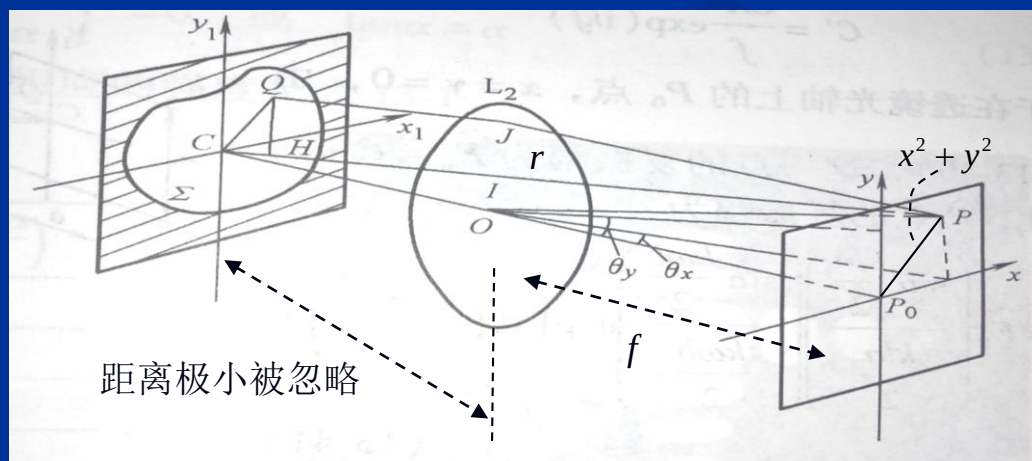
f 取代 z



$$\tilde{E}(x, y) = \frac{C'A'}{f} \exp jk \left(f + \frac{x^2 + y^2}{2f} \right) \iint_{\Sigma} \exp \left[-jk(\theta_x x_1 + \theta_y y_1) \right] dx_1 dy_1$$

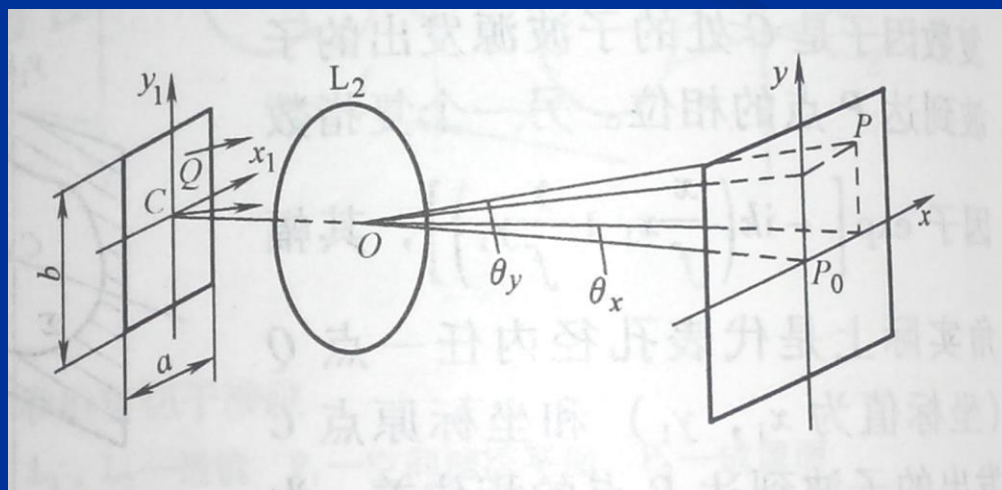
$$r = \left[(x^2 + y^2) + f^2 \right]^{\frac{1}{2}} = f \left[1 + \frac{(x^2 + y^2)}{f^2} \right]^{\frac{1}{2}} \approx f + \frac{(x^2 + y^2)}{2f}$$

$$k \left(\frac{x}{f} x_1 + \frac{y}{f} y_1 \right) = k(x_1 \cos \theta_x + y_1 \cos \theta_y) = k \vec{C}_Q \cdot \vec{C}_P = \varphi_{Q_P} - \varphi_{Q_C} = \Delta_{CH}$$



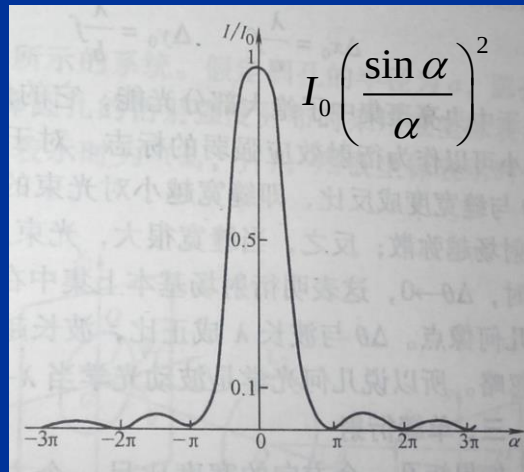
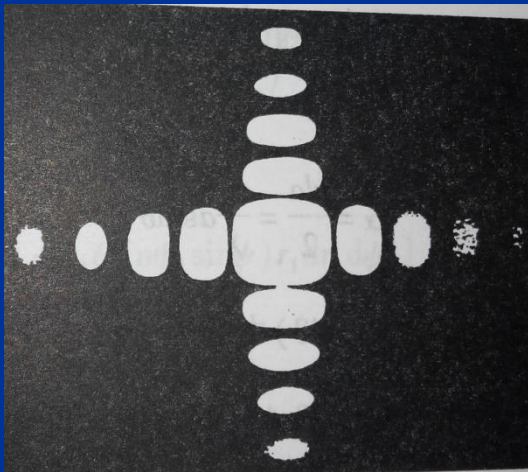
1) 夫琅和费衍射：矩孔

$$\begin{aligned}\tilde{E}(x, y) &= \frac{C'A'}{f} \exp jk \left(f + \frac{x^2 + y^2}{2f} \right) \iint_{\Sigma} \exp \left[-jk(\theta_x x_1 + \theta_y y_1) \right] dx_1 dy_1 \\ &= C'' \exp jk \left(\frac{x^2 + y^2}{2f} \right) \int_{-a/2}^{a/2} \int_{-b/2}^{b/2} \exp \left[-jk(\theta_x x_1 + \theta_y y_1) \right] dx_1 dy_1 \\ &= C'' ab \frac{\sin \frac{ka\theta_x}{2}}{\frac{ka\theta_x}{2}} \frac{\sin \frac{kb\theta_y}{2}}{\frac{kb\theta_y}{2}} \exp jk \left(\frac{x^2 + y^2}{2f} \right)\end{aligned}$$



$$I_P = |\tilde{E}_P|^2 = \left| C'' ab \frac{\sin \frac{ka\theta_x}{2} \sin \frac{kb\theta_y}{2}}{\frac{ka\theta_x}{2} \frac{kb\theta_y}{2}} \exp jk \left(\frac{x^2 + y^2}{2f} \right) \right|^2 = \left| \tilde{E}_0 \frac{\sin \frac{ka\theta_x}{2} \sin \frac{kb\theta_y}{2}}{\frac{ka\theta_x}{2} \frac{kb\theta_y}{2}} \right|^2$$

$$= |\tilde{E}_0|^2 \left| \frac{\sin \frac{ka\theta_x}{2} \sin \frac{kb\theta_y}{2}}{\frac{ka\theta_x}{2} \frac{kb\theta_y}{2}} \right|^2 = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin \beta}{\beta} \right)^2$$

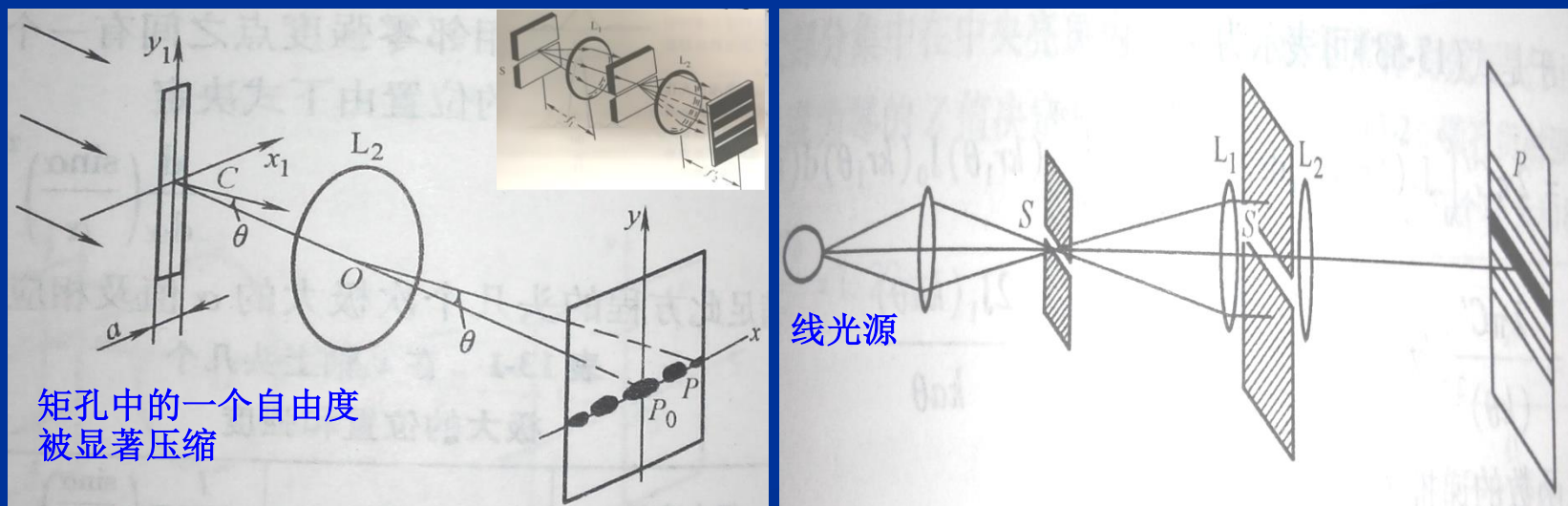


极大序号	α	$\frac{I}{I_0} = \left(\frac{\sin \alpha}{\alpha} \right)^2$
0	0	1
1	$1.43\pi = 4.493$	0.04718
2	$2.459\pi = 7.725$	0.01694
3	$3.470\pi = 10.90$	0.00834
4	$4.479\pi = 14.07$	0.00503

2) 夫琅和费衍射：单缝

$$I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

单缝衍射因子

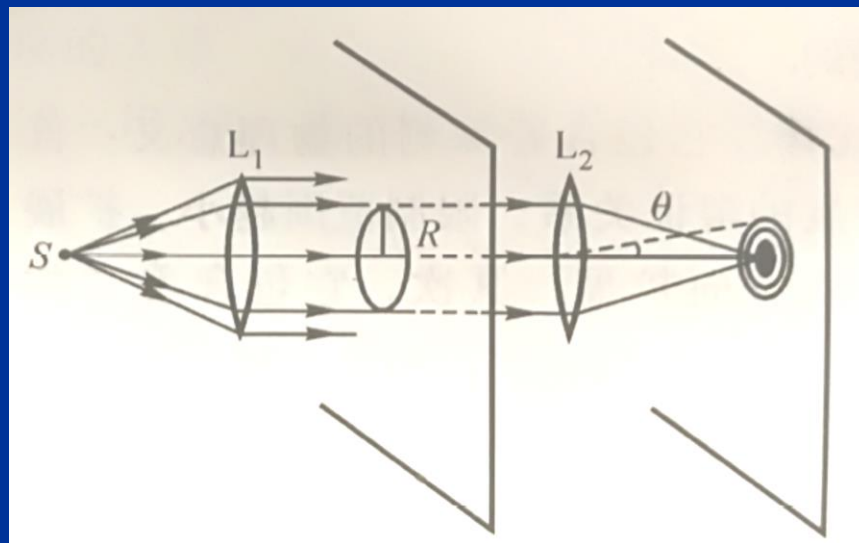


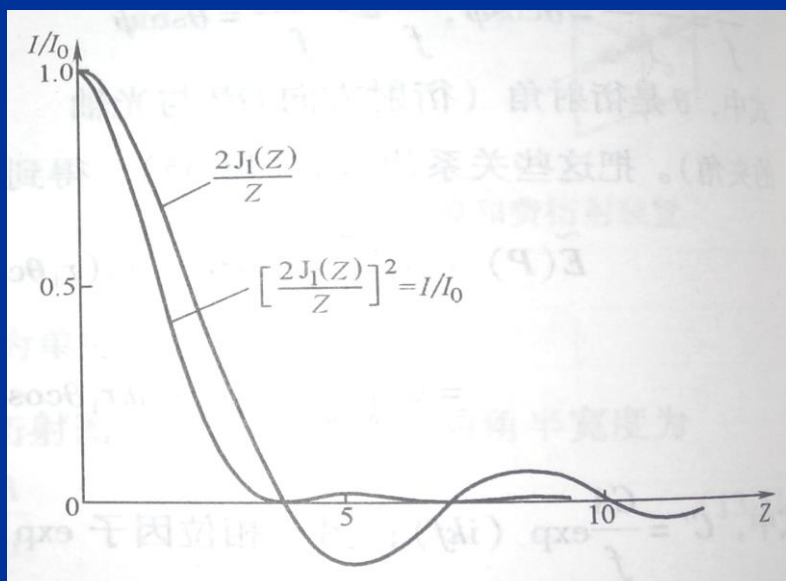
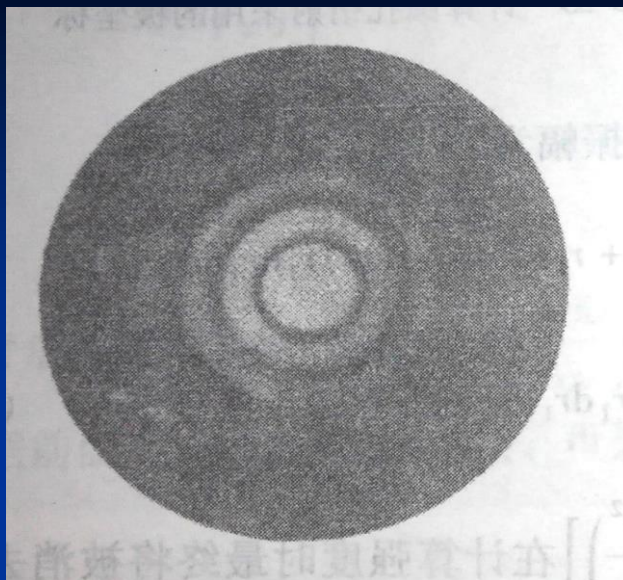
3) 夫琅和费衍射：圆孔

$$I_P = (\pi a^2)^2 |C''|^2 \left[\frac{2J_1(ka\theta)}{ka\theta} \right]^2$$

$$x = r \cos \psi, y = r_1 \sin \psi$$

$$x_1 = r_1 \cos \psi_1, y_1 = r_1 \sin \psi_1$$





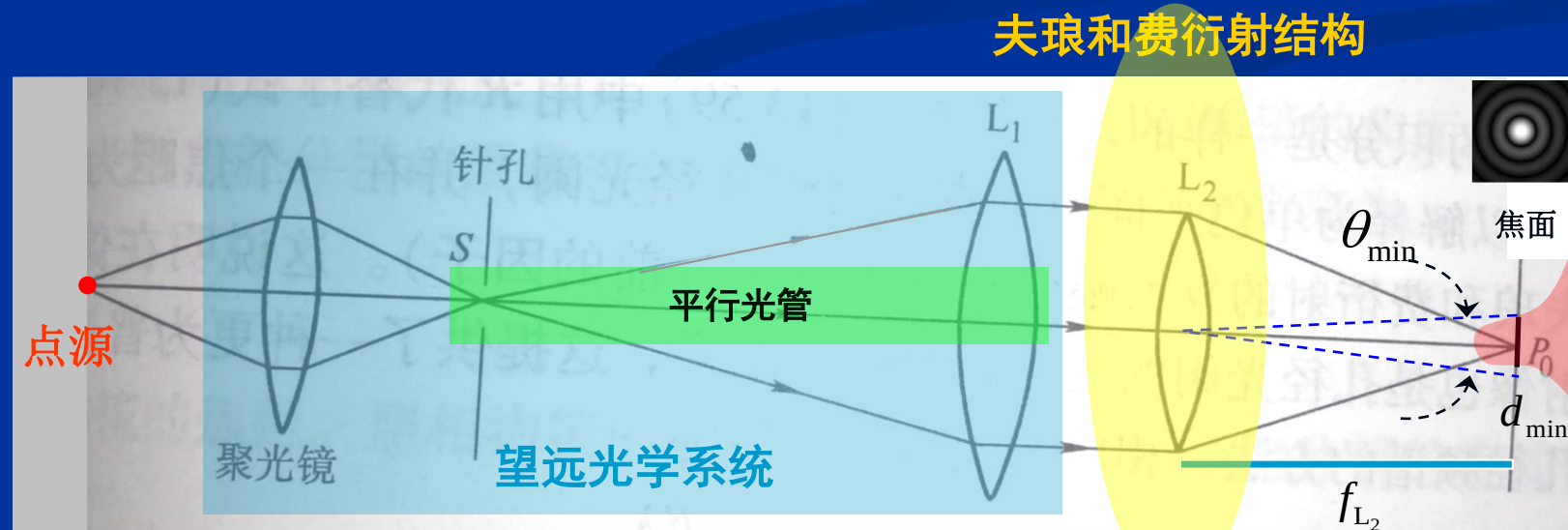
极大和极小	Z	$\frac{I}{I_0} = \left[\frac{2J_1(Z)}{Z} \right]^2$
中央极大	0	1
极小	$1.22\pi = 3.833$	0
次极大	$1.635\pi = 5.136$	0.0175
极小	$2.233\pi = 7.016$	0
次极大	$2.679\pi = 8.417$	0.0042
极小	$3.238\pi = 10.174$	0
次极大	$3.699\pi = 11.620$	0.0016

10.4 光学成像系统的衍射和分辨本领

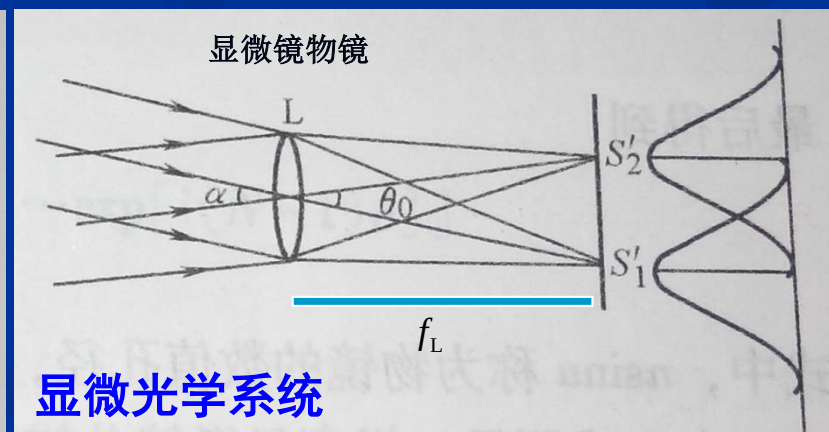
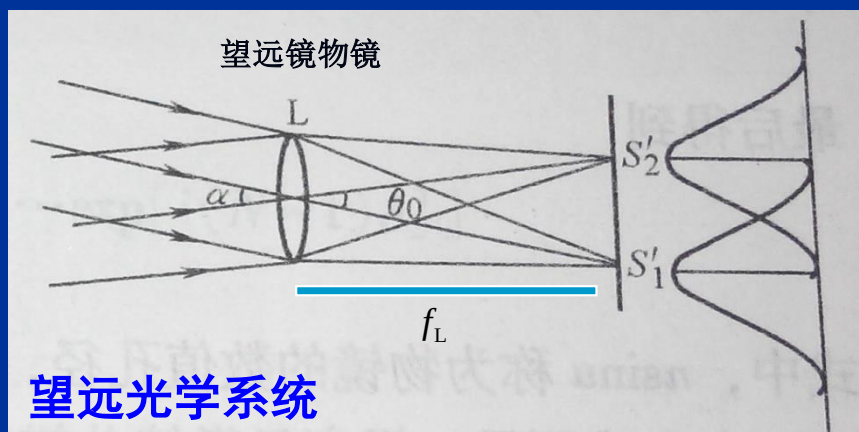
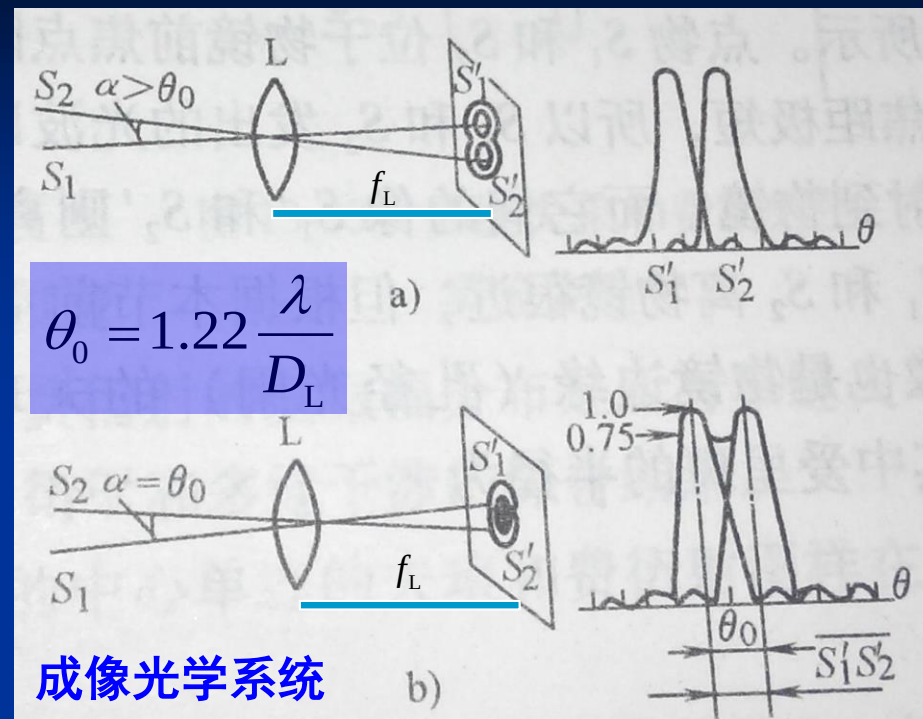
➤ 点物→理想光学系统→点像/衍射斑/艾里斑

➤ 透镜：视为→夫琅和费衍射结构

➤ 点像/衍射斑/艾里斑尺寸：
几何尺寸 $d_{\min} = f_{L_2} \cdot \theta_{\min} = f_{L_2} \cdot 1.22 \frac{\lambda}{D_{L_2}}$ 角尺寸



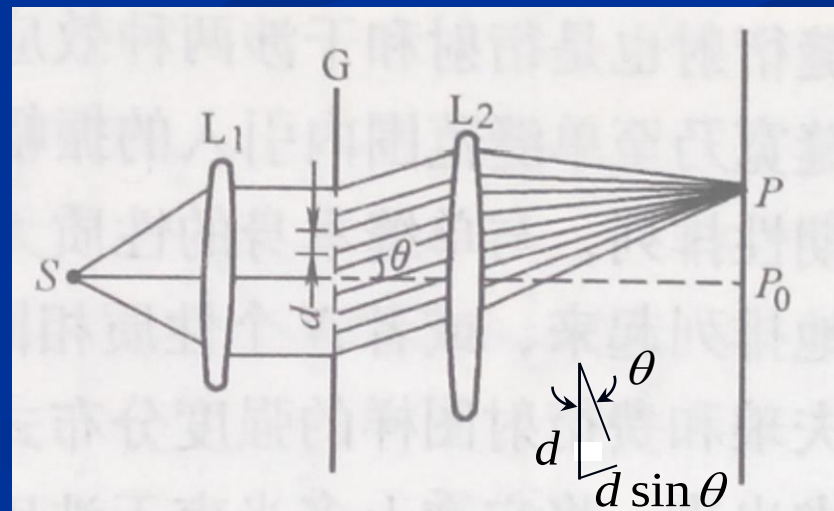
角分辨本领 θ_0



10.5 多缝夫琅和费衍射

- 中心单缝: $\tilde{E}_P = A \left(\frac{\sin \alpha}{\alpha} \right)$
- 相邻单缝在P点的衍射光其相差:

$$\delta = \frac{2\pi}{\lambda} d \sin \theta$$

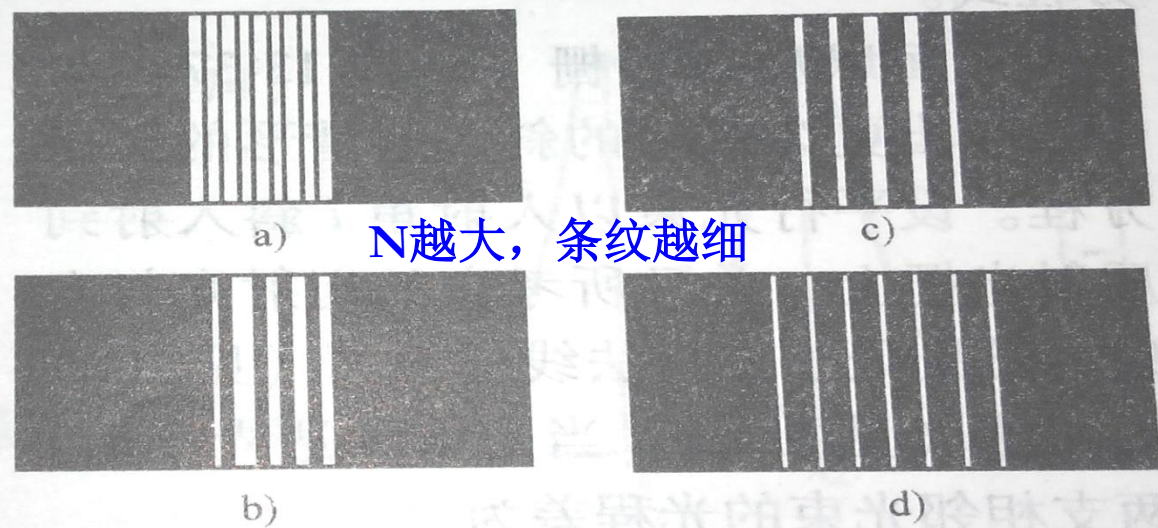
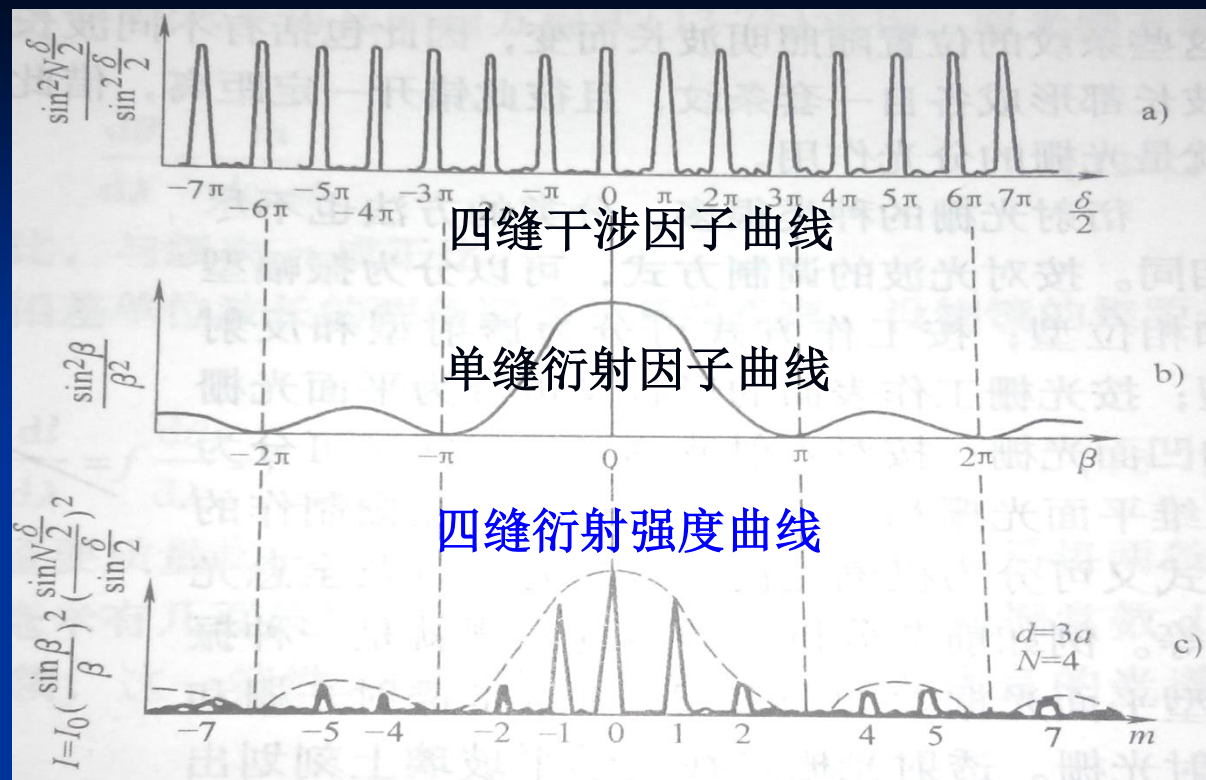


$$\tilde{E}_P = A \left(\frac{\sin \alpha}{\alpha} \right) \left[1 + \exp(j\delta) + \exp(j2\delta) + \dots + \exp(j(N-1)\delta) \right]$$

$$= A \left(\frac{\sin \alpha}{\alpha} \right) \left(\frac{\sin \frac{N}{2} \delta}{\sin \frac{\delta}{2}} \right) \exp \left(j(N-1) \frac{\delta}{2} \right)$$

$$I_P = I_0 A \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(\frac{\sin \frac{N}{2} \delta}{\sin \frac{\delta}{2}} \right)^2$$

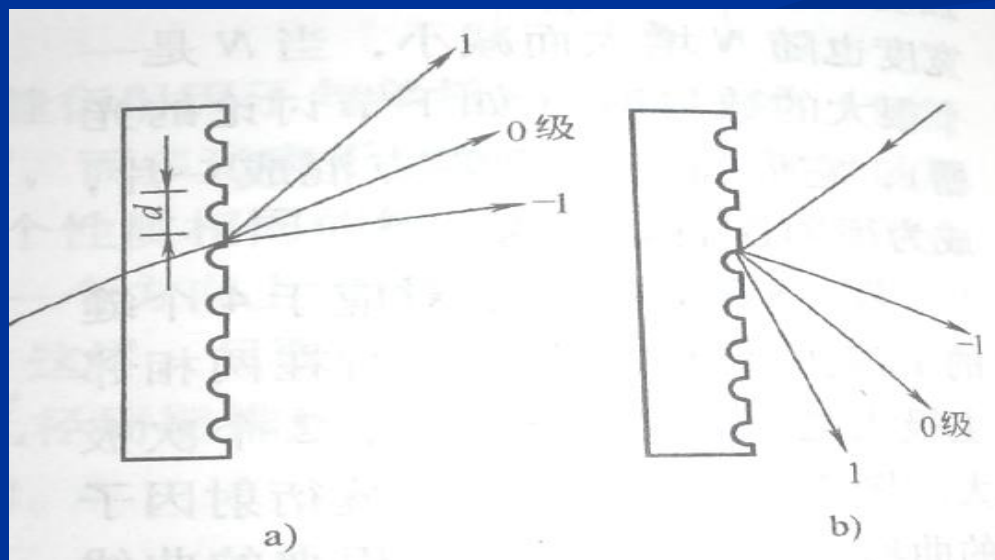
多缝干涉因子



类似FP效应

10.6 衍射光栅

- 衍射光栅： 对入射光的振幅、相位进行空间周期性调制
→ 元件、结构 → 分光、波谱成分按空间分离排布
- 衍射光栅的夫琅和费衍射图样：
光栅光谱
- 典型分类： 振幅型/相位型、透射型/反射型
平面型/三维光栅、机制光栅/复制光栅/全息光栅

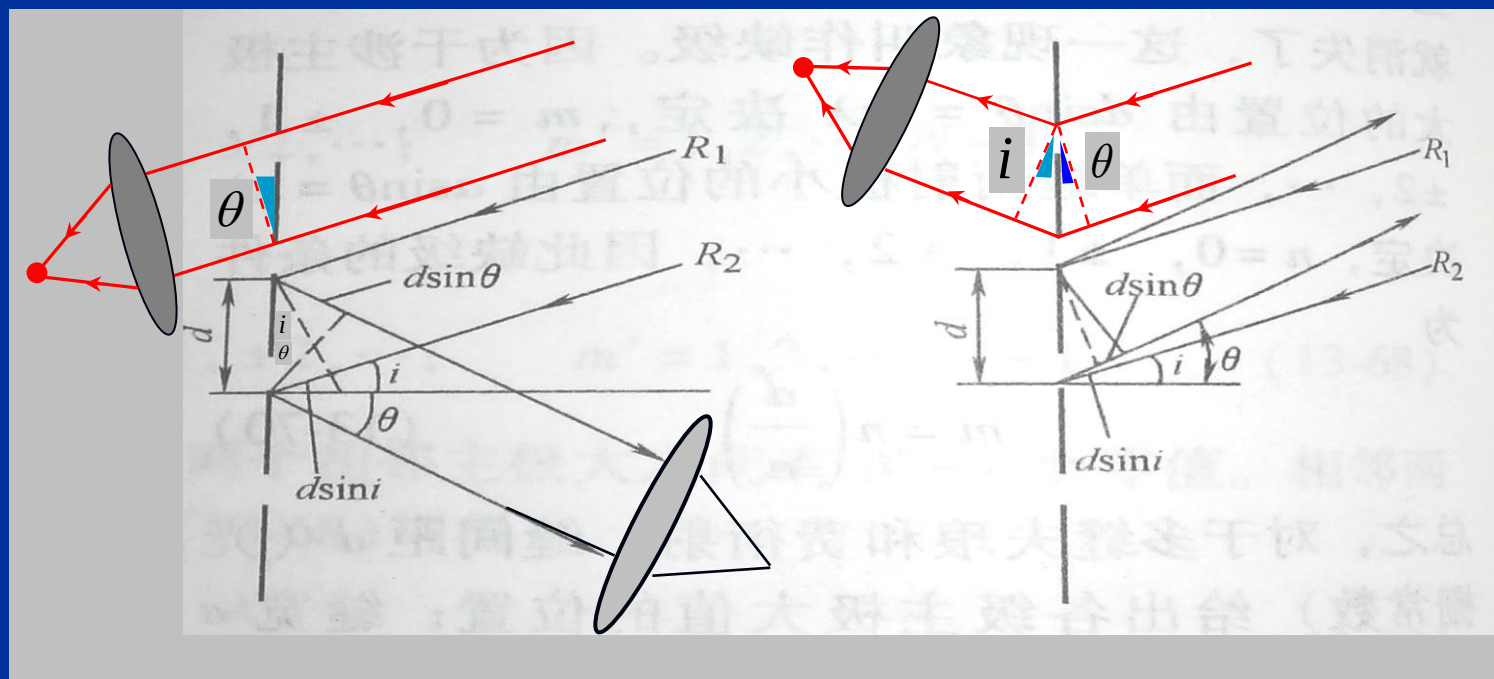


光栅方程

- 基于菲涅耳波带片
- 光栅方程: $\Delta = d \sin i \pm d \sin \theta = m\lambda, m = 0, \pm 1, \pm 2, \dots$

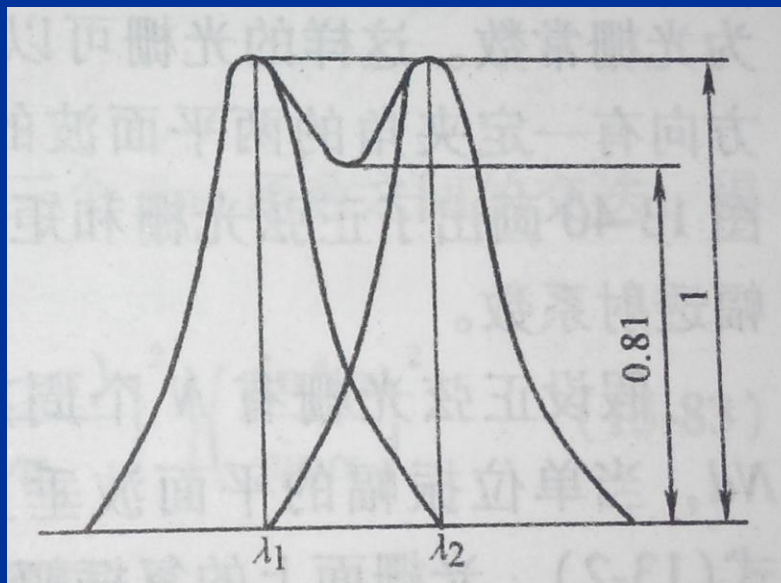
衍射光、入射光同侧/反射: 取+号

异侧/透射: 取-号

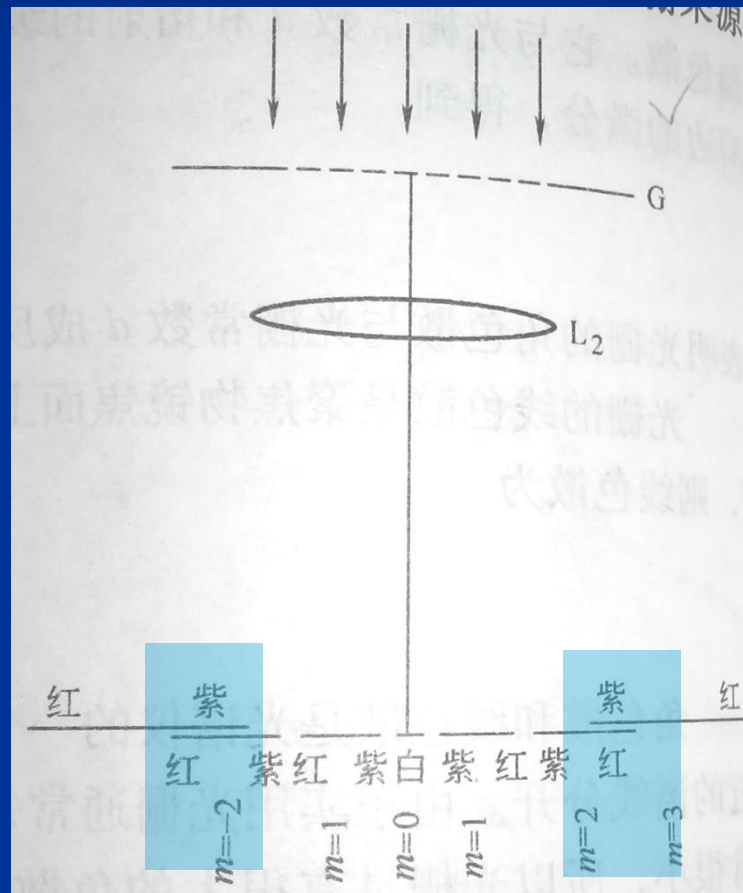


光栅色分辨本领:

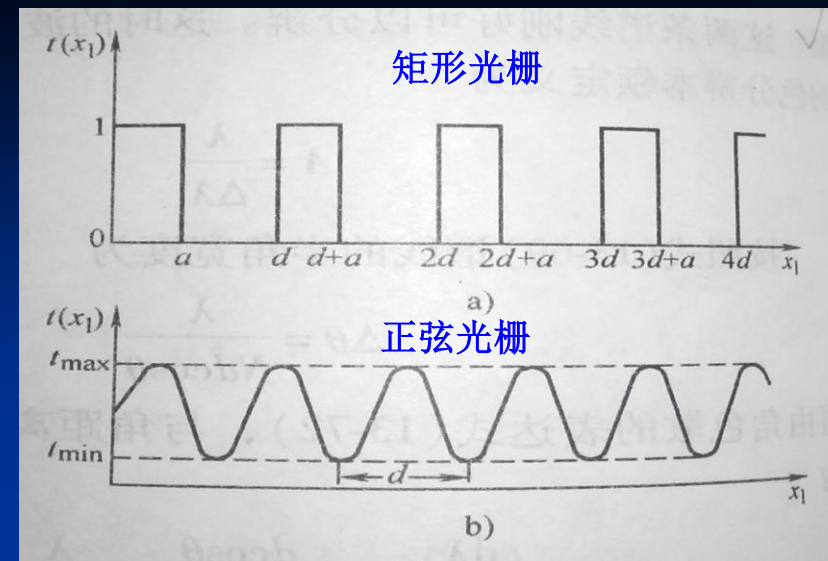
$$A = \frac{\lambda}{\Delta\lambda} = mN, m = 0, \pm 1, \pm 2, \dots$$



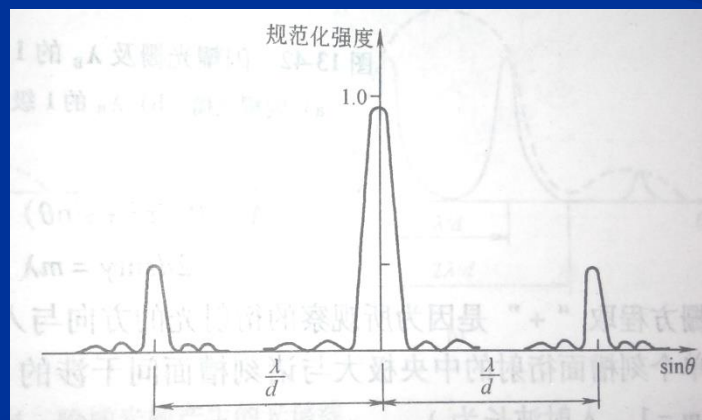
光栅不重叠区: $\Delta\lambda = \frac{\lambda}{m}$



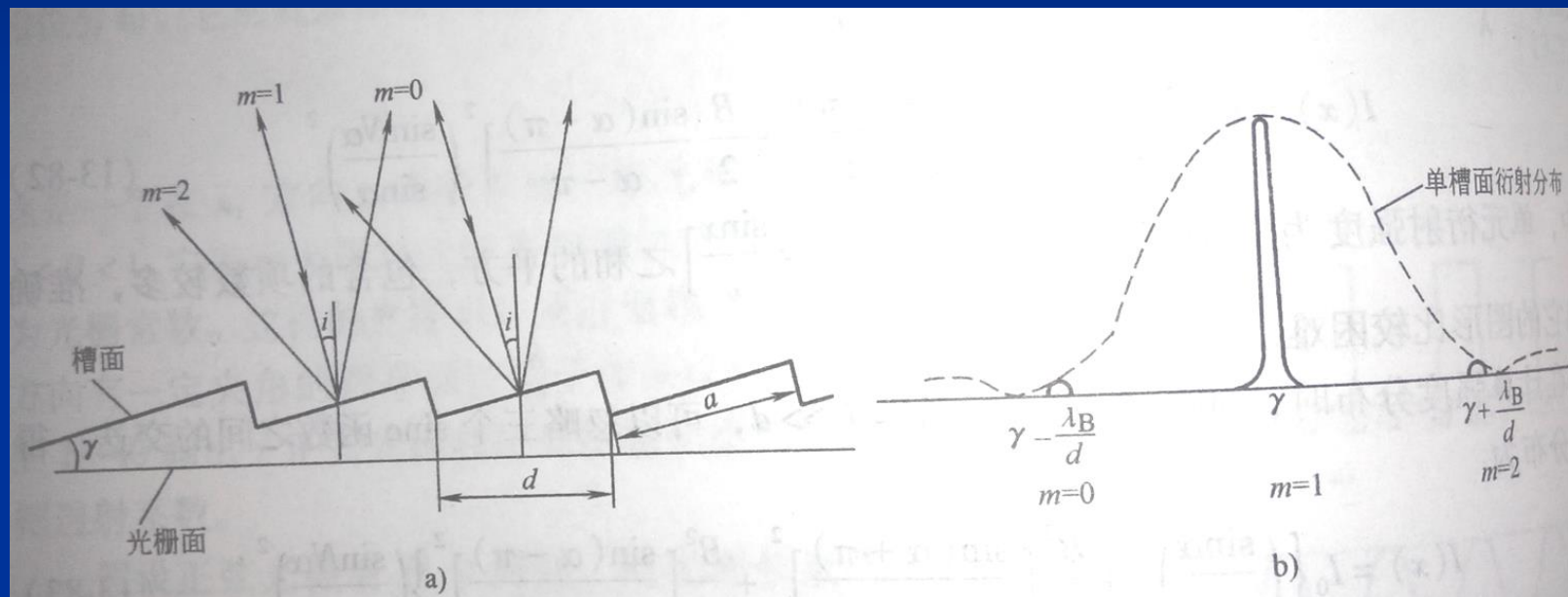
$$\tilde{E}(x_1) = \begin{cases} 1 + B \cos \frac{2\pi}{d} x_1, & x_1 \leq |Nd/2| \\ 0, & x_1 > |Nd/2| \end{cases}$$



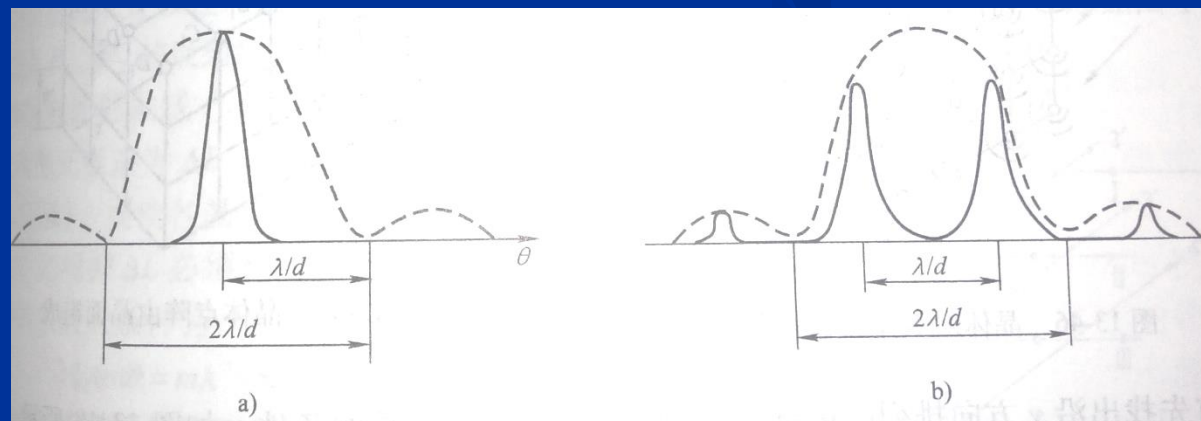
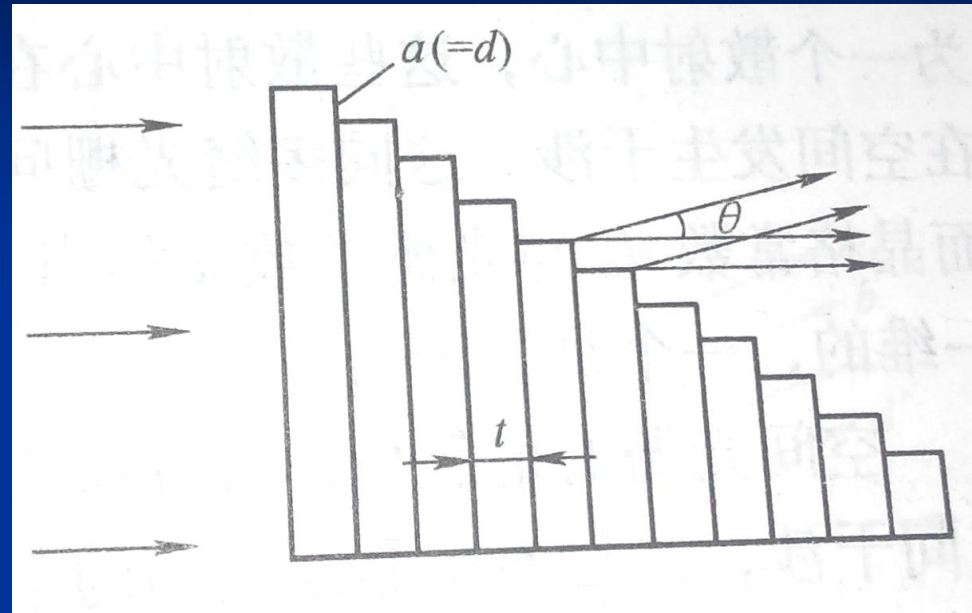
$$I(x_1) = I_0 \left[\frac{\sin \alpha}{\alpha} + \frac{B \sin(\alpha + \pi)}{2(\alpha + \pi)} + \frac{B \sin(\alpha - \pi)}{2(\alpha - \pi)} \right]^2 \left(\frac{\sin N\alpha}{\alpha} \right)^2$$



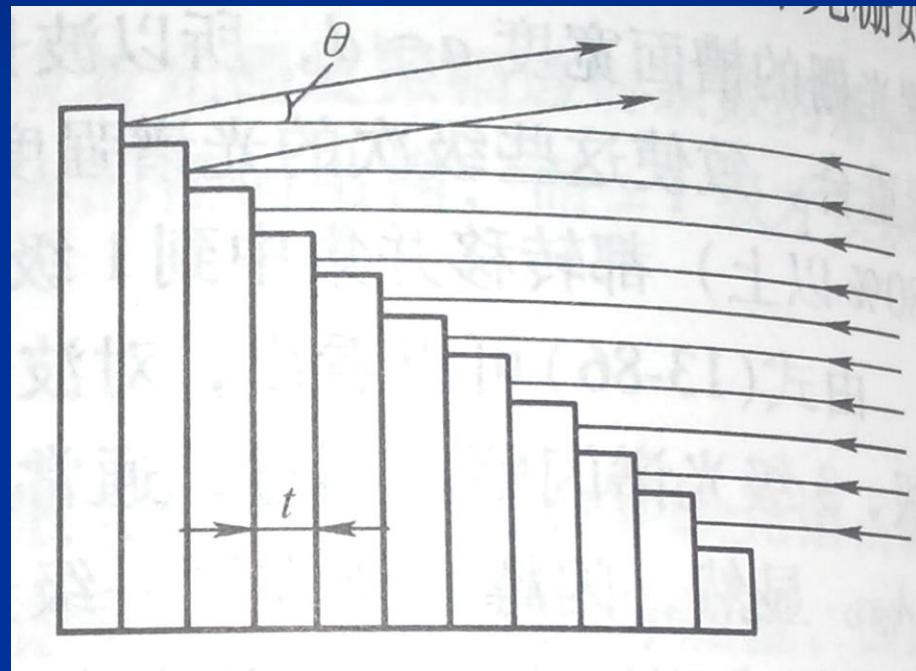
闪耀光栅



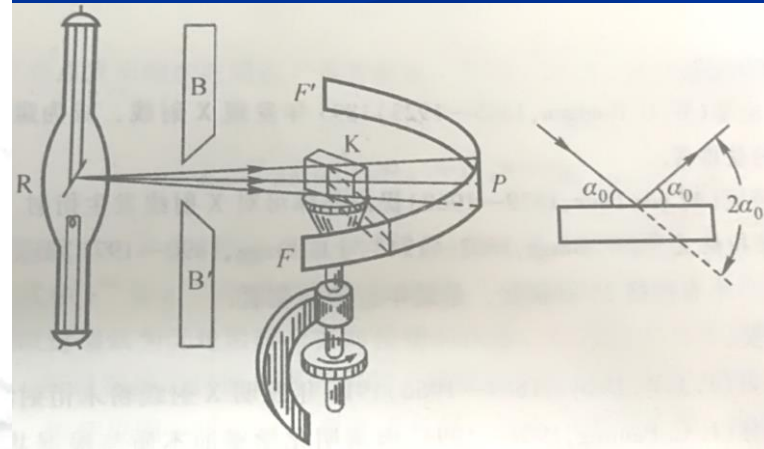
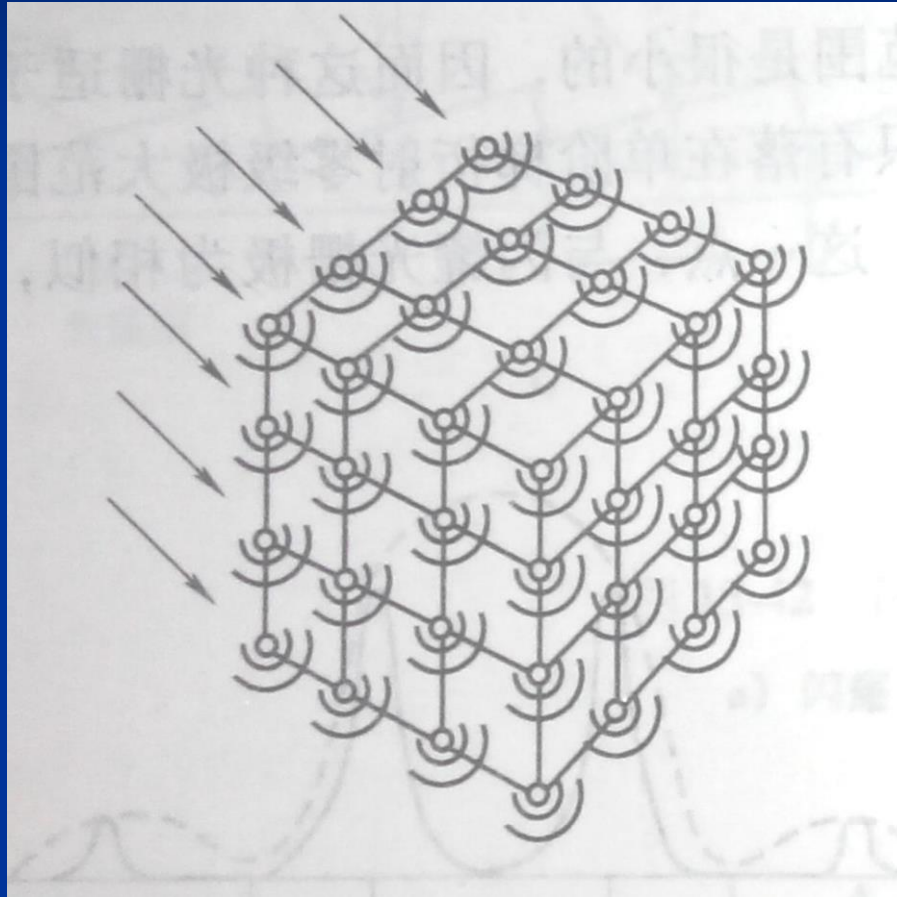
透射式阶梯光栅

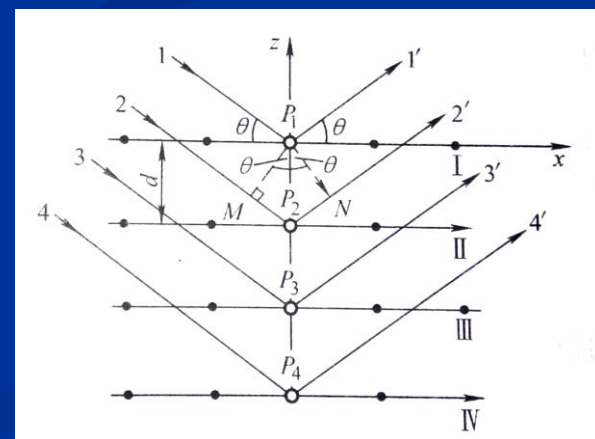
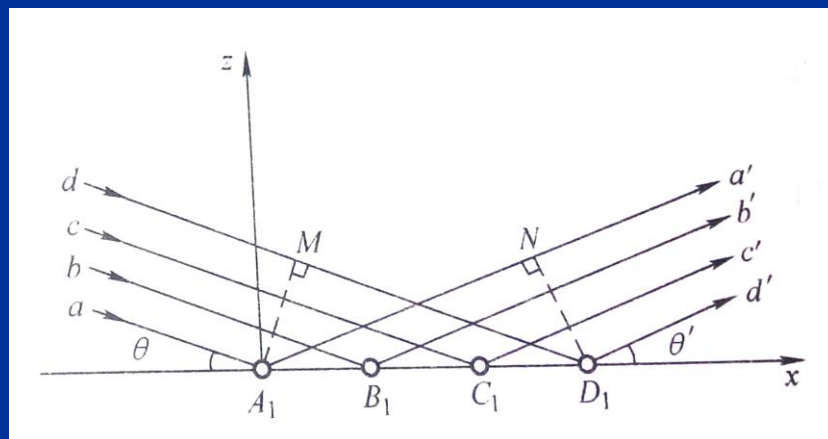
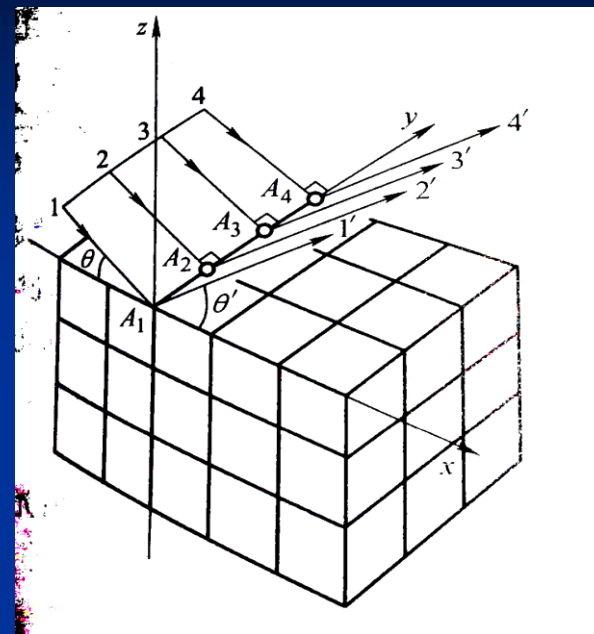
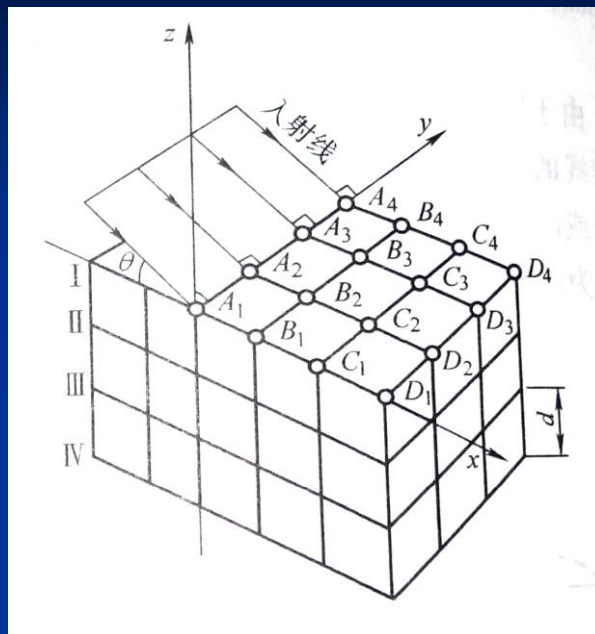


反射式阶梯光栅



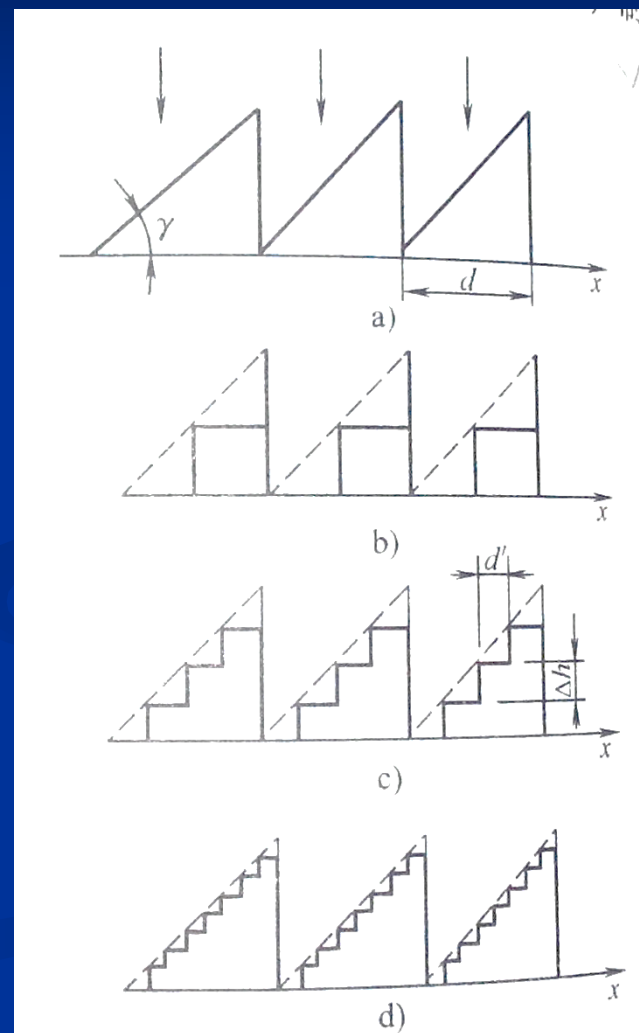
晶体光栅

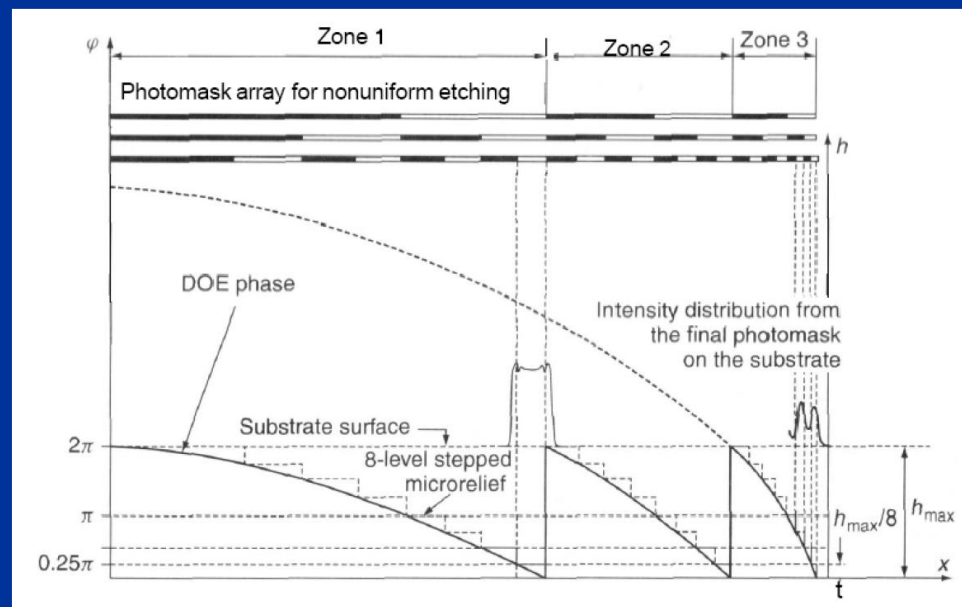
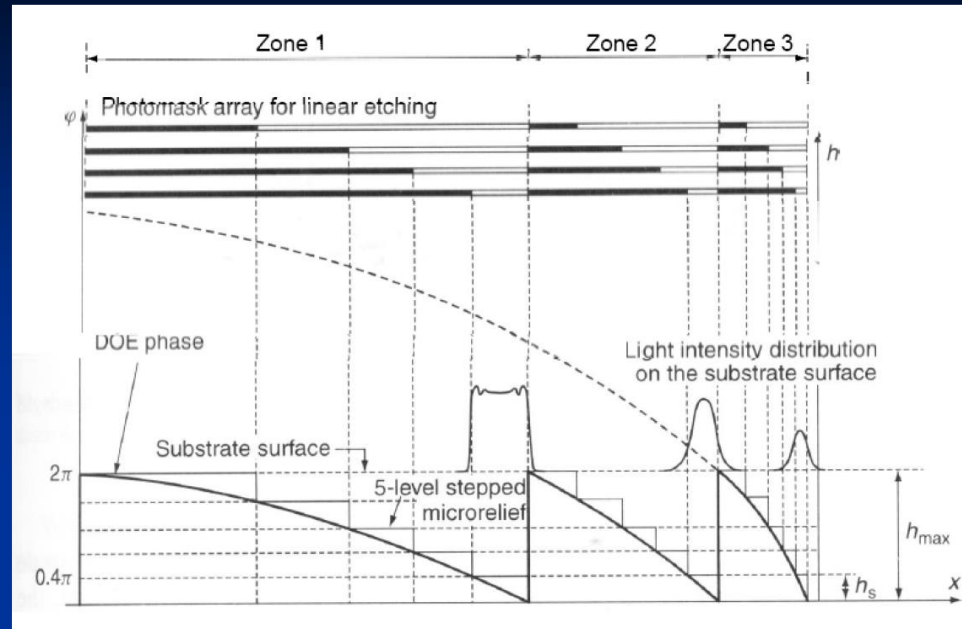




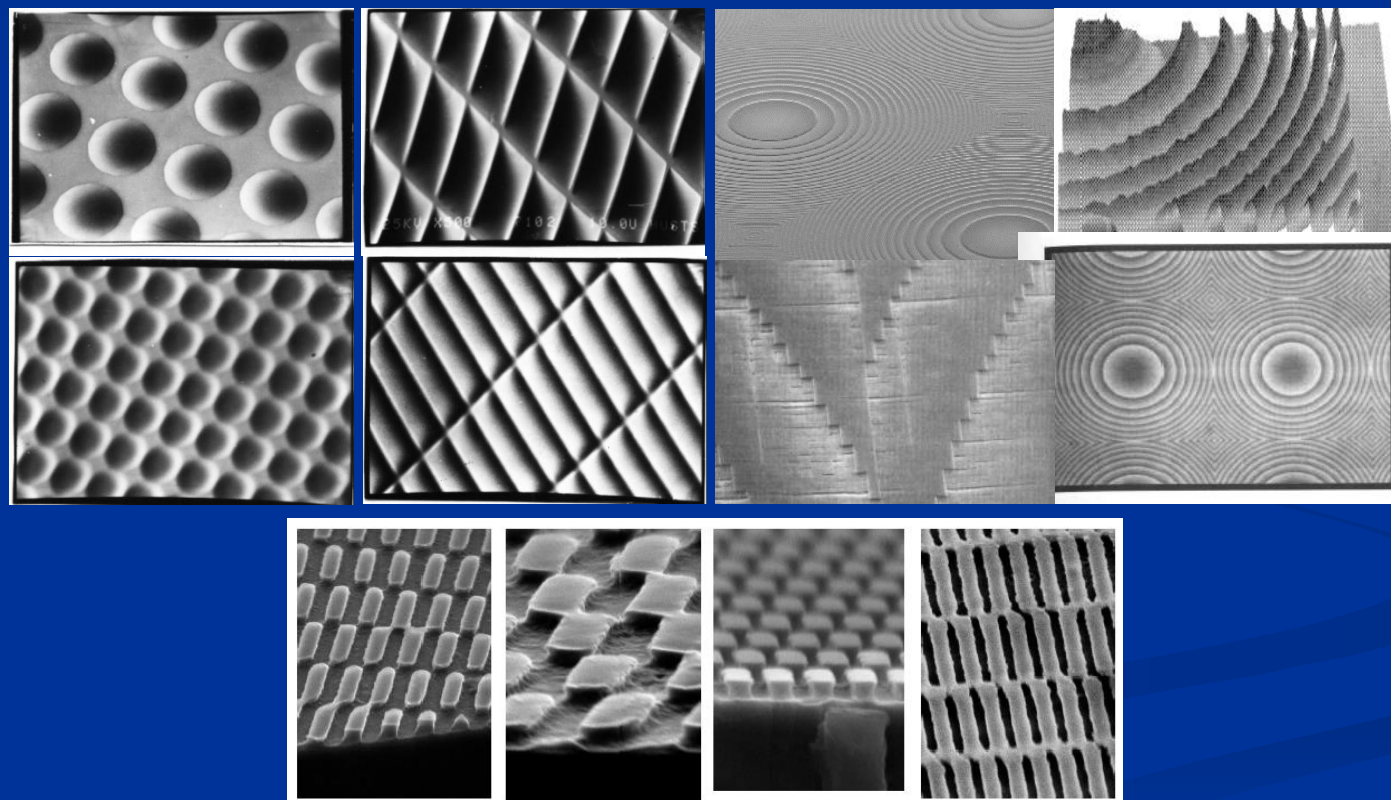
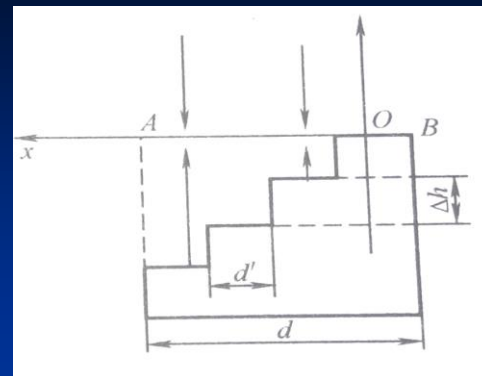
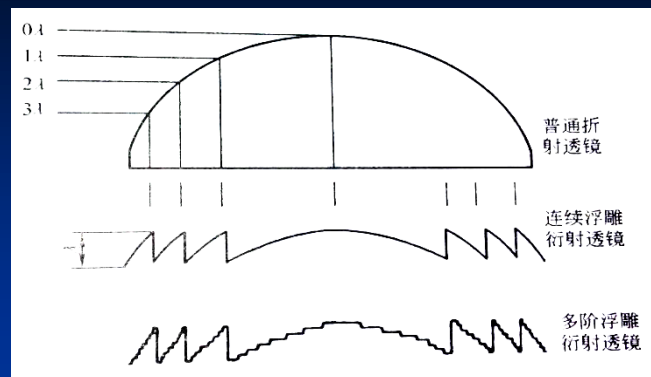
10.7 二元光学

- 基于微电子工艺，将每个周期单元制成 $N=2^n$ ($n=1, 2, 3, \dots$) 个微台阶，相邻微台阶对光波产生 $2\pi/N$ 相差.
- 台阶越多，越接近锯齿形相位轮廓.
- 光栅常数: $d' = d/N$

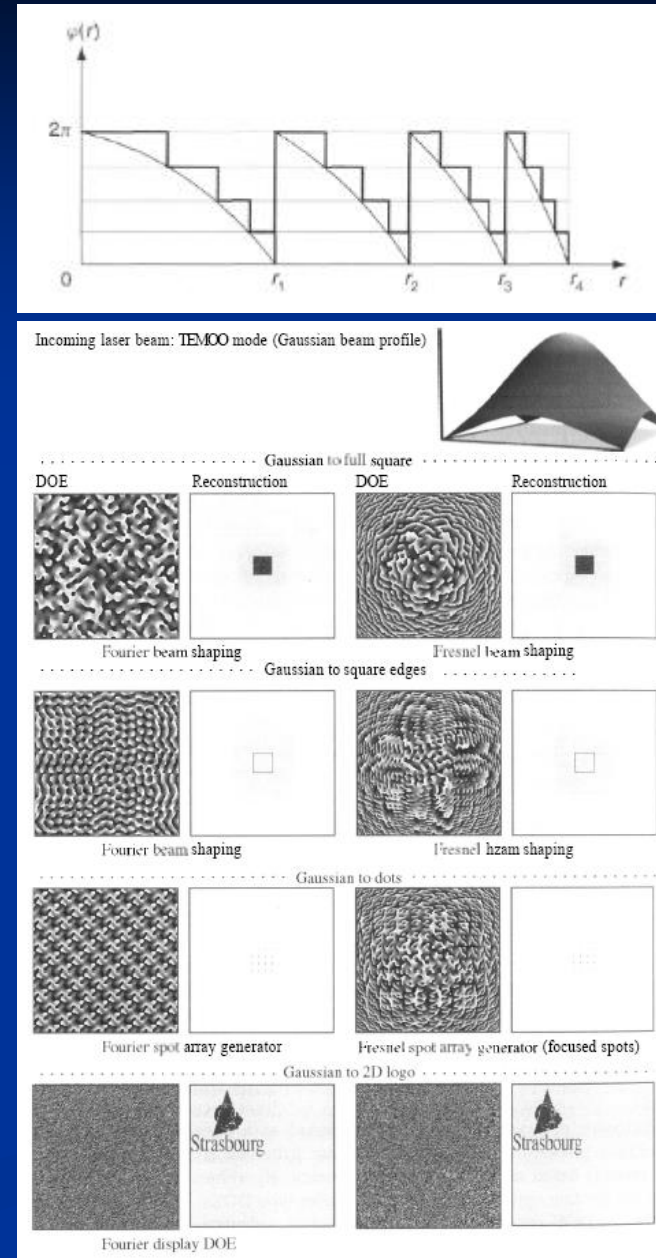
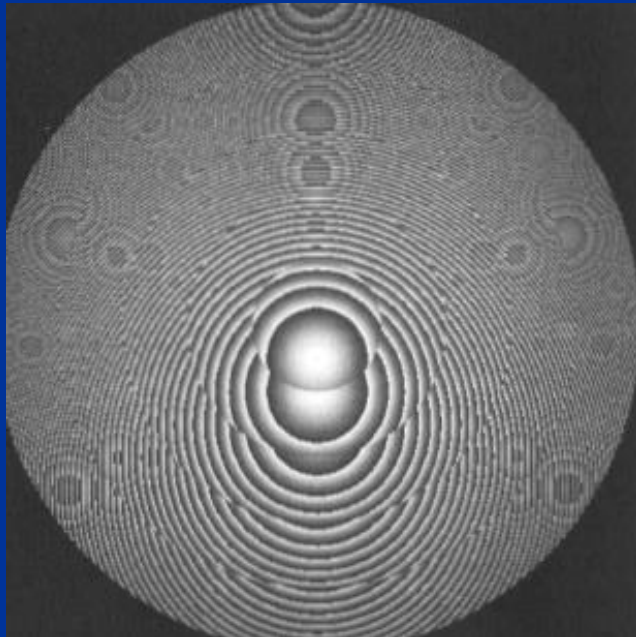
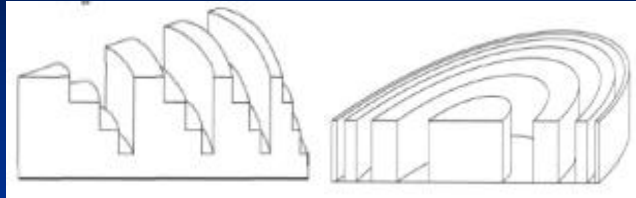


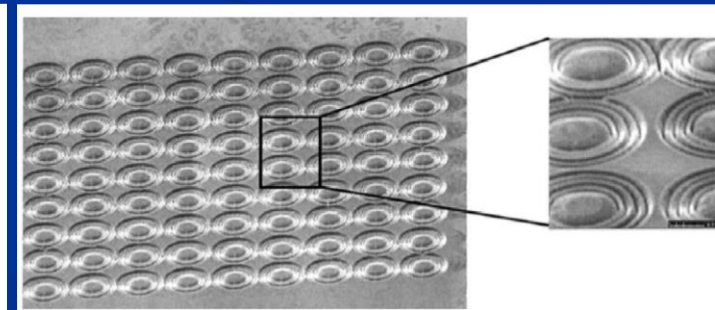
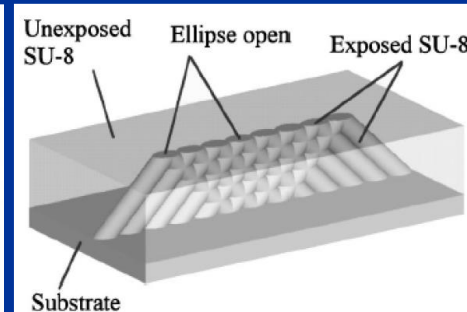
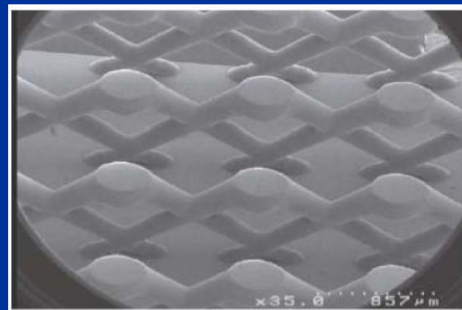
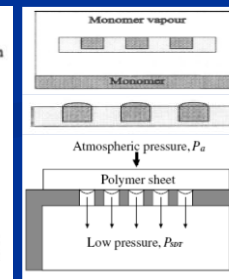
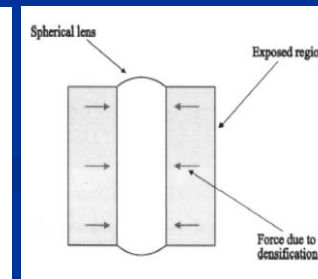
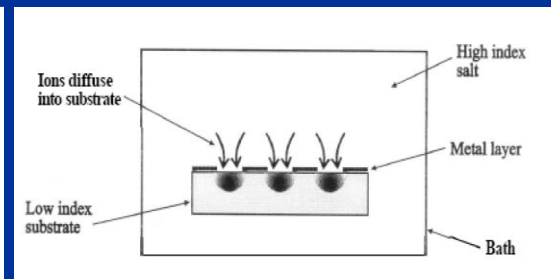
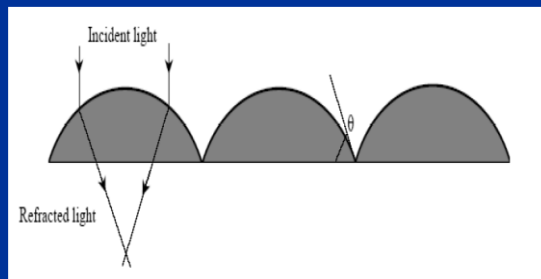
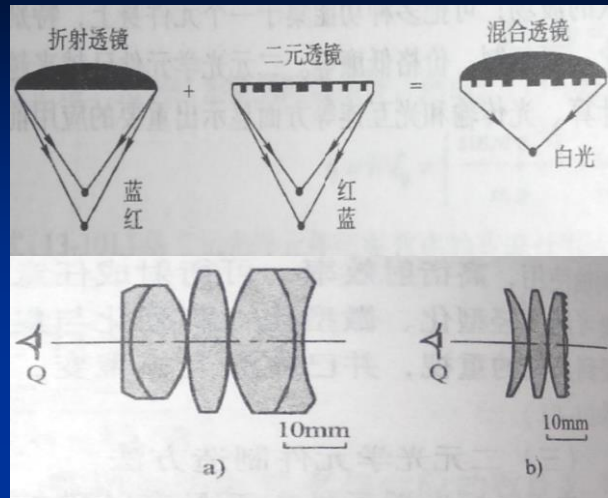


传统衍射结构

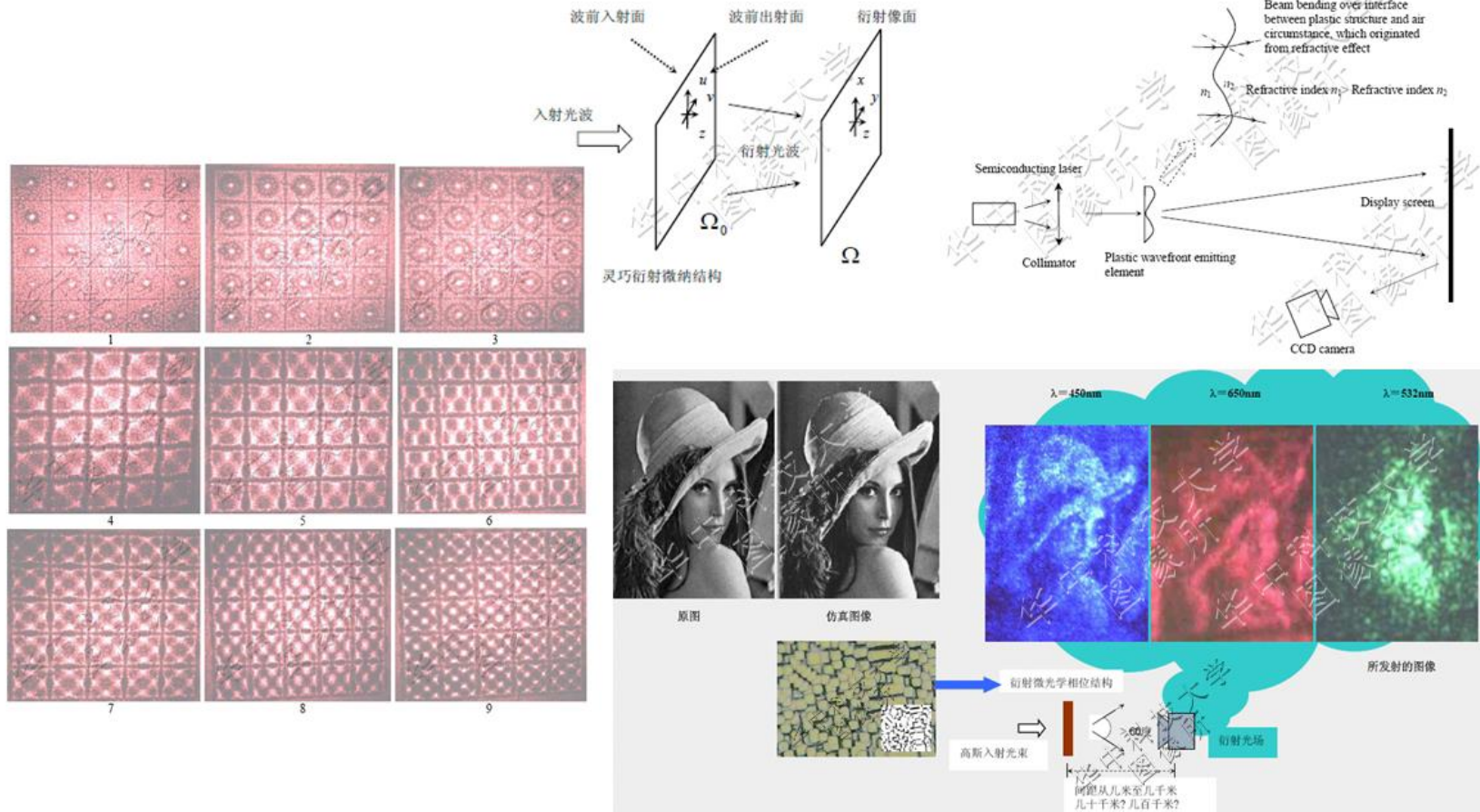


复合：传统衍射结构与超表面





10.8 衍射图像生成



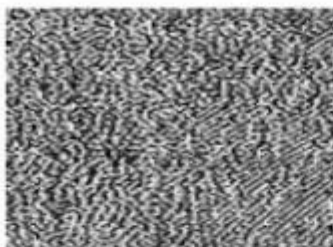
波前

强度图像

Original



Design:
Phasemask

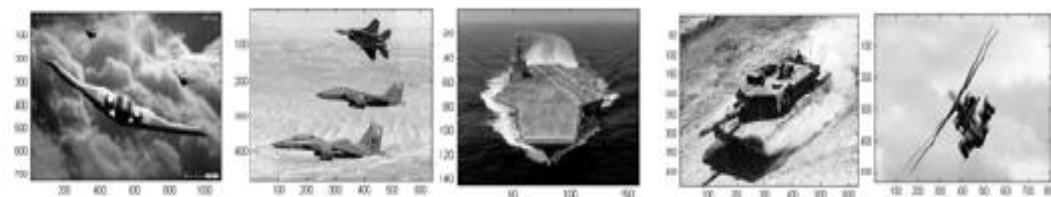


Realization:
Photoresist

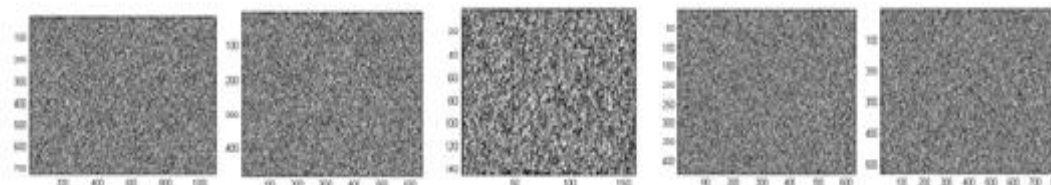


Farfield

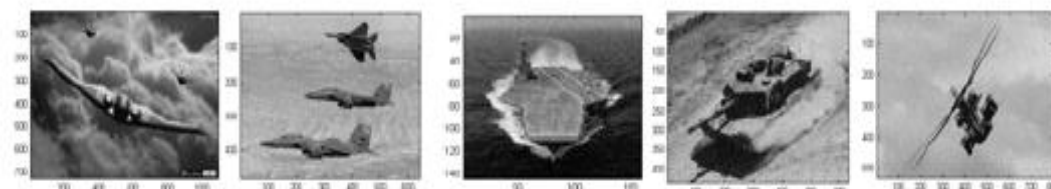




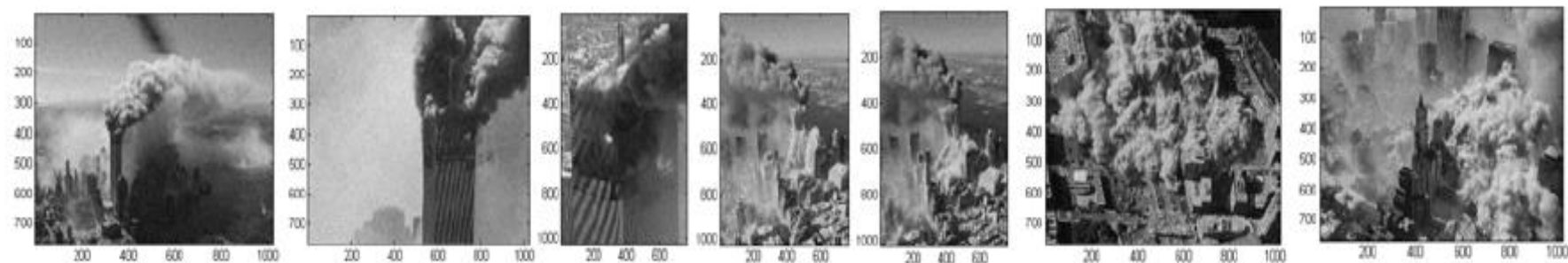
各类目标图像



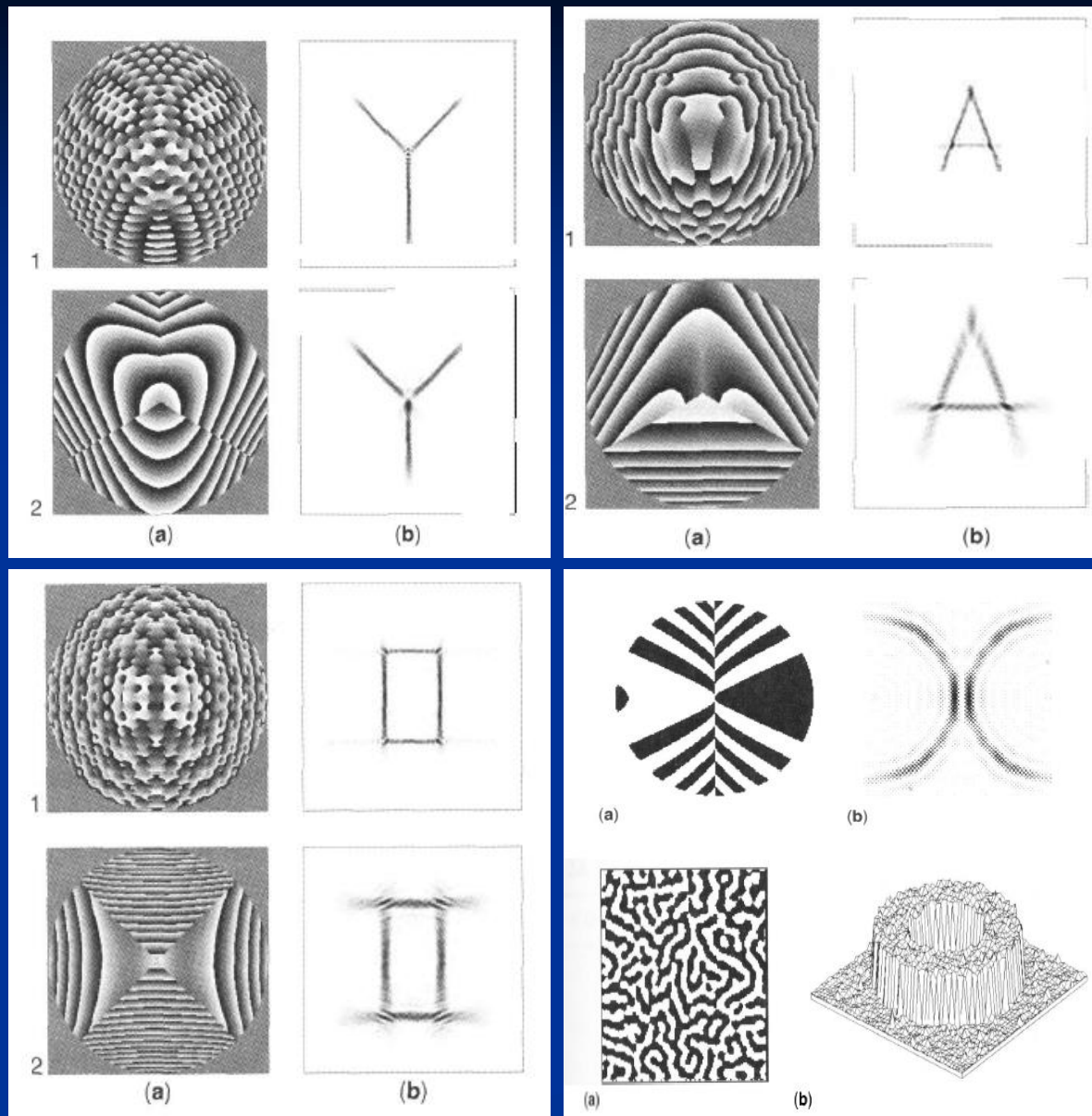
DOE 版图



所生成的图像

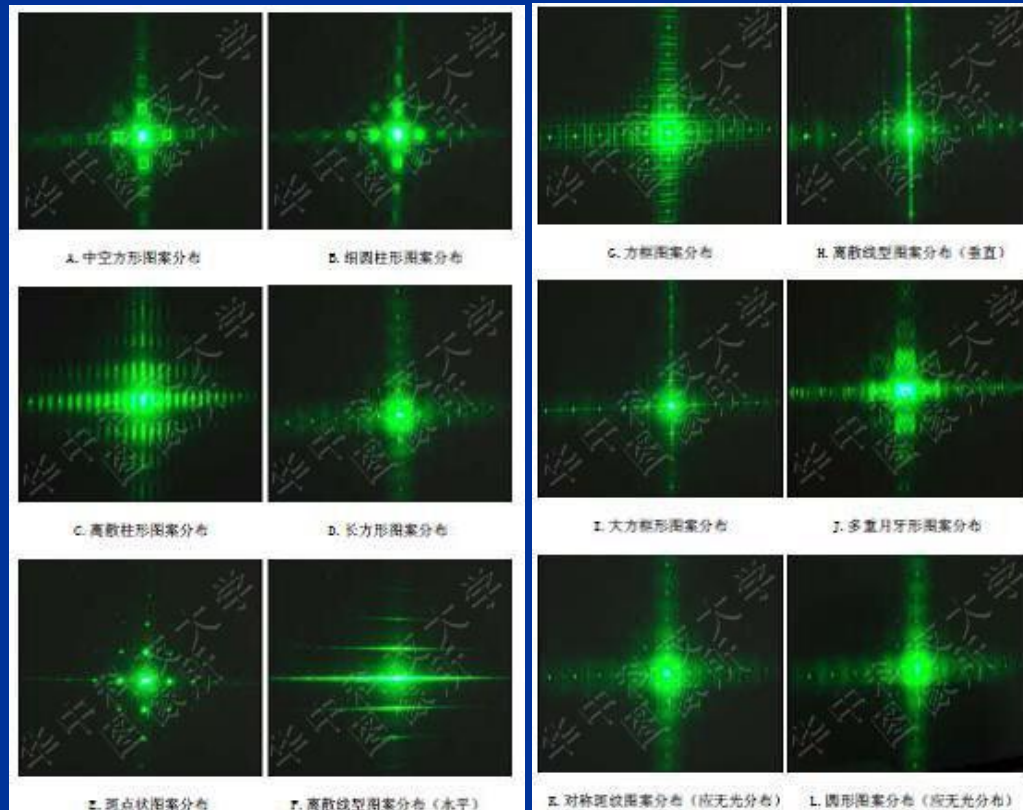
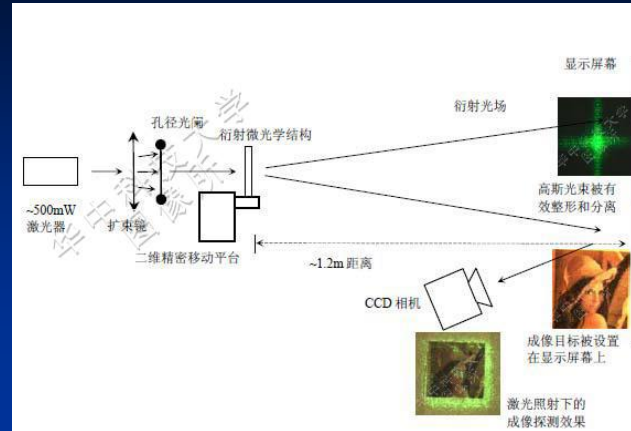


视频仿真图像



发射文字

光束衍射整形



本章小结

解决衍射

表征、特征、典型应用、等问题

- 衍射模型

改造基尔霍夫衍射关系

- 利用

多光束干涉

- 典型应用

分析焦斑特征、成像分辨能力、选谱